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CHAPTER 1

INTRODUCTION TO BLOCKCHAIN-BASED PRIVACY-PRESERVING AUTHENTICATION IN INTERNET OF VEHICLES

1.1 Background

The rapid evolution of intelligent transportation systems has led to the emergence of the Internet of Vehicles (IoV), an advanced paradigm that integrates vehicles, roadside infrastructure, cloud platforms, and users through wireless communication technologies. IoV enables a wide range of applications such as traffic management, accident prevention, autonomous driving, infotainment services, and real-time navigation. However, the large-scale exchange of sensitive vehicular data, including vehicle identity, location, speed, and driver behavior, has introduced serious security and privacy concerns. Due to the high mobility, dynamic topology, and open wireless communication environment of IoV networks, they are highly vulnerable to various cyber threats such as impersonation attacks, Sybil attacks, replay attacks, false data injection, and unauthorized tracking. Traditional centralized security architectures rely on trusted third parties and centralized servers, which suffer from single-point-of-failure, scalability issues, and increased latency. These limitations make conventional security solutions inadequate for next-generation IoV environments.

To address these challenges, blockchain technology has emerged as a promising solution due to its decentralized architecture, immutability, transparency, and cryptographic security. Blockchain eliminates the need for a centralized authority by enabling distributed trust management among vehicles and infrastructure. However, while blockchain enhances trust and data integrity, it introduces challenges such as high computational cost, storage overhead, and privacy leakage if not carefully designed. In addition, the growing sophistication of cyber-attacks necessitates intelligent security mechanisms capable of detecting malicious behavior in real time. Machine learning and deep learning techniques have proven effective in identifying abnormal patterns and malicious activities in dynamic environments like IoV.

Furthermore, ensuring data privacy during communication and storage requires robust cryptographic mechanisms with optimized key management. This thesis focuses on developing efficient, secure, and privacy-preserving authentication protocols for IoV networks by integrating blockchain technology with intelligent learning models and optimized cryptographic techniques.

1.2 Internet of Vehicles (IoV) Security and Privacy

The Internet of Vehicles is an extension of Vehicular Ad Hoc Networks (VANETs) that incorporates cloud computing, edge computing, artificial intelligence, and blockchain technologies. In IoV, vehicles continuously exchange safety and non-safety messages with other vehicles (V2V), roadside units (V2I), pedestrians (V2P), and cloud servers (V2C). While this connectivity improves transportation efficiency and safety, it also exposes vehicles to serious privacy risks. As shown in Figure 1.1, IoV enables multiple communication modes among vehicles, infrastructure, pedestrians, and cloud services, which significantly increases security and privacy challenges.

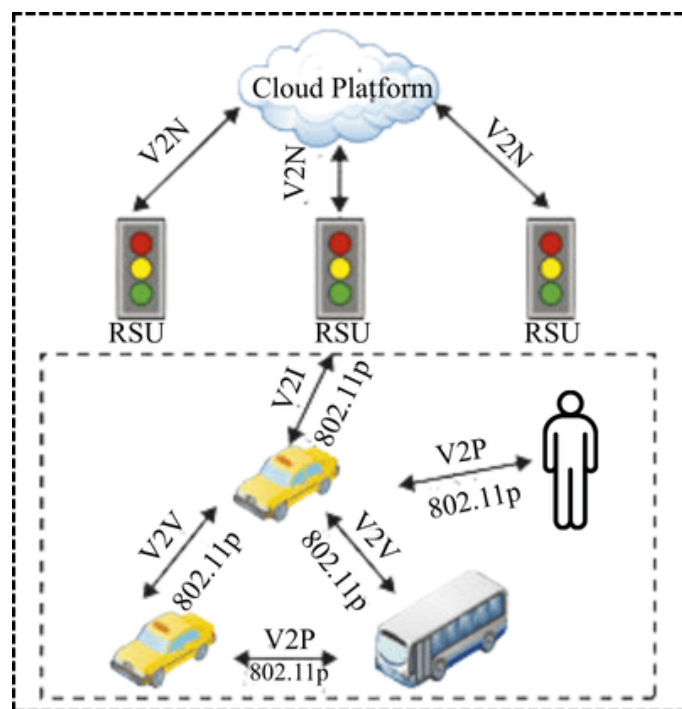


Figure 1.1 General Internet of Vehicles (IoV) communication architecture

Figure 1.1 presents a general Internet of Vehicles (IoV) communication architecture illustrating different communication modes such as Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Network

(V2N). Vehicles equipped with on-board units communicate with roadside units (RSUs), cloud platforms, and nearby entities using wireless technologies. This architecture highlights the highly dynamic and distributed nature of IoV networks, which necessitates secure, scalable, and privacy-preserving authentication mechanisms.

Vehicular data often contains personally identifiable information such as real-time location, travel routes, and driver behavior. Unauthorized access or manipulation of this data can lead to privacy violations, traffic disruption, and even life-threatening situations. Therefore, IoV systems must ensure:

- Authentication of legitimate vehicles and infrastructure
- Data integrity to prevent message tampering
- Conditional privacy, allowing anonymity with traceability
- Resistance to malicious attacks

Conventional cryptographic solutions alone are insufficient due to scalability constraints and the inability to detect intelligent attacks. Hence, advanced solutions combining blockchain, machine learning, and optimized encryption are required.

1.3 Blockchain-Based Authentication in IoV

Blockchain technology provides a decentralized and tamper-proof ledger that securely records transactions among untrusted entities. In IoV networks, blockchain enables secure authentication, data integrity, and trust establishment without relying on a centralized authority. Key Advantages of Blockchain in IoV:

- Decentralization: Eliminates single-point-of-failure
- Immutability: Prevents unauthorized data modification
- Transparency and Traceability: Enables auditing and accountability
- Smart Contracts: Automate authentication and access control

Despite these advantages, blockchain-based IoV systems face challenges such as increased latency, high storage requirements, and privacy leakage due to transparent

ledgers. Therefore, blockchain must be combined with intelligent authentication mechanisms and privacy-preserving cryptography to achieve practical deployment. As shown in Figure 1.2, integrating blockchain with IoV communication enables decentralized transaction management, secure data sharing, and reliable authentication across different vehicular scenarios.

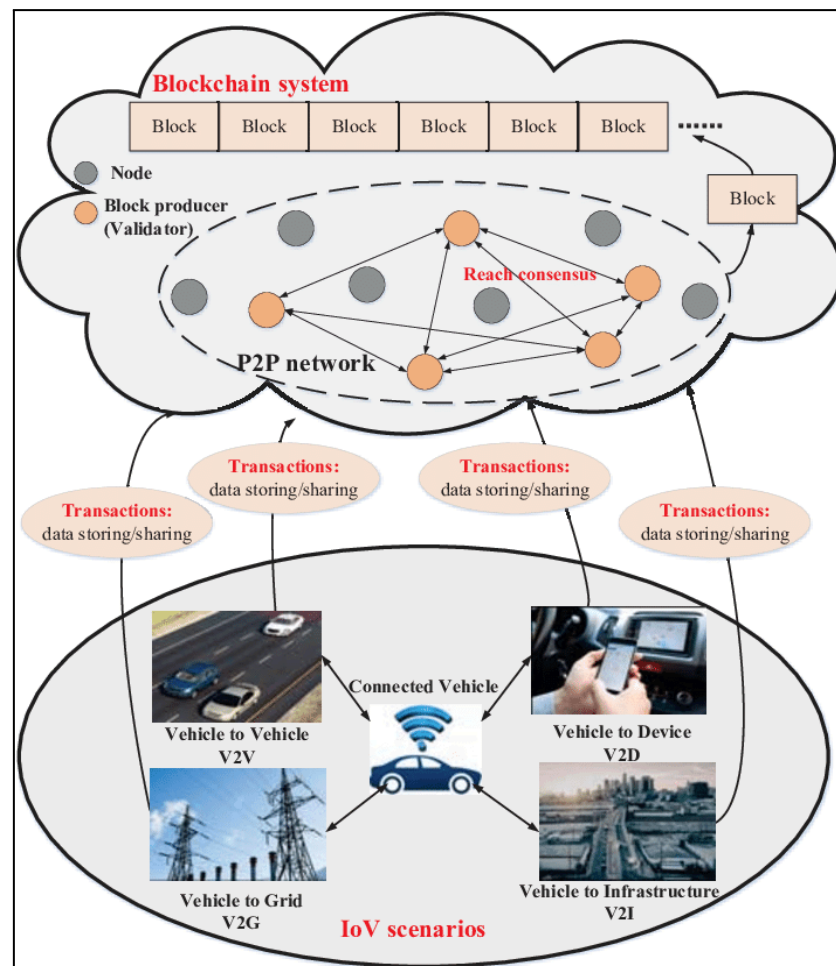


Figure 1.2 Blockchain-based Internet of Vehicles communication and data-sharing architecture

Figure 1.2 illustrates a blockchain-based Internet of Vehicles architecture where vehicular communication scenarios such as Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Device (V2D), and Vehicle-to-Grid (V2G) generate transactions that are securely stored and shared through a decentralized blockchain system. The peer-to-peer network, validators, and consensus mechanism

ensure secure authentication, data integrity, and trusted information exchange among IoV entities.

1.4 Machine Learning and Deep Learning for Malicious Attack Detection

Machine learning techniques play a vital role in detecting malicious activities in IoV networks by analyzing traffic patterns and behavioral anomalies. Traditional rule-based systems fail to adapt to evolving attack strategies, whereas learning-based models can dynamically learn from data.

1.4.1 Machine Learning-Based Authentication

In the initial phase of this research, a Hybrid Adaptive Network (HANet) combining Multi-Layer Perceptron (MLP) and Ridge Classifier was employed to detect malicious nodes efficiently. Parameter optimization using Opposition Beluga Whale Optimization (OBWO) further improved detection accuracy while reducing computational overhead.

1.4.2 Deep Learning-Based Authentication

To overcome the limitations of conventional machine learning, this research further introduces a Cascaded and Dilated Residual Recurrent Neural Network (CD-RRNN). The CD-RRNN effectively captures temporal dependencies in vehicular data, mitigates vanishing gradient problems, and enhances malicious attack detection accuracy in highly dynamic IoV environments.

1.5 Privacy Preservation Using Optimized Cryptographic Techniques

Preserving privacy in Internet of Vehicles (IoV) communication is a critical requirement, as vehicular data often contains sensitive information such as vehicle identity, location, and movement patterns. Unauthorized disclosure of such data can lead to serious privacy violations, tracking, and misuse. Therefore, IoV systems require strong encryption mechanisms supported by efficient key management strategies to ensure confidentiality during data transmission and storage.

Asymmetric cryptographic techniques are widely used in secure communication due to their ability to provide strong security guarantees. Among them, El-Gamal encryption is recognized for its robustness against cryptographic attacks and suitability for secure data exchange. However, the traditional El-Gamal encryption scheme involves high computational complexity, particularly when keys are not optimally selected. This increased complexity can lead to higher encryption and decryption time, making it less suitable for resource-constrained and real-time IoV environments. To overcome these limitations, this research introduces an Optimized Key El-Gamal Encryption (OKEE) scheme, in which encryption keys are optimally generated using the Modified Manta-Ray Foraging Optimization (MMRFO) algorithm. The MMRFO algorithm efficiently explores the search space to select optimal cryptographic keys, thereby reducing computational overhead while maintaining strong security properties. By optimizing key generation, the proposed approach significantly improves encryption efficiency and enhances resistance against cryptographic and brute-force attacks.

Furthermore, integrating optimized cryptographic techniques with blockchain technology strengthens overall privacy preservation in IoV networks. Blockchain provides a secure and tamper-resistant platform for storing encrypted transactions, while OKEE ensures that sensitive vehicular information remains confidential even in a decentralized environment. This combined approach effectively balances privacy, security, and performance requirements in dynamic vehicular networks.

The integration of optimized cryptography with blockchain offers several advantages:

- Secure data transmission across vehicular communication channels
- Confidentiality of vehicular identities and sensitive information
- Reduced computational overhead, suitable for real-time IoV applications
- Enhanced scalability for large-scale vehicular networks

1.6 Challenges in Blockchain-Based IoV Authentication

Despite notable progress in securing Internet of Vehicles (IoV) networks, several critical challenges continue to hinder the effective deployment of blockchain-based authentication mechanisms. These challenges arise due to the unique characteristics of vehicular environments and the inherent complexity of integrating blockchain, intelligent learning models, and cryptographic techniques.

1.6.1 High Mobility and Dynamic Network Topology

One of the most prominent challenges in IoV authentication is the high mobility of vehicles and the continuously changing network topology. Vehicles frequently join and leave the network at high speeds, resulting in short communication durations and unstable connections. Traditional authentication mechanisms often rely on repeated handshakes or centralized verification, which become inefficient in such dynamic environments. Frequent re-authentication increases latency and communication overhead, making it unsuitable for time-critical IoV applications such as collision avoidance and emergency response systems.

1.6.2 Computational Overhead and Resource Constraints

Blockchain-based authentication involves cryptographic operations, consensus mechanisms, and ledger maintenance, all of which introduce significant computational overhead. Vehicular on-board units (OBUs) typically have limited processing power, memory, and energy resources. Performing complex encryption, decryption, and blockchain validation tasks can overload these devices, leading to increased response time and reduced system efficiency. Therefore, designing lightweight authentication and encryption mechanisms that maintain strong security without exhausting vehicular resources remains a major challenge.

1.6.3 Privacy Leakage and Data Exposure Risks

Although blockchain ensures transparency and immutability, these same properties can pose privacy risks in IoV systems. Vehicular communications often include sensitive information such as location, identity, and movement patterns. Storing or sharing such data on transparent ledgers may unintentionally expose private

information to unauthorized entities. Achieving a balance between transparency and privacy is difficult, as complete anonymity can prevent accountability, while full traceability may compromise user privacy. Ensuring conditional privacy, where legitimate users remain anonymous while malicious nodes can be identified when necessary, is a complex research challenge.

1.6.4 Accuracy in Malicious Attack Detection

IoV networks are vulnerable to a wide range of sophisticated attacks, including impersonation, Sybil, replay, and false data injection attacks. These attacks are often adaptive and intelligent, making them difficult to detect using static or rule-based security systems. Accurate detection requires adaptive learning models capable of identifying abnormal behavior patterns in real time. However, achieving high detection accuracy while minimizing false positives and false negatives is challenging, especially in highly dynamic vehicular environments with large volumes of data.

1.6.5 Scalability in Large-Scale IoV Networks

IoV networks are expected to scale to millions of vehicles communicating simultaneously. Authentication frameworks must therefore be scalable and capable of handling large volumes of transactions without performance degradation. Blockchain consensus mechanisms, if not optimized, can introduce delays and bottlenecks as the network grows. Ensuring fast authentication, efficient data storage, and minimal latency in large-scale IoV deployments is a significant challenge that requires careful architectural design and optimization.

1.6.6 Trade-off Between Security, Privacy, and Performance

Strengthening security and privacy often increases computational complexity and communication delay, while lightweight solutions may weaken protection against attacks. Achieving an optimal balance between security strength, privacy preservation, computational efficiency, and system scalability is one of the most difficult aspects of IoV authentication design. This trade-off becomes even more critical in real-time vehicular applications, where delays can directly impact safety.

1.7 Significance of the Research

This research proposes two progressive authentication frameworks for IoV networks:

1. A machine learning–based blockchain authentication model using HANet with OBWO optimization
2. An advanced deep learning and cryptography–integrated blockchain framework using CD-RRNN and OKEE with MMRFO

The proposed solutions significantly improve authentication accuracy, reduce attack detection latency, enhance privacy preservation, and minimize computational overhead. By integrating intelligent learning models with optimized cryptographic techniques, this research provides a robust, scalable, and secure IoV authentication architecture suitable for real-world deployment.

1.8 Objectives of the Research

The objectives of the work are:

- To design a secure blockchain-based authentication system for Internet of Vehicles (IoV) by using machine learning techniques to detect malicious vehicles, while ensuring data integrity, vehicle anonymity, and controlled traceability in the network.
- To improve the security and efficiency of IoV communication by applying deep learning models and optimized cryptographic methods, so as to reduce computational overhead, increase attack detection accuracy, and provide stronger privacy protection in dynamic vehicular environments.

1.9 Organization Of The Thesis

The thesis is organized as follows:

Chapter 1 introduces the research background, motivation, key challenges in Internet of Vehicles (IoV) security, and the objectives of the proposed work.

Chapter 2 presents a comprehensive review of existing security, privacy, and authentication techniques in IoV, with particular emphasis on blockchain-based and intelligent learning-based approaches, and identifies the research gaps addressed in this thesis.

Chapter 3 describes the machine learning-based blockchain authentication framework proposed in this research. This chapter details the use of smart contract-enabled blockchain for secure data storage, the Hybrid Adaptive Network (HANet) for malicious attack detection, and the optimization of model parameters using the Opposition Beluga Whale Optimization (OBWO) algorithm.

Chapter 4 presents the advanced deep learning and cryptography-integrated authentication protocol. This chapter explains the Cascaded and Dilated Residual Recurrent Neural Network (CD-RRNN) for effective malicious attack detection, along with the Optimized Key El-Gamal Encryption (OKEE) scheme using Modified Manta-Ray Foraging Optimization (MMRFO) to ensure enhanced privacy preservation and computational efficiency.

Chapter 5 provides the experimental setup, performance evaluation, and comparative analysis of the proposed frameworks with existing state-of-the-art IoV authentication methods, using standard performance metrics.

Chapter 6 concludes the thesis by summarizing the major contributions, discussing limitations, and outlining potential directions for future research in blockchain-based IoV security.

1.10 Summary

This chapter introduced the significance of secure and privacy-preserving authentication in Internet of Vehicles (IoV) networks. The shortcomings of conventional centralized security solutions were discussed, highlighting issues such as scalability and single-point-of-failure. The need for decentralized blockchain-based authentication frameworks was emphasized. The role of machine learning and deep learning techniques in detecting malicious activities was outlined. The importance of

optimized cryptographic methods for ensuring privacy and efficiency was also presented. The integration of blockchain, intelligent learning models, and cryptography was identified as an effective solution to IoV security challenges. The objectives and scope of the research were summarized. The next chapter presents a detailed literature review of existing IoV security and blockchain-based authentication approaches.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The Internet of Vehicles (IoV) represents a critical component of modern intelligent transportation systems, enabling vehicles to exchange information with surrounding entities to improve safety, efficiency, and driving intelligence. Through continuous communication among vehicles, roadside units, cloud services, and edge platforms, IoV supports applications such as collision avoidance, traffic optimization, and autonomous decision-making. However, the open and highly dynamic nature of vehicular communication significantly increases exposure to security threats and privacy risks. Authentication plays a vital role in ensuring that only legitimate vehicles and infrastructure entities participate in IoV communication. Without strong authentication mechanisms, malicious nodes can inject false messages, impersonate trusted vehicles, or disrupt network operations, potentially leading to severe safety consequences. Conventional authentication schemes, which rely heavily on centralized authorities and static trust models, are often inadequate in large-scale and rapidly changing vehicular environments due to high latency, limited scalability, and vulnerability to single-point failures.

To overcome these challenges, decentralized security frameworks based on blockchain technology have been widely investigated. Blockchain enables distributed trust establishment, immutable record keeping, and secure verification of transactions without requiring a central authority. Nevertheless, blockchain-based IoV systems introduce new challenges, including computational overhead, storage complexity, and privacy exposure when sensitive data is recorded on transparent ledgers. These limitations necessitate the integration of privacy-preserving mechanisms and efficient authentication designs. In recent years, intelligent learning techniques such as machine learning and deep learning have been adopted to enhance IoV security by identifying malicious behaviors through data-driven analysis. These models can adapt to evolving attack patterns and improve detection accuracy in dynamic environments.

Additionally, optimized cryptographic techniques have been explored to protect vehicular data while minimizing computational cost, making them suitable for real-time IoV applications.

This chapter critically reviews existing research on IoV authentication, blockchain-enabled security frameworks, privacy-preserving mechanisms, intelligent attack detection models, and optimized cryptographic solutions. The reviewed studies are analyzed to highlight their contributions and limitations, ultimately identifying open research challenges that motivate the proposed solutions presented in subsequent chapters.

2.2 Blockchain and Intelligent Learning-Based Security Frameworks for IoV

2.2.1 Blockchain-Based Authentication Mechanisms

Xiaoliang Wang et al. (2019) proposed one of the early decentralized authentication mechanisms for IoV using blockchain consensus. Their work addressed the problem of malicious and selfish vehicle behavior caused by the open and self-organized nature of vehicular networks. By replacing centralized trust authorities with blockchain consensus, the proposed mechanism enhanced resistance to impersonation and message manipulation attacks. Simulation results demonstrated improved trust establishment; however, the framework did not explicitly address privacy preservation or computational overhead, limiting its suitability for large-scale deployments.

Chen Mao et al. (2021) designed an IoV authentication platform based on Hyperledger Fabric, leveraging blockchain's modular architecture to replace traditional certificate authority-based authentication. Their scheme integrated cryptographic identity verification with blockchain endorsement, ensuring tamper-proof authentication records. While the approach improved trust and data integrity, it relied on permissioned blockchain infrastructure, which may limit scalability in highly dynamic vehicular environments. H. Chai et al. (2022) introduced CyberChain, a CyberTwin-empowered blockchain framework for IoV authentication. Unlike conventional approaches, CyberChain decouples blockchain consensus from

the physical vehicular network by executing consensus operations in cyberspace. A privacy-preserving parallel Pedersen commitment mechanism and a diffused PBFT consensus algorithm were proposed to reduce latency and storage overhead. Simulation results showed significant performance improvements, though the framework mainly focused on authentication efficiency rather than intelligent attack detection.

2.2.2 Privacy-Preserving Authentication and Conditional Traceability

Privacy preservation is a critical concern in IoV systems, as vehicular messages often contain sensitive identity and location information. Xia Feng et al. (2022) addressed this issue by proposing PBAG, a blockchain-based authentication protocol with global-updated commitments. Instead of querying the blockchain during every authentication, vehicles use pre-updated commitments, significantly reducing verification time. The scheme supports anonymity, unlinkability, and conditional traceability, but depends on frequent global updates, which may introduce synchronization challenges. C. Lin et al. (2021) presented a blockchain-based conditional privacy-preserving authentication (BCPPA) protocol using Ethereum. The scheme simplified certificate management by integrating blockchain with key derivation techniques and supported batch verification to reduce authentication delay. Despite improved efficiency, the reliance on public blockchain platforms introduces transaction cost concerns.

Pravin Mundhe et al. (2022) proposed a hybrid blockchain-based conditional privacy-preserving authentication scheme to reduce certificate revocation list (CRL) overhead. By storing pseudo-identities and encrypted real identities on the blockchain, the scheme enabled efficient revocation while maintaining anonymity. Experimental evaluation demonstrated reduced computation cost, though blockchain storage growth remained a challenge. Kirti A. Yadav and Vijayakumar (2022) designed a lightweight privacy-preserving signature-based authentication protocol using elliptic curve cryptography. Their approach reduced trusted authority burden by involving roadside

units in authentication assistance. While computationally efficient, the scheme focused primarily on signature authentication without intelligent attack detection.

2.2.3 Blockchain Integrated with Machine Learning and Deep Learning

R. Kumar et al. (2022) proposed P2SF-IoV, a privacy-preserving secured framework integrating blockchain with deep learning-based intrusion detection. Blockchain ensured secure data transmission, while a deep learning module detected attacks using IoT-Botnet and ToN-IoT datasets. The framework achieved higher detection accuracy than existing solutions but incurred additional learning overhead. Shujuan Wang et al. (2022) introduced a blockchain-based trust management system where vehicles' credibility was evaluated using a deep learning verification model. A Proof-of-Trust consensus algorithm incentivized honest behavior while penalizing malicious vehicles. The framework effectively detected false information dissemination, though training complexity increased with network scale.

Gunasekaran Manogaran et al. (2023) proposed a token-based authorization and authentication scheme combining blockchain with random forest learning. Frequent token updates were minimized to reduce overhead, improving communication efficiency. However, classical machine learning models may struggle with highly dynamic attack patterns. Ali et al. (2021) provided a comprehensive review of machine learning and deep reinforcement learning techniques for secure IoV communication, emphasizing edge and fog-based authentication mechanisms and privacy-aware offloading strategies.

2.2.4 Federated Learning and Blockchain for IoV Security

Y. Lu et al. (2020) proposed an asynchronous federated learning architecture integrated with a hybrid blockchain framework to address privacy and communication efficiency issues in IoV data sharing. Their approach combines a permissioned blockchain with a local Directed Acyclic Graph (DAG) to securely store and verify learning model updates. To handle intermittent vehicular connectivity, deep reinforcement learning was employed for intelligent node selection during training.

The framework improved learning accuracy and convergence speed while reducing communication overhead, making it suitable for dynamic IoV environments. M. Fan et al. (2022) introduced a decentralized anonymous federated learning authentication scheme named DAFL to overcome the limitations of centralized authentication in federated learning systems. The scheme utilizes DAG-based blockchain and accumulator-based storage to enable lightweight identity authentication with batch verification. Pseudonym-based communication ensures anonymity while allowing identity traceability in case of malicious behavior. Experimental results demonstrated reduced authentication overhead and improved convergence performance compared to traditional federated learning approaches.

N. Wang et al. (2022) proposed BPFL, a blockchain-based privacy-preserving federated learning framework designed to protect both client-side and server-side data in IoV. The framework integrates homomorphic encryption with an improved Multi-Krum filtering mechanism to securely aggregate encrypted local models. A reputation-based incentive mechanism was also introduced to encourage honest participation from vehicles. Although the scheme enhanced privacy and verifiability, the use of encryption increased computational complexity. B. Ghimire and Rawat (2022) presented a comprehensive analysis of blockchain-enabled federated learning as a secure and verifiable collaborative learning solution for IoV networks. Their study highlighted how blockchain can address federated learning vulnerabilities such as model poisoning and reverse engineering attacks. By enabling auditable and tamper-resistant model updates, blockchain improves trust among participating vehicles. The work emphasized the need for lightweight blockchain integration to ensure practical deployment in resource-constrained vehicular environments.

2.2.5 Cryptographic Enhancements and Key Management

Mahmoud A. Shawky et al. (2023) proposed a blockchain-assisted secret key reconciliation scheme to reduce the computational overhead of public-key authentication in vehicular networks. Their approach uses public-key cryptography only for initial authentication, followed by symmetric key-based authentication for

subsequent communications. Blockchain smart contracts were employed to securely publish reconciliation information. Performance analysis showed significant reductions in authentication delay and packet loss. D. S. Gupta et al. (2022) designed a quantum-resistant certificateless authentication protocol for IoV using lattice-based cryptography combined with blockchain-assisted batch verification. The scheme eliminates the reliance on traditional certificate authorities while resisting quantum attacks. Blockchain ensures trust and traceability during batch authentication. Formal security analysis demonstrated resistance to replay, impersonation, and forgery attacks with reduced computation and storage overhead.

Eddine et al. (2021) introduced EASBF, an efficient blockchain-based authentication scheme for fog computing-enabled IoV systems. The scheme employs elliptic curve cryptography and one-way hash functions to provide confidentiality, anonymity, and perfect forward secrecy. Blockchain is used to manage certificates and consensus securely. Security verification using AVISPA and performance evaluation confirmed the scheme's robustness and efficiency. A. K. Sutrala et al. (2020) proposed a conditional privacy-preserving batch authentication protocol based on elliptic curve cryptography. The scheme allows vehicles and roadside units to authenticate multiple messages simultaneously, significantly reducing verification time. Formal and informal security analysis proved resistance against common vehicular attacks. The protocol demonstrated lower communication and computational overhead compared to existing methods.

2.2.6 Optimization and Intelligent Model Enhancement

Changting Zhong et al. (2022) proposed the Beluga Whale Optimization (BWO) algorithm, inspired by the natural foraging behaviors of beluga whales. The algorithm incorporates adaptive exploration, exploitation, and whale-fall strategies to balance global and local search. Extensive benchmark testing showed superior convergence speed and scalability compared to existing metaheuristic algorithms. These properties make BWO suitable for optimizing learning parameters and cryptographic key selection.

Weiguo Zhao et al. (2020) introduced the Manta Ray Foraging Optimization (MRFO) algorithm, which mimics the chain, cyclone, and somersault foraging behaviors of manta rays. The algorithm demonstrated strong performance across benchmark functions and real-world engineering problems. Its ability to efficiently balance exploration and exploitation makes MRFO effective for optimizing complex IoV security models, particularly in cryptographic and deep learning applications.

2.2.7 Intrusion Detection and Deep Learning Models

Yufeng Wang et al. (2019) proposed a residual recurrent neural network (R2N2) model for detecting false data injection attacks in smart grids. The model combines linear prediction with nonlinear residual learning to accurately identify malicious data patterns. An adaptive threshold mechanism was employed to improve detection accuracy. The approach demonstrated superior performance compared to conventional prediction-based detection techniques. Nujud Al-Aql AlShammari (2024) introduced a hybrid intrusion detection model combining Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks for vehicle CAN systems. The hybrid model effectively detected DoS, fuzzy, and impersonation attacks by learning temporal dependencies in vehicular data. Experimental results showed high detection accuracy, indicating the suitability of deep learning models for securing in-vehicle communication networks.

Abin Oommen Philip and Saravanaguru (2023) proposed a blockchain-based framework for traffic incident detection and evidence management in IoV. The framework integrates cooperative event correlation with LSTM and Bayesian models to detect and resolve conflicting traffic reports. Verified events are securely stored on the blockchain to ensure immutability and accountability. The results demonstrated improved reliability and efficiency in incident management.

2.2.8 Trust, Reputation, and Incentive-Based Approaches

Bowei Zhang et al. (2023) proposed a reputation-based incentive mechanism using deep reinforcement learning to suppress selfish behavior in vehicular networks. Blockchain was employed to securely record vehicle behaviors, ensuring tamper-proof reputation management. The DRL-based model dynamically learned optimal strategies to discourage attacks. Simulation results showed improved cooperation and reduced attack motivation. Lai et al. (2021) developed a trust-based privacy-preserving friend matching scheme for social IoV environments. The scheme uses Bloom filters to securely match vehicle interests while protecting sensitive attribute information. Trust evaluation mechanisms were employed to eliminate malicious vehicles. Performance evaluation showed efficient privacy protection and scalability.

Q. Mei et al. (2022) proposed a blockchain-enabled privacy-preserving authentication scheme for transportation of cyber-physical systems using pairing-free ring signatures. The scheme supports batch authentication and unconditional anonymity while simplifying key management. Blockchain ensures reliable authentication without relying on centralized authorities. Security proofs and simulation results confirmed feasibility and efficiency. Gupta et al. (2022) introduced blockchain-based game-theoretic and lightweight branched blockchain authentication frameworks for IoV. Their approaches addressed physical cloning attacks, scalability issues, and computational overhead by combining PUFs, dynamic consensus, and lightweight blockchain design. Comparative analysis demonstrated improved security and reduced transaction costs. Wu et al. (2023) proposed a blockchain and mobile edge computing-based time-sensitive authentication mechanism for IoV. The scheme offloads authentication tasks to edge nodes while using blockchain for secure identity verification. Performance evaluation showed reduced authentication delay and lower computation overhead, making the approach suitable for large-scale IoV deployments.

2.3 Inference From The Literature Review

Table 2.1 Comparative Literature Review of Recent IoV Security Works

S.No	Author and Year	Methodology Used	Limitations
1	Chai et al., 2022	Used blockchain with CyberTwin concept to provide secure and privacy-preserving vehicle authentication.	System design is complex and difficult to deploy in real IoV environments.
2	Xie et al., 2023	Proposed a blockchain-based handover authentication scheme with privacy protection and key agreement.	Handover process increases system complexity during high vehicle mobility.
3	Feng et al., 2022	Designed a blockchain authentication scheme using global commitment to reduce verification delay.	Requires frequent updates of global commitment information.
4	Lin et al., 2021	Developed an Ethereum-based blockchain authentication scheme with batch verification support.	Public blockchain may cause higher delay and transaction cost.
5	Shen et al., 2022	Introduced blockchain-assisted authentication for edge-based IoV systems to reduce cloud load.	Depends heavily on trusted edge nodes for authentication.
6	R. Kumar et al., 2022	Combined blockchain and deep learning to detect intrusions and secure IoV communication.	Deep learning model may require high training time and large datasets.

7	Shujuan Wang et al., 2022	Proposed blockchain-based trust management using deep learning to identify malicious vehicles.	Trust evaluation accuracy depends on data quality and training process.
8	Manogaran et al., 2023	Used blockchain with token-based authorization and machine learning for authentication.	Machine learning model may not detect highly advanced attacks effectively.
9	N. Wang et al., 2022	Proposed blockchain-based federated learning to preserve data privacy in IoV.	Encryption increases computation overhead and communication delay.
10	Gupta et al., 2022	Designed a blockchain-based authentication scheme resistant to quantum attacks.	Cryptographic operations are computationally heavy for vehicles.

2.4 Research gap

Although numerous studies have explored security and privacy solutions for Internet of Vehicles (IoV) networks, existing approaches still face several unresolved limitations. Blockchain-based authentication methods successfully eliminate centralized control and improve data integrity; however, many of these solutions introduce excessive computation, communication delay, and storage overhead, which restrict their applicability in fast-moving vehicular environments. Similarly, machine learning and deep learning techniques have been applied to detect malicious behavior in IoV networks, but most models are developed without considering rapid topology changes and real-time processing requirements. As a result, their performance may degrade under highly dynamic traffic conditions.

In addition, current research lacks an effective integration of optimized learning models and efficient cryptographic mechanisms within a single IoV security framework. Many encryption-based solutions rely on static key generation methods,

leading to increased encryption time and reduced scalability. Likewise, intelligent detection models are often used without parameter optimization, resulting in unnecessary computational cost and suboptimal detection accuracy. There is also limited work that simultaneously ensures secure authentication, privacy preservation, attack detection accuracy, and low system overhead. These shortcomings indicate a clear research gap and justify the need for a lightweight, blockchain-enabled IoV authentication framework that combines optimized deep learning techniques with efficient cryptographic key management.

2.5 Summary

This chapter reviewed existing research on security and privacy in Internet of Vehicles (IoV) networks. Blockchain-based authentication approaches were discussed for their role in decentralization and data integrity. The use of machine learning and deep learning techniques for malicious attack detection was examined. Various cryptographic methods for privacy preservation were also reviewed. Optimization techniques were highlighted for improving efficiency and performance. Several limitations, including high overhead and lack of integration, were identified. These gaps indicate the need for a unified and optimized IoV security framework. The next chapter introduces the first proposed work based on a machine learning-driven blockchain authentication framework.

2.6 References

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CHAPTER 3

BLOCKCHAIN-BASED PRIVACY-PRESERVING AUTHENTICATION FRAMEWORK USING HYBRID ADAPTIVE NETWORK FOR INTERNET OF VEHICLES

3.1 Preamble

The rapid growth of intelligent transportation systems has led to the development of the Internet of Vehicles (IoV), where vehicles, roadside units, and cloud servers communicate continuously to support applications such as traffic safety, autonomous driving, and real-time navigation. While IoV improves transportation efficiency and user experience, it also introduces serious security and privacy challenges due to its open wireless environment, high vehicle mobility, and large-scale data exchange. Traditional centralized authentication mechanisms depend on trusted authorities, which suffer from scalability issues, high latency, and single-point-of-failure problems, making them unsuitable for dynamic IoV scenarios. Blockchain technology has emerged as a promising solution by enabling decentralized, tamper-resistant, and transparent authentication without relying on a central entity. However, blockchain alone cannot effectively handle intelligent and evolving cyber-attacks such as impersonation and false data injection. To overcome this limitation, intelligent learning models are required to detect malicious behavior accurately. In this context, this chapter presents an efficient privacy-preserving authentication scheme that integrates blockchain technology with a Hybrid Adaptive Network (HANet) to provide secure, scalable, and intelligent authentication in IoV networks. The proposed framework ensures data integrity, vehicle anonymity, and resistance to malicious attacks while maintaining low computational overhead, thereby making it suitable for real-time vehicular environments.

3.2 Background and Motivation

The Internet of Vehicles (IoV) represents a large-scale, heterogeneous communication environment where vehicles, roadside units (RSUs), infrastructure, and users exchange information through wireless networks to enable intelligent transportation

services. This interconnected ecosystem supports applications such as traffic monitoring, accident prevention, and autonomous driving. With the advancement of machine learning and deep learning techniques, data collected by RSUs and vehicles can be effectively utilized to build intelligent systems for vehicular misbehavior detection, trust evaluation, and anomaly identification, as highlighted by Kumar et al. (2022) and Wang et al. (2022). However, traditional machine learning techniques often struggle to operate efficiently in highly distributed and dynamic IoV environments due to limited adaptability and generalization issues.

Security and privacy remain major concerns in IoV networks because vehicles continuously transmit sensitive information such as location, identity, and driving patterns. Conventional IoV security solutions are primarily centralized and rely on trusted authorities, which introduce single-point-of-failure problems and scalability limitations, as discussed by Wang et al. (2019) and Qureshi et al. (2022). These centralized approaches are vulnerable to attacks such as unauthorized tracking, identity hijacking, replay attacks, and the injection of false information, which can mislead vehicles and compromise road safety. Moreover, centralized trust models may fail to provide timely authentication and accountability in high-mobility vehicular environments.

Another critical challenge is ensuring conditional privacy, where vehicles remain anonymous during normal operation but can be traced and penalized when malicious behavior is detected. Existing trust-based approaches typically depend on third-party authorities to validate transactions, which may themselves become untrustworthy or compromised, as highlighted by Mundhe et al. (2022) and Chaudhary and Singh (2021). As IoV networks grow in scale and complexity, traditional security mechanisms become insufficient due to ineffective malicious node traceability, insecure communication channels, and inefficient data transmission mechanisms.

Blockchain technology has emerged as a promising solution to address these challenges by enabling decentralized, tamper-resistant, and transparent data

management. Blockchain distributes trust among participating nodes using consensus mechanisms, eliminating the need for centralized authorities, as demonstrated by Wang et al. (2019), Chai et al. (2022), and Feng et al. (2022). In a blockchain-based IoV environment, transactions are recorded through a peer-to-peer network, and each node maintains a synchronized copy of the distributed ledger. Before any transaction is permanently stored, it is validated by multiple nodes, ensuring data integrity and transparency. The immutability property of blockchain ensures that any modification to stored data is computationally infeasible, thereby strengthening resistance against data tampering and insider attacks.

Despite these advantages, blockchain-based IoV systems may suffer from communication overhead, storage consumption, and latency issues, particularly when handling large volumes of vehicular data. Federated Learning (FL) has been proposed as an effective solution to mitigate these limitations by allowing vehicles to train local models without sharing raw data, as demonstrated by Lu et al. (2020), Fan et al. (2022), and Wang et al. (2022). FL enhances privacy preservation and reduces network congestion while enabling collaborative learning. However, FL-based IoV systems are still vulnerable to malicious model updates, inference attacks, and inefficient aggregation processes.

Machine learning techniques provide intelligent solutions for analyzing large-scale IoV data and identifying abnormal traffic patterns. Hybrid learning models are particularly effective in distinguishing malicious and normal traffic behavior due to their ability to capture complex data patterns, as discussed by Ali et al. (2021) and Nie et al. (2020). Motivated by these observations, this research integrates blockchain technology with a hybrid adaptive machine learning model to ensure secure, efficient, and privacy-preserving authentication in IoV networks. The proposed framework combines smart contract-enabled blockchain with an intelligent Hybrid Adaptive Network (HANet) to detect malicious activities, secure data transmission, reduce latency, and optimize resource utilization in dynamic vehicular environments. The key

contributions of the proposed privacy-preserving authentication scheme for Internet of Vehicles are summarized as follows:

- A blockchain-based privacy-preserving authentication framework is designed to ensure secure data authentication and reliable communication among vehicles, RSUs, and infrastructure while preventing malicious access.
- A smart contract-enabled blockchain mechanism is implemented to restrict unauthorized third-party participation, enhance data integrity, and automate trust verification within the IoV network.
- A Hybrid Adaptive Network (HANet), combining Multilayer Perceptron (MLP) and Ridge Classifier, is developed to accurately identify malicious and legitimate traffic patterns in IoV communications.
- An Optimized Beluga Whale Optimization (OBWO) algorithm is proposed to fine-tune feature selection and model parameters, enhancing detection accuracy and system efficiency based on the principles introduced by Zhong et al. (2022).
- Extensive performance evaluation and comparative analysis are conducted against existing classifiers and optimization techniques using multiple metrics to validate the effectiveness, scalability, and robustness of the proposed framework.

3.3 Architecture of the Proposed System

The Internet of Vehicles (IoV) is a highly dynamic ad-hoc communication network designed to improve traffic efficiency, road safety, and intelligent transportation services. Although IoV shares several characteristics with conventional Internet of Things (IoT) systems, it operates in open wireless environments where network topology changes rapidly due to vehicle mobility. This dynamic nature significantly increases the complexity of data processing, communication reliability, and service continuity. As a result, IoV networks are more vulnerable to security and privacy threats such as sensitive data exposure, identity impersonation, message forgery, and data manipulation, which may directly endanger passenger safety and property.

Blockchain technology offers an effective solution to these challenges by enabling decentralized trust management through distributed consensus mechanisms. In vehicular environments where vehicles cannot inherently trust neighboring nodes, blockchain ensures that transmitted data cannot be altered once verified and recorded. By eliminating reliance on centralized authorities, blockchain reduces the risk of single-point failures and improves system transparency. However, due to high vehicle speed, frequent handovers, and uncertain communication conditions, reliable and secure data transmission in IoV remains a critical challenge. In addition, traditional centralized key management approaches increase the likelihood of key exposure and unauthorized access. Therefore, there is a strong need for a blockchain-enabled IoV architecture that supports efficient privacy preservation and secure authentication. The overall architecture of the proposed blockchain-based IoV system is illustrated in Figure 3.1.

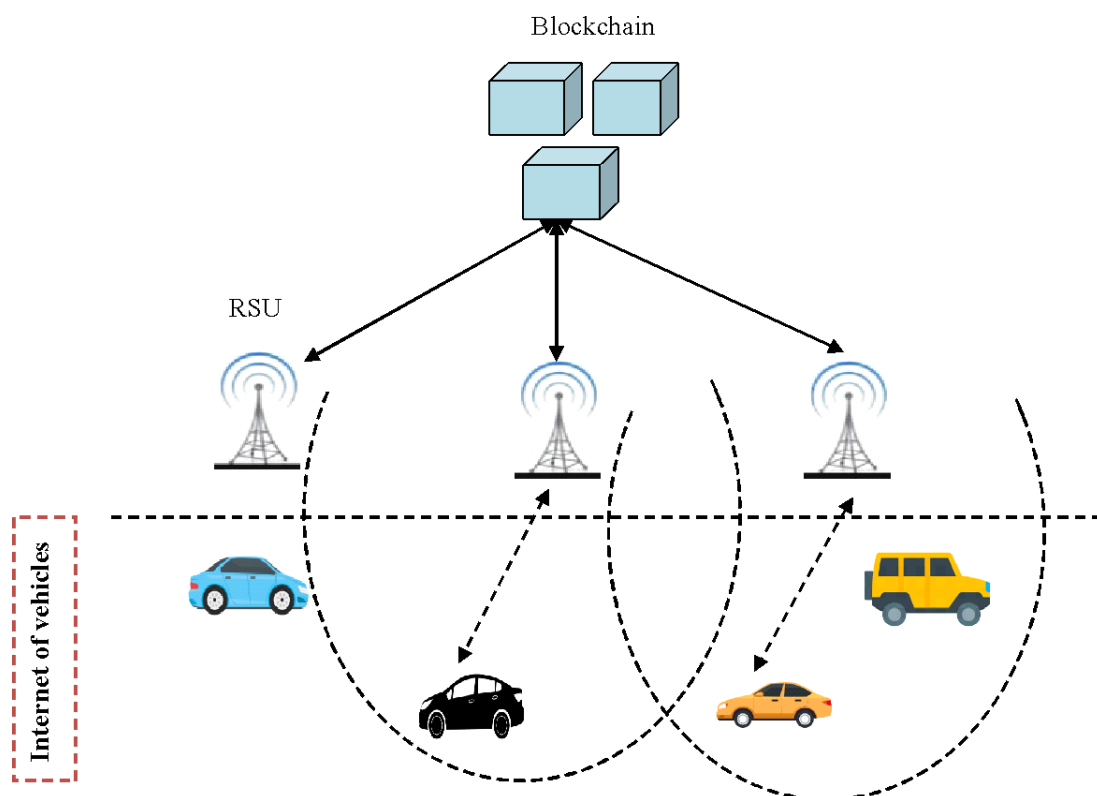


Figure 3.1: Architecture of the Proposed Blockchain-Enabled Privacy-Preserving Authentication Framework for IoV

A smart contract-enabled blockchain framework integrated with machine learning is proposed to achieve secure, efficient, and anonymous conditional privacy-preserving authentication in Internet of Vehicles (IoV) networks. In the proposed system, vehicles are required to register their genuine identity information with the roadside infrastructure before participating in data transmission and message broadcasting. Blockchain technology is employed to store and protect vehicular data in tamper-resistant blocks, ensuring transparency and trust among participating entities. Alongside blockchain, a machine learning-based intrusion detection mechanism is incorporated to recognize and mitigate malicious activities within the IoV network.

To ensure secure data communication, the system combines machine learning with a smart contract-assisted blockchain mechanism, enabling automated access control and secure authentication. Feature selection and attack detection are carried out using the proposed Hybrid Adaptive Network (HANet), which integrates a Multilayer Perceptron (MLP) with a Ridge classifier to accurately distinguish between legitimate and malicious traffic patterns. Furthermore, the Optimal Beluga Whale Optimization (OBWO) algorithm is utilized to optimally tune the selected features, enhancing classification accuracy and reducing computational overhead. The performance of the proposed approach is evaluated against existing machine learning-based authentication methods for IoV, demonstrating its effectiveness in terms of security, privacy, and efficiency. The overall architecture of the proposed system is illustrated in Figure 3.2.

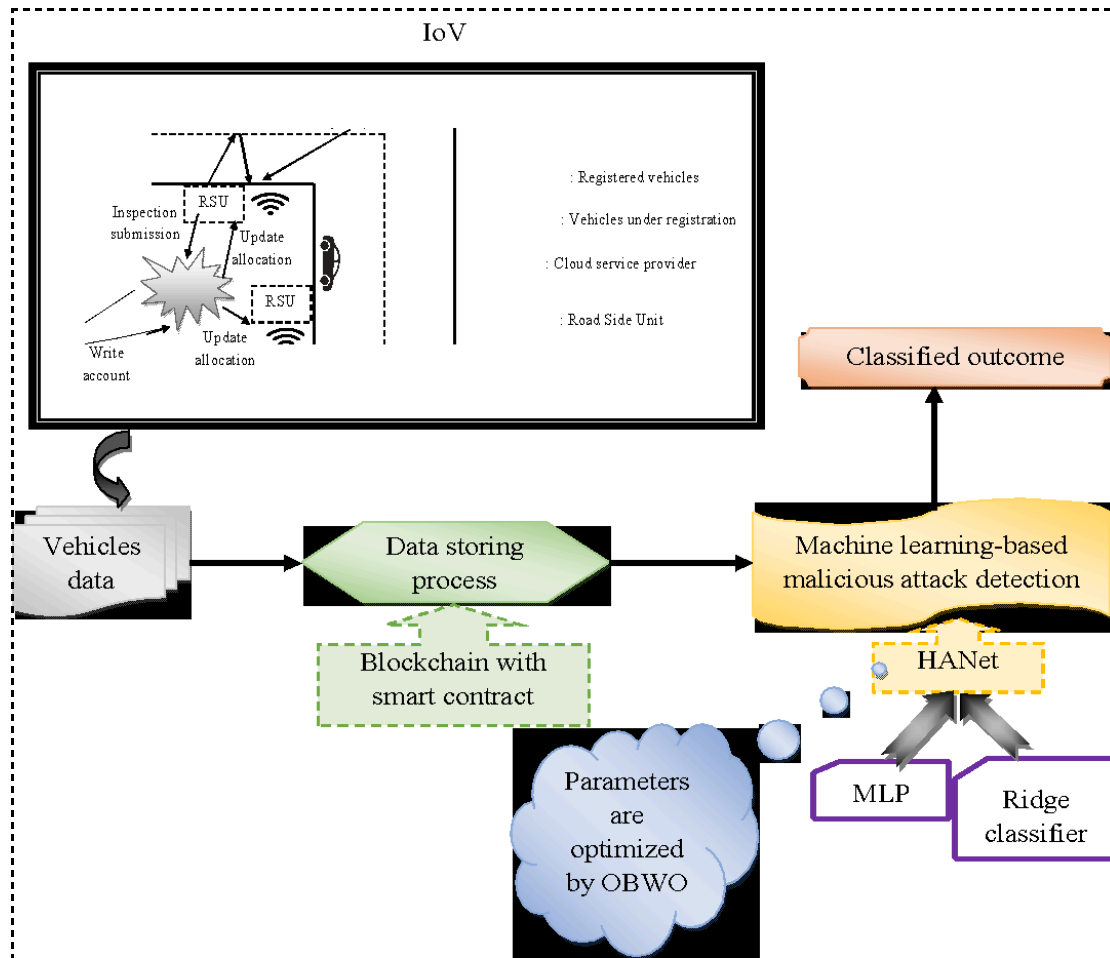


Figure 3.2: Blockchain-Assisted Secure and Anonymous Authentication Mechanism for IoV

3.3.1 Information on Vehicle Attributes

To develop and evaluate the proposed blockchain-based privacy-preserving authentication framework for the Internet of Vehicles (IoV), a publicly available Internet of Vehicles dataset is utilized. The dataset is obtained from the Kaggle repository and contains real-world vehicular attributes relevant to intelligent transportation environments (Harunachiro Magombe, 2023). This dataset provides structured information describing vehicle behavior and movement patterns, which are essential for analyzing network activities and identifying abnormal or malicious actions within IoV systems.

The dataset includes multiple vehicular parameters such as speed, acceleration, location-related indicators, collision-related factors, and communication

characteristics. These attributes are widely used in IoV research to support traffic monitoring, collision detection, and security analysis. In the proposed system, the dataset serves as the primary input for training and testing the intelligent learning models integrated with the blockchain-based authentication mechanism. Let the individual vehicle data records extracted from the dataset be represented as V_d , where $V_d = 1, 2, \dots, D$. Here, each V_d corresponds to a single vehicle instance containing a set of measured attributes. The complete dataset is denoted by D , which represents the collection of all vehicle records used during the experimental evaluation. This structured representation enables effective feature processing, optimized attribute selection, and accurate detection of malicious behavior in the IoV environment.

The extracted vehicle attributes are further processed by the proposed Hybrid Adaptive Network (HANet) and optimized using the Improved Beluga Whale Optimization (OBWO) algorithm to enhance authentication accuracy, reduce computational overhead, and strengthen privacy preservation in dynamic vehicular networks.

3.4 Smart Contract-Enabled Blockchain and OBWO for Privacy-Preserving Authentication

Blockchain technology plays a vital role in ensuring secure and reliable authentication in Internet of Vehicles (IoV) environments by providing tamper-resistant data storage and decentralized trust management. To establish a strong foundation for the proposed authentication framework, an overview of the blockchain architecture and its operational principles is presented in Section 3.4.1. This section explains how distributed ledger technology enhances data integrity and mitigates malicious activities.

The secure management of vehicular information using smart contracts is discussed in Section 3.4.2, where the mechanism for storing, validating, and accessing

authentication data on the blockchain is detailed. Smart contracts automate authentication operations while eliminating the need for third-party intervention.

Furthermore, Section 3.4.3 introduces the proposed Optimized Beluga Whale Optimization (OBWO) algorithm, which is employed for parameter tuning and privacy enhancement within the authentication process. This section also explains how optimized cryptographic and learning parameters contribute to privacy preservation and efficient authentication in dynamic IoV networks.

The overall architecture of the proposed smart contract-assisted blockchain-based privacy-preserving authentication system for IoV is illustrated in Figure 3.3, highlighting the interaction between vehicles, blockchain components, and intelligent optimization modules.

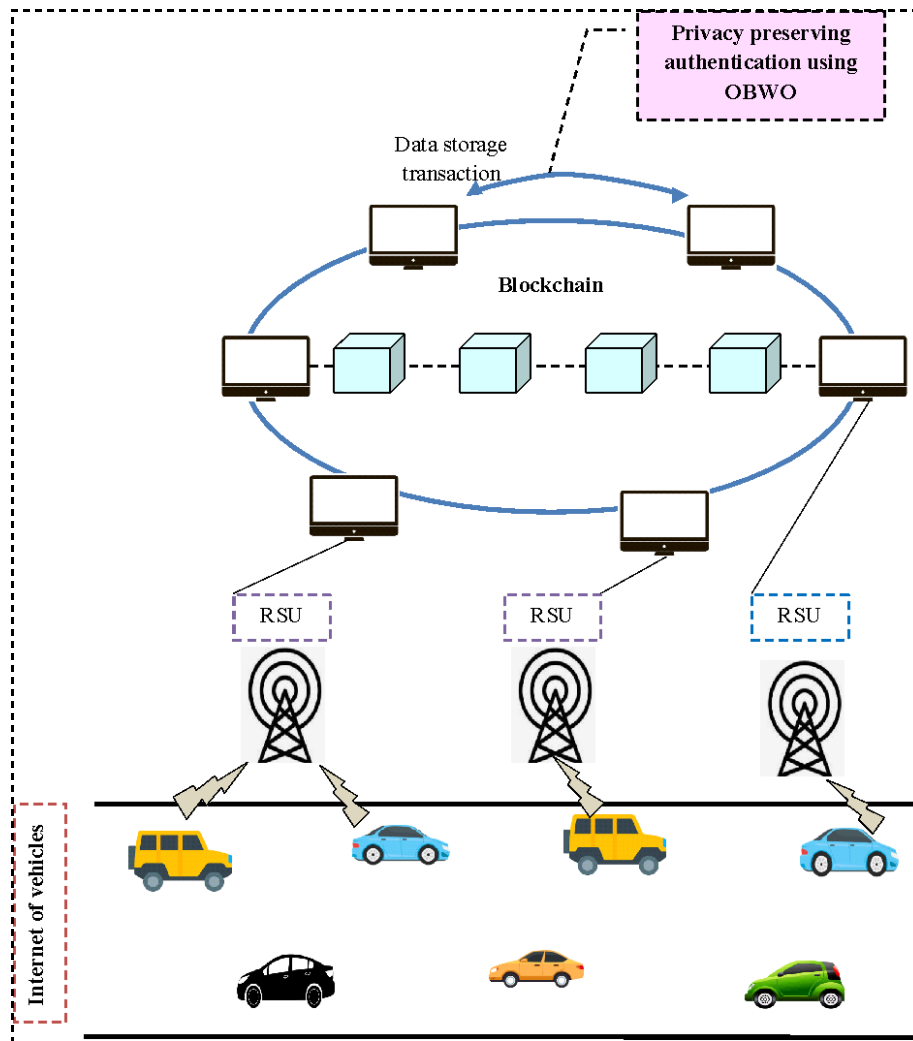


Figure 3.3: System Model of Smart Contract-Assisted Privacy-Preserving Authentication in IoV

3.4.1 The Infrastructure of Blockchain Technology

Blockchain technology, first introduced as the core mechanism behind Bitcoin in 2008, represents a decentralized and autonomous framework for managing and validating time-stamped transactions without reliance on a central authority. It operates as a peer-to-peer (P2P) distributed network in which all participating nodes maintain identical copies of a shared ledger, ensuring consistency and resilience against data tampering. As highlighted by X. Wang et al. (2019) and K. N. Qureshi et al. (2022), blockchain integrates multiple advanced technologies, including asymmetric cryptography, distributed ledger management, smart contracts, and incentive mechanisms, to establish trust in open and untrusted environments such as the Internet of Vehicles (IoV).

In a blockchain network, user identities are typically represented through cryptographic public keys, which provide anonymity while preserving authentication and accountability. Transaction data are organized into fundamental units known as blocks, each of which is cryptographically linked to its predecessor, forming an immutable chain. Once a block is validated and appended to the ledger through a consensus process, its contents cannot be altered without modifying all subsequent blocks, a task that is computationally infeasible. This property ensures data integrity, transparency, and traceability, as emphasized by H. Chai et al. (2022) and Xia Feng et al. (2022).

Depending on access control and management strategies, blockchains can be broadly categorized into public chains, consortium (permissioned) chains, and private chains, each differing in trust assumptions, consensus mechanisms, and degrees of decentralization. Despite these variations, core blockchain characteristics—such as immutability, encryption, chronological data ordering, openness, and auditability—remain consistent across implementations. Structurally, blockchain systems are commonly organized into layered architectures comprising the data layer, network layer, consensus layer, incentive layer, contract layer, and application layer, as discussed by Chen Mao et al. (2021). This layered infrastructure enables blockchain to support secure, scalable, and privacy-aware applications, making it particularly suitable for authentication and data management in IoV environments.

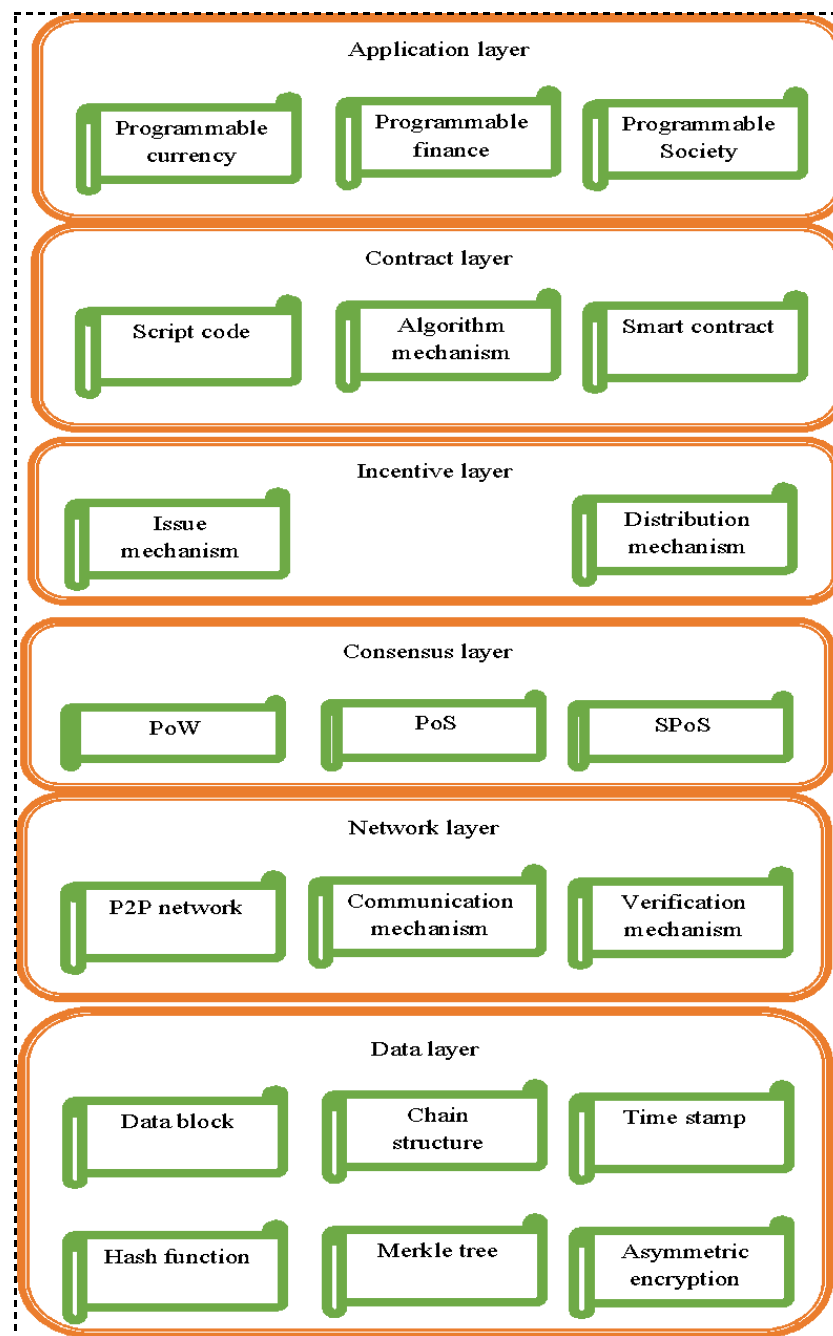


Figure 3.4: Layered architecture of blockchain technology for the proposed privacy-preserving authentication framework in Internet of Vehicles (IoV)

3.4.2 Data Storage Using Smart Contracts

In the proposed framework, the collected vehicular data V_d are securely maintained within a blockchain environment through the use of smart contracts. A smart contract refers to a self-executing program embedded in blockchain blocks, which automatically performs predefined operations once specified conditions are satisfied. The primary purpose of incorporating smart contracts into the blockchain is to enable autonomous and trustworthy service execution without dependence on centralized authorities or external trusted parties.

By leveraging smart contracts, the proposed system eliminates the involvement of intermediaries such as administrators, control centers, and centralized service managers, thereby reducing security vulnerabilities and operational overhead. The blockchain enforces the execution of predefined rules and access conditions in a transparent and tamper-resistant manner, ensuring that data handling and authentication procedures are carried out consistently across the network. Furthermore, smart contracts facilitate automated decision-making and verification, enabling a fully decentralized and self-managed authentication process.

In the proposed privacy-preserving authentication scheme, vehicular data are stored and managed through smart contract logic to ensure secure access, integrity, and confidentiality. This approach strengthens the overall functionality of the blockchain by providing reliable data storage, preventing unauthorized modifications, and supporting efficient authentication in dynamic IoV environments.

3.4.3 Opposition Beluga Whale Optimization (OBWO)

A detailed examination of existing optimization techniques reveals that, although many algorithms demonstrate acceptable performance, they often suffer from limitations related to slow convergence and an improper balance between exploration and exploitation. In large-scale learning environments, some methods also become unstable when handling limited or imbalanced training samples, which can indirectly compromise data privacy and model reliability. To overcome these challenges, nature-inspired metaheuristic algorithms have gained attention due to their adaptability and computational efficiency. Among them, the Beluga Whale Optimization (BWO) algorithm has shown strong capability in addressing both unimodal and multimodal optimization problems with relatively low computational cost. The BWO algorithm is inspired by the natural behaviors of beluga whales, marine mammals well known for their advanced communication and coordinated movement patterns. By simulating these behaviors, the original BWO algorithm has been successfully applied to various optimization tasks and has demonstrated competitive performance compared to other metaheuristic approaches. However, the conventional BWO relies on random selection strategies during population initialization and update phases, which can lead to suboptimal convergence and reduced solution accuracy, particularly in complex and high-dimensional problem spaces.

To address these shortcomings, an enhanced version referred to as Opposition Beluga Whale Optimization (OBWO) is introduced. The OBWO algorithm incorporates opposition-based learning principles to improve population diversity and guide the search process more effectively toward optimal regions. This enhancement enables the algorithm to better capture complex structural relationships in the learning space and strengthens the propagation of relevant features within the model. As a result, the OBWO-driven parameter tuning process improves convergence speed while minimizing the objective cost function. In the proposed privacy-preserving authentication framework for IoV, OBWO is employed to optimally tune model parameters, thereby enhancing authentication accuracy and system robustness. The improved convergence behavior and scalability of OBWO contribute to reliable performance even in dynamic and large-scale vehicular environments. To formalize the whale selection mechanism in the OBWO process, a newly derived mathematical expression is introduced, as presented in Eq. 3.1.

$$S = \frac{(bfit^{1.5})}{(wfit \times mfit)} \quad (3.1)$$

Here, the average fitness value and the optimal fitness value obtained from the BWO algorithm are denoted as $bfit$ and $mfit$, respectively. In addition, the term $wfit$ represents the least favorable (worst) fitness value achieved during the optimization process.

The original Beluga Whale Optimization algorithm proposed by Changting Zhong et al. (2022) is a population-based metaheuristic inspired by the natural foraging behavior of beluga whales. The optimization process is organized into three major operational phases, namely exploration, exploitation, and whale-fall, which are modeled based on the biological behaviors of pair swimming, prey hunting, and whale fall, respectively. During the exploration stage, beluga whales are randomly distributed across the search space to enhance global search capability and avoid premature convergence. The exploitation stage focuses on refining candidate solutions by intensifying the local search around promising regions of the solution space. Additionally, the whale-fall mechanism introduces probabilistic position updates that help diversify the population and prevent stagnation in local optima.

Due to its population-driven nature, the BWO algorithm represents each beluga whale as an individual search agent, where each agent corresponds to a potential solution to the optimization problem. These agents iteratively update their positions throughout the optimization process to improve solution quality. The positions of all search agents are

mathematically modeled in a matrix representation, which captures the multidimensional search space and is formally expressed in Eq. 3.2.

$$H = \begin{bmatrix} h_{1,1} & h_{1,2} & \boxtimes & h_{1,e} \\ h_{2,1} & h_{2,2} & \boxtimes & h_{2,e} \\ \boxtimes & \boxtimes & \boxtimes & \boxtimes \\ h_{o,1} & h_{o,2} & \boxtimes & h_{o,e} \end{bmatrix} \quad (3.2)$$

Here, the parameter o represents the size of the beluga whale population, whereas the parameter e corresponds to the population size of the design variables. The key fitness values for each beluga whale are calculated using the expression provided in Eq. (3.3).

$$G_h = \begin{bmatrix} g(h_{1,1}, h_{1,2}, \dots, h_{1,e}) \\ g(h_{2,1}, h_{2,2}, \dots, h_{2,e}) \\ \boxtimes \\ g(h_{o,1}, h_{o,2}, \dots, h_{o,e}) \end{bmatrix} \quad (3.3)$$

The transition between the exploration and exploitation stages in the BWO algorithm is governed by the balance factor C_g . This balance factor, which controls the shift between global search and local refinement, is mathematically defined in Eq. 3.4.

$$C_g = C_0 \left(\frac{1-u}{2U_{\max}} \right) \quad (3.4)$$

In this formulation, the current iteration count is denoted by u , while the maximum number of iterations is represented by $U_{\max}$. The parameter C_0 is a random value that varies between 0 and 1 during each execution of the algorithm. When the balance factor satisfies $C_g > 0.5$, the algorithm operates in the exploration stage; otherwise, when $C_g \leq 0.5$, the exploitation stage is activated.

The exploration phase of the BWO algorithm is inspired by the pair-swimming behavior of beluga whales. During this stage, the positions of the search agents are updated based on this cooperative swimming mechanism, and the corresponding position update rule is expressed in Eq. 3.5.

$$\begin{cases} H_{j,k}^{u+1} = H_{j,Q_k}^u + (H_{s,Q_1}^u - H_{j,Q_k}^u)(1 + s_1)\sin(2\pi s_2), & k = \text{even} \\ H_{j,k}^{u+1} = H_{j,Q_k}^u + (H_{s,Q_1}^u - H_{j,Q_k}^u)(1 + s_1)\cos(2\pi s_2) & k = \text{odd} \end{cases} \quad (3.5)$$

The updated position of the j^{th} beluga whale along the k^{th} dimension is represented as $H_{j,k}^{u+1}$. The parameter Q_k ($k = 1, 2, \dots, e$) denotes a randomly selected index from the e -dimensional search space. The current positions of the j^{th} and s^{th} beluga whales are denoted by H_{j,Q_k}^u and H_{s,Q_k}^u , respectively, where H_{j,Q_k}^u indicates the position of the j^{th} whale in the Q_k^{th} dimension.

In the standard BWO method, beluga whales are selected randomly, which can adversely affect solution accuracy and overall optimization performance. To overcome this limitation, the proposed OBWO strategy introduces a controlled random selection mechanism based on Eq. 3.1. The variables s_1 and s_2 represent random values generated within the ranges $(0, 1)$, $\sin(2\pi s_2)$, and $\cos(2\pi s_2)$. These stochastic parameters enhance the diversity of search movements and strengthen the exploratory capability of the algorithm.

Following the foraging (preying) behavior of beluga whales, the exploitation phase of the BWO algorithm is executed to refine promising solutions. In this stage, the Lévy flight strategy is employed to enable effective local search and avoid premature convergence. The mathematical formulation governing the exploitation phase is provided in Eq. 3.6.

$$H_j^{u+1} = s_3 H_{bst}^u - s_4 H_j^u + D_1 \cdot F_L \cdot (H_s^u - H_j^u) \quad (3.6)$$

The new position of a beluga whale after updating is denoted by H_j^{u+1} , while the optimal position identified among all beluga whales is represented as H_{best}^u . The current positions of the selected beluga whale and a randomly chosen whale are expressed as H_j^u and H_s^u , respectively. The random variables s_3 and s_4 are independently generated within the interval $[0, 1]$ to introduce stochastic behavior during the search process.

The parameter $D_j = 2s_4 \left(\frac{1-u}{U_{max}} \right)$ defines the adaptive jump magnitude, which controls the intensity of the Lévy flight movement. This factor plays a crucial role in regulating the step size during exploitation and helps balance global exploration with local refinement. The Lévy flight function itself is denoted by F_L , and its mathematical formulation is provided in Eq. 3.7.

$$F_L = 0.05 \times \frac{v \times \sigma}{|w|^{1/\alpha}} \quad (3.7)$$

$$\sigma = \left(\frac{\Gamma(1+\alpha) \times \sin(\pi\alpha/2)}{\Gamma((1+\alpha)/2) \times \alpha \times 2^{(\alpha-1)/2}} \right)^{1/\alpha} \quad (3.8)$$

Here, the attributes v and w are assumed to follow a random distribution and are generated as arbitrary integers, while the variable represents a predefined constant, typically set close to the value 1.5.

During their natural foraging and migration activities, beluga whales are exposed to several external threats, including human interference, predators such as polar bears, and larger marine species. Although many whales successfully evade these dangers, a fraction of the population fails to survive and eventually sinks to the ocean floor. This phenomenon is

modeled in the optimization process as the “whale fall” mechanism, which is mathematically represented in Eq. 3.9.

$$H_j^{u+1} = s_5 H_j^u - s_6 H_s^u + s_7 H_{step} \quad (3.9)$$

The randomly generated parameters are represented as s_5, s_6 , and s_7 each taking values within the interval $[0, 1]$. The movement magnitude of the search agent is defined by the step size, denoted as H_{step} , which is mathematically expressed in Eq. 3.10.

$$H_{step} = (u_c - l_c) \exp(-D_2 u / U_{\max}) \quad (3.10)$$

In this formulation, the parameter D_2 corresponds to the step control factor, which is associated with both the population size and the whale fall phenomenon. The permissible range of the optimization variables is bounded by the lower and upper limits, represented as l_c and u_c , respectively.

Eq. 3.11 formulates the likelihood of the whale falling F_w .

$$F_w = \frac{0.1 - 0.05u}{U_{\max}} \quad (3.11)$$

The probability of the whale fall event is gradually adjusted from 0.1 at the initial stage of execution to 0.5 at the final stage. This indicates that as the beluga whales move closer to the optimal food source during the optimization process, their exposure to threats is significantly reduced. The detailed operational steps of the proposed algorithm are presented in Algorithm 1, while the corresponding procedural flow is illustrated in Figure 3.5.

Algorithm 1: Proposed OBWO Procedure

Step 1: Initialize the population matrix and set the maximum number of iterations.

Step 2: Compute the objective (fitness) function for all search agents.

Step 3: For iteration $u = 1$ to U_{max} :

For each beluga whale $j = 1$ to N_{pop} :

Generate random values using Eq. 3.1.

Execute the proposed OBWO updating strategy.

Calculate the balance factor using Eq.3.4.

Update the whale position in the exploration phase using Eq.3.5.

Update the whale position in the exploitation phase using Eq. 3.6.

Apply the Lévy flight mechanism using Eq. 3.7.

Update the whale-fall position using Eq. 3.9.

End For

Step 4: End For

Step 5: Repeat the above steps until the stopping criterion or optimal solution is achieved.

Step 6: Output the best solution obtained by the algorithm.

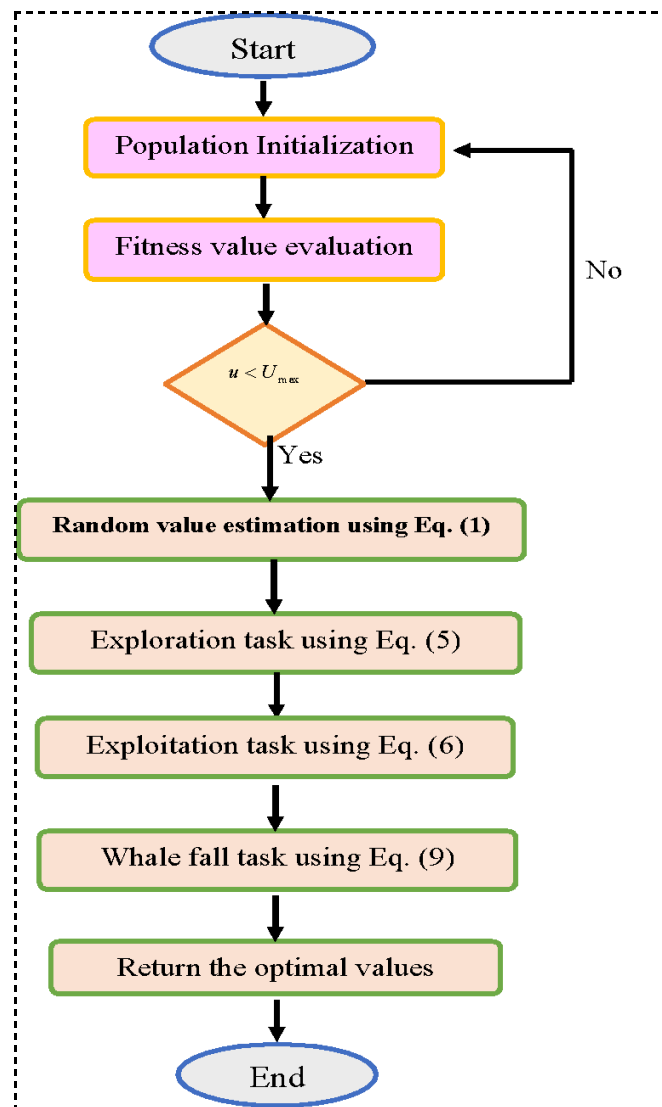


Figure 3.5: Flowchart of the proposed Opposition Beluga Whale Optimization (OBWO) algorithm

3.5 Hybrid Adaptive Network–Based Authentication Using Smart Contracts for the Internet of Vehicles

Authentication plays a vital role in ensuring that vehicular data are accessed, shared, and stored in a secure and trustworthy manner within Internet of Vehicles (IoV) environments. To strengthen authentication while preserving user privacy, this research integrates intelligent learning techniques with smart contract–enabled blockchain mechanisms. In particular, a Hybrid Adaptive Network (HANet) is developed by combining a Multilayer Perceptron (MLP) model with a Ridge classifier to achieve reliable and efficient authentication. The

architectural and operational principles of the MLP and Ridge models are discussed in detail in Subsections 3.5.1 and 3.5.2, respectively, while Subsection 3.5.3 formulates the objective function designed to enhance authentication accuracy and precision.

Recent advancements in deep learning have enabled more robust security solutions by facilitating intelligent decision-making over large-scale and dynamic data. Such models have been increasingly adopted to improve privacy protection and authentication in complex networked systems. Nevertheless, many existing deep learning approaches suffer from high training complexity, long convergence time, and susceptibility to overfitting, which limit their practical applicability in real-time IoV scenarios. For instance, convolutional neural networks (CNNs), although effective in feature extraction, often fail to capture long-term dependency patterns and introduce additional computational overhead.

To overcome these challenges, this research adopts an MLP–Ridge hybrid framework. The MLP model ensures fast learning and rapid prediction by efficiently capturing nonlinear relationships within vehicular traffic data, whereas the Ridge classifier provides regularization to control overfitting and improve generalization performance. By integrating these complementary models, the proposed HANet achieves a balanced trade-off between learning efficiency, classification robustness, and computational cost, making it well suited for secure and privacy-preserving authentication in IoV networks.

3.5.1 Multilayer Perceptron

A Multilayer Perceptron (MLP) is a widely used form of artificial neural network that is well suited for handling complex and nonlinear data relationships. In the proposed authentication framework, the collected vehicular data v_d are provided as inputs to the MLP model. The network architecture is composed of three fundamental layers: an input layer, one or more

hidden layers, and an output layer. The input layer represents a data sample $v_d = (v_{d,1}, v_{d,2}, \dots, v_{d,h+1})$, ($1 \leq d \leq m$), where m denotes the total number of training samples. An additional bias term $v_{d,h+1} = -1$ is included to improve learning flexibility, and h_1 represents the number of hidden neurons.

During the training process, each input vector is multiplied by a corresponding weight matrix $w_l \in R^{(h+1) \times (h_1+1)}$, producing a linear combination that is propagated to the hidden layer. The output of each layer is then passed through a nonlinear activation function to capture complex patterns present in the data. This forward propagation continues across layers until the final output layer generates the classification result z_d^* . Through iterative weight updates, the MLP effectively learns discriminative features from vehicular data, enabling accurate identification of normal and malicious behavior in the IoV authentication process.

The activation function used in the MLP model is mathematically defined in Eq. 3.12:

$$f(v) = (1 + e^{-v})^{-1} \quad (3.12)$$

Since Paillier encryption does not directly support the exponentiation operations required by common activation functions, an approximation strategy is adopted. To address this limitation, the activation function is approximated using the Taylor series expansion, which enables efficient computation while preserving privacy. The Taylor-based approximation of the activation function is formulated in Eq. 3.13.

$$f(v) = (1 + e^{-v})^{-1} \approx 0.5 + 0.25v + 0.02v^3 \quad (3.13)$$

To minimize the loss function during the training process, the gradient descent optimization algorithm is employed. This algorithm iteratively updates the model parameters in the

direction that reduces the error. The mathematical formulation of the gradient descent update rule is presented in Eq. 3.14.

$$\min_w U(w) = \frac{1}{2} \sum_{k=1}^{h_3} (z_{d,k}^* - z_{d,k})^2 \quad (3.14)$$

Here, the number of neurons in the output layer is represented by h_3 . The error is computed as the difference between the target output z_d and the actual output z_d^* . In this context, z_d^* denotes the value predicted by the MLP model using the weight parameter w and the input vector v_d . The input sample v_d is then updated according to the rule defined in Eq. 3.15.

$$w \leftarrow w - \mu \Delta w \quad (3.15)$$

The learning rate $\mu \in [0, 1]$ is denoted by the corresponding parameter. The overall structure of the MLP model is depicted in Figure.3.6.

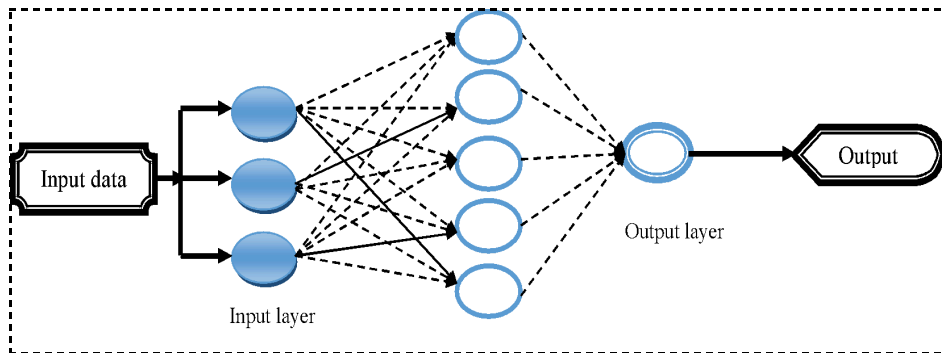


Figure 3.6: Architecture of the Multilayer Perceptron (MLP) model used in the proposed blockchain-based privacy-preserving authentication scheme for IoV

3.5.2 Ridge Classifier

The Ridge classifier is a linear learning model that is well suited for handling high-dimensional data and multicollinearity issues, which are common in IoV environments. In the proposed authentication framework, the Ridge classifier operates as a complementary component to the MLP model and uses the raw vehicular data V_d as its input. The overall working procedure of the Ridge classifier is divided into three major stages: initialization, training (fit), and prediction.

Initialization stage: In this phase, the key parameters that govern the behavior of the classifier are defined. The regularization parameter α is introduced to control the magnitude of model coefficients, thereby preventing overfitting and stabilizing the classification process. The parameter `max_iter` specifies the maximum number of iterations allowed for the optimization solver to converge. In addition, the solver type is selected to efficiently compute the model coefficients; common solvers include options such as `cholesky`, `lsqr`, and `sparse` variants, which are chosen based on the size and characteristics of the training data.

Training (fit) stage: During the training phase, the Ridge classifier learns from the labeled vehicular data. The input feature matrix is represented as X , while the corresponding class labels are stored in vector y . Each row of the matrix X corresponds to a feature vector associated with a specific vehicle instance. By applying regularized least squares optimization, the classifier estimates a coefficient vector that captures the relationship between the input features and the target classes. The inclusion of the regularization term ensures that the learned model remains robust even in the presence of noisy or redundant attributes.

Prediction stage: In the final stage, the trained coefficient vector is utilized to classify new or unseen vehicular data. The Ridge classifier computes a linear combination of the input features and the learned coefficients to generate decision scores, which are then mapped to the corresponding authentication classes. This stage enables efficient and reliable identification of legitimate and malicious entities within the IoV network.

The structural workflow of the Ridge classifier adopted in the proposed privacy-preserving authentication mechanism is illustrated in Figure 3.7.

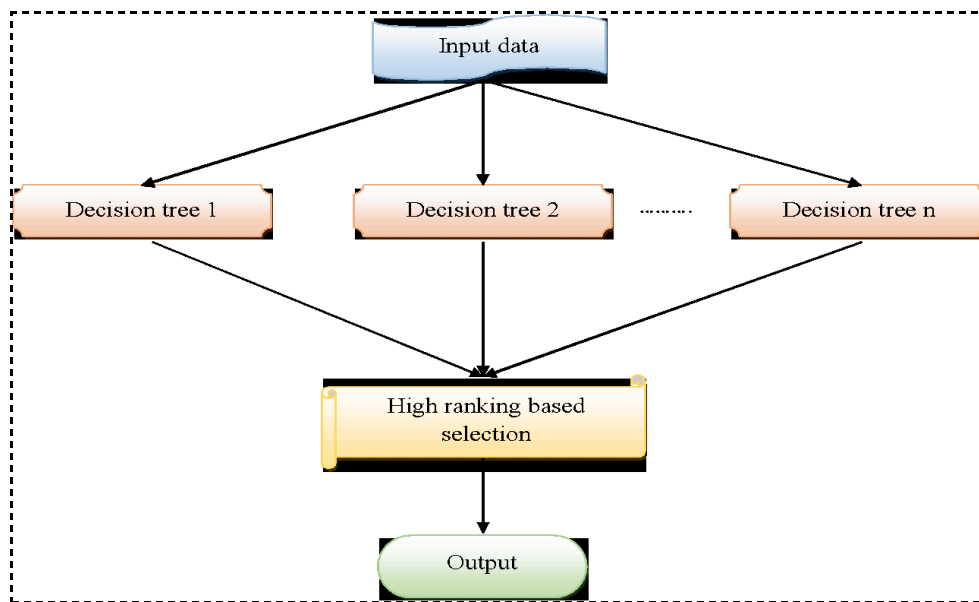


Figure 3.7 Structural overview of the Ridge classifier employed in the blockchain-based privacy-preserving authentication framework for IoV

3.5.3 Machine Learning-Based Authentication Using Smart Contracts

Internet of Vehicles environments are exposed to a wide range of security threats, including cyber intrusions, Sybil attacks, and the participation of compromised or malicious entities, which seriously affect the reliability of vehicular communications. Prior studies have emphasized that traditional rule-based and cryptographic security mechanisms alone are insufficient to handle such intelligent and evolving attacks in IoV networks (Ali et al., 2021). As a result, intelligent data-driven techniques have gained attention for strengthening authentication and attack detection. Machine learning has emerged as a powerful tool in this context, as it enables systems to learn complex patterns from data and identify abnormal behaviors with high accuracy.

Recent research has shown that integrating machine learning with blockchain can significantly improve the security and trustworthiness of distributed systems (Kumar et al., 2022; Wang et al., 2022). Blockchain provides an immutable and decentralized ledger, while machine learning enhances decision-making by analyzing transaction behaviors and network traffic. When combined with smart contracts, this integration enables automated, tamper-resistant, and secure validation of vehicular transactions without relying on centralized authorities. However, existing approaches still face limitations in terms of detection accuracy, computational efficiency, and adaptability to large-scale IoV environments.

To address these challenges, this research proposes a Hybrid Adaptive Network (HANet) for malicious activity detection and authentication in blockchain-enabled IoV systems. The HANet model integrates a Multilayer Perceptron (MLP) with a Ridge classifier to leverage the strengths of both learning paradigms. The MLP component effectively captures nonlinear and complex patterns in vehicular data, while the Ridge classifier mitigates overfitting and handles correlated features, as highlighted in earlier machine learning studies (Yadav and Vijayakumar, 2022). Through this hybrid design, every transaction in the IoV network is verified and authenticated before being recorded on the blockchain via smart contracts.

During the training phase, the proposed HANet learns to distinguish between legitimate and malicious behaviors, enabling early detection of unauthorized access attempts. The model is capable of processing large volumes of vehicular data efficiently, which is essential for real-time IoV applications. To further enhance performance, key network parameters are optimally tuned using the proposed Opposition Beluga Whale Optimization (OBWO) algorithm, inspired by the work of Zhong et al. (2022). The primary objective of this learning-based authentication framework is to maximize detection accuracy and robustness while minimizing false alarms, as formally defined by the objective function in Eq. 3.16.

$$ob = \arg \max_{\{h^{MLP}, ep^{MLP}, se^{MLP}, h^{rg}, ep^{rg}\}} \left[A + P + NPV + \frac{1}{FPR} \right] \quad (3.16)$$

Here, the number of hidden neurons used in the MLP model is represented as h^{MLP} , while the total number of training epochs in the MLP is denoted by ep^{MLP} . The parameter se^{MLP} indicates the number of steps executed per epoch during MLP training. Similarly, the hidden neuron count in the Ridge classifier is expressed as h^{rg} , and its corresponding epoch count is denoted by ep^{rg} .

In this work, the hidden neurons in the MLP are varied within the range of 5 to 255, and the number of epochs is selected between 5 and 50. The steps per epoch for the MLP are adjusted from 50 to 250. For the Ridge classifier, the hidden neuron count also ranges from 5 to 255, while the number of epochs lies between 5 and 50. These parameter ranges are chosen to ensure balanced learning performance and computational efficiency.

Furthermore, the symbol A represents the accuracy metric, while P denotes precision. The Negative Predictive Value is indicated by NPV , and the False Positive Rate is represented by FPR . These evaluation metrics are computed using Eqs. 3.17–3.20 to assess the effectiveness of the proposed authentication model.

Accuracy measures the similarity between the predicted outcomes and the actual data and is defined as:

$$A = \frac{op + qn}{op + qn + sr + ut} \quad (3.17)$$

Precision reflects the proportion of correctly identified positive instances among all predicted positives and is calculated as:

$$P = \frac{op}{op + qn} \quad (3.18)$$

Negative Predictive Value (NPV) indicates the probability that a negative prediction truly corresponds to a non-malicious instance and is given by:

$$NPV = \frac{op}{op + ut} \quad (3.19)$$

Finally, the False Positive Rate (FPR) quantifies the proportion of normal instances incorrectly classified as malicious and is expressed as:

$$FPR = \frac{sr}{sr + op} \quad (3.20)$$

Here, the measures of false negative and false positive correspond to incorrectly classified attack and normal instances, respectively, while true negative and true positive represent correctly identified normal and malicious events.

Finally, the output scores generated by the MLP and Ridge classifier models are averaged to obtain a final decision for malicious attack detection. This ensemble-based decision mechanism improves robustness by leveraging the complementary strengths of both classifiers. With the support of OBWO-based parameter optimization, the proposed Hybrid

Adaptive Network (HANet) effectively detects malicious activities in IoV networks while ensuring reliable and privacy-preserving authentication.

3.6. Results and Discussion

The proposed privacy-preserving authentication framework was implemented using the Python programming environment to ensure accurate and reliable performance evaluation. A comprehensive set of experiments was conducted to assess the effectiveness of the developed model under realistic IoV conditions. To demonstrate its superiority, the performance of the proposed approach was systematically compared with several existing authentication and security techniques using standard evaluation metrics. This comparative analysis highlights the strengths of the proposed framework in terms of detection accuracy, efficiency, and robustness, thereby validating its suitability for secure and privacy-aware Internet of Vehicles environments.

6.1 Simulation Setup

The proposed blockchain-enabled privacy-preserving authentication framework for the Internet of Vehicles (IoV) was implemented using the Python programming environment to evaluate its effectiveness under realistic conditions. The experimental analysis was carried out using the Internet of Vehicles Dataset obtained from the Kaggle repository, which is widely used for IoV-related research. The dataset consists of approximately 1000 instances with seven attributes, enabling a balanced evaluation of security and performance aspects. The selected attributes include the number of traffic lanes, driver status, environmental conditions, vehicle speed, inter-vehicle distance, and braking capability, all of which are critical factors influencing vehicular behavior and network dynamics.

For the machine learning and optimization process, key parameters were carefully selected to ensure fair and consistent evaluation. The maximum number of iterations was set to 50, and the population size was fixed at 10 for the optimization process, while the chromosome length was maintained at 50 in the proposed scheme. To validate the effectiveness of the developed Opposition Beluga Whale Optimization-based Hybrid Adaptive Network (OBWO-HANet), its performance was compared with several optimization-assisted authentication models, including Grey Wolf Optimizer-based HANet proposed by Faris et al. (2018), Eurasian Oystercatcher Optimiser-based HANet introduced by Salim et al. (2022), Lion Optimization Algorithm-based HANet developed by Yazdani and Jolai (2016), and the conventional Beluga Whale Optimization-based HANet proposed by Changting Zhong et al. (2022).

In addition, multiple classification models were employed to evaluate the robustness of the authentication mechanism, including Convolutional Neural Networks as discussed by Nie et al. (2020), Multi-Layer Perceptron models referenced by Yongbiao Li et al. (2015), Ridge Classifier techniques studied by Anagh Singh and Shiva Prakash (2016), and ensemble-based learning approaches presented by Rajkumar Gaur et al. (2022). Furthermore, the security performance of the proposed framework was assessed against several blockchain-based authentication approaches, including blockchain-based reputation and consensus authentication mechanisms proposed by Xiaoliang Wang et al. (2019), batch authentication schemes introduced by Palak Bagga et al. (2020), consortium blockchain architectures discussed by Chen Mao et al. (2021), and smart contract-enabled authentication frameworks presented by Gunasekaran Manogaran et al. (2023). This comprehensive simulation setup ensures a fair and thorough evaluation of the proposed privacy-preserving authentication scheme.

3.6.2 Performance Metrics

To assess the performance of the proposed privacy-preserving authentication framework for IoV, several standard evaluation metrics are employed. These metrics are commonly used to measure classification accuracy, error rates, and reliability in malicious attack detection systems.

The F1-score represents the harmonic mean of precision and recall, providing a balanced measure of detection performance, as defined in Eq. (3.21). Sensitivity evaluates the system's ability to correctly identify malicious instances, while Specificity measures how accurately legitimate vehicles are recognized without false alarms, as given in Eqs. (3.22) and (3.23), respectively.

$$F1\ S = 2 \times \frac{qn \times sr}{qn + sr} \quad (3.21)$$

$$Sen = \frac{op}{op + qn} \quad (3.22)$$

$$spec = \frac{sr}{sr + ut} \quad (3.23)$$

The False Negative Rate (FNR) quantifies the proportion of attacks that are incorrectly classified as normal, whereas the False Discovery Rate (FDR) estimates the ratio of false detections among all detected attacks, as shown in Eqs. (3.24) and (3.25). Recall indicates the effectiveness of the system in identifying actual malicious activities and is computed using Eq. (3.26).

$$FNR = \frac{ts}{ut + op} \quad (3.24)$$

$$FDR = \frac{sr}{op + sr} \quad (3.25)$$

$$Re = \frac{qn}{qn + ut} \quad (3.26)$$

Finally, the Matthews Correlation Coefficient (MCC) provides a comprehensive performance measure by jointly considering true and false classifications, making it suitable for imbalanced datasets. The MCC formulation is presented in Eq. (3.27).

$$MCC = \frac{ut \times qn - sr \times op}{\sqrt{(ut + sr)(ts + qn)(op + sr)(qn + ut)}} \quad (3.27)$$

3.6.3 Confusion Matrix Analysis of the Proposed Privacy-Preserving Authentication Task

The effectiveness of the proposed privacy-preserving authentication framework is evaluated using a confusion matrix, as illustrated in Fig. 3.8. The confusion matrix provides a detailed view of the classification outcomes by presenting the correctly and incorrectly identified instances for both legitimate and malicious vehicles. Through this analysis, the overall accuracy of the proposed approach is quantitatively assessed. The results demonstrate that the developed authentication scheme achieves a high level of correctness in identifying malicious activities while minimizing false classifications, thereby confirming the reliability and accuracy of the proposed model.

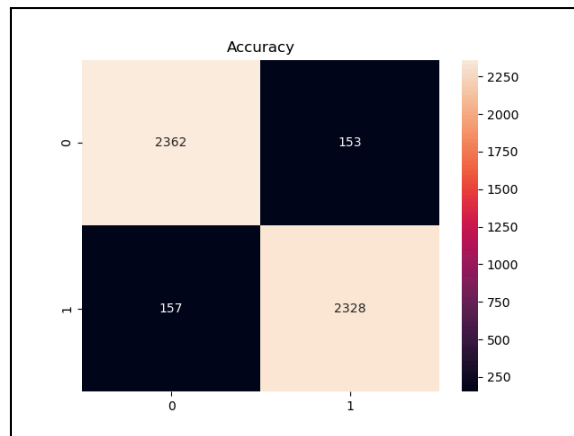


Figure 3.8: Confusion matrix analysis of the proposed blockchain-based privacy-preserving authentication scheme for Internet of Vehicles

3.6.4 ROC Analysis of the Proposed Privacy-Preserving Authentication Scheme Using Different Classifiers

Figure 3.9 presents the Receiver Operating Characteristic (ROC) analysis of the proposed blockchain-based privacy-preserving authentication scheme for the Internet of Vehicles, compared with several existing classification models. The evaluation is performed by examining the trade-off between the true positive rate and the false positive rate. As observed in Figure 3.10, when the false positive rate is fixed at 0.6, the proposed approach outperforms the conventional classifiers by noticeable margins, achieving improvements of 31.6% over the Convolutional Neural Network, 33.3% over the Multi-Layer Perceptron, 35% over the Ridge classifier, and 36.6% over the ensemble-based method. These results clearly indicate that the proposed authentication framework provides superior detection capability and more reliable classification performance compared to existing approaches.

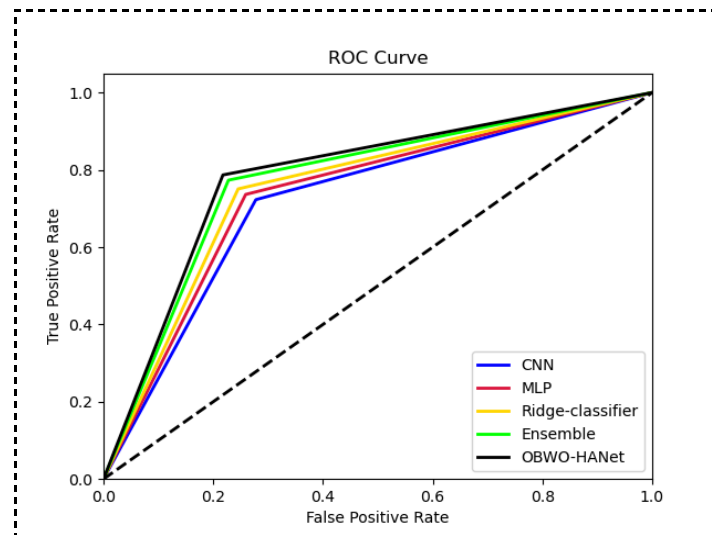


Figure 3.9: ROC performance comparison of the proposed blockchain-based privacy-preserving authentication scheme with conventional classifiers in Internet of Vehicles

3.6.5 Convergence Analysis of the Proposed OBWO Algorithm Compared with Existing Optimization Methods

The convergence behavior of the proposed Opposition-Based Beluga Whale Optimization (OBWO) algorithm is analyzed and compared with several conventional optimization techniques, as illustrated in Figure 3.10. Convergence performance is evaluated by observing the variation of the objective function across successive iterations. At an iteration count of 30, the proposed OBWO algorithm demonstrates significantly faster and more stable convergence, achieving improvements of 95% over GWO-HANet, 94.6% over EOO-HANet, 95% over LOA-HANet, and 93.6% over the standard BWO-HANet approach. These results confirm that the OBWO algorithm converges more efficiently than the existing optimization methods, making it highly suitable for parameter tuning in privacy-preserving IoV authentication systems.

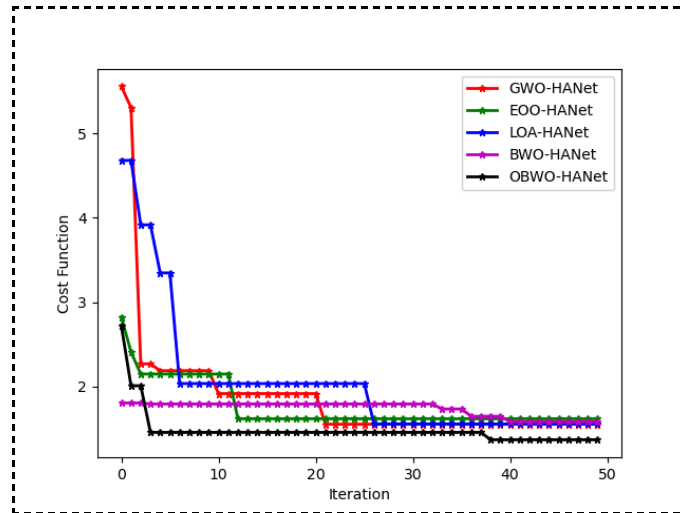


Figure 3.10: Convergence performance comparison of the proposed OBWO algorithm with conventional optimization techniques

3.6.6 Statistical Analysis of the Proposed OBWO Algorithm in Comparison with Conventional Methods

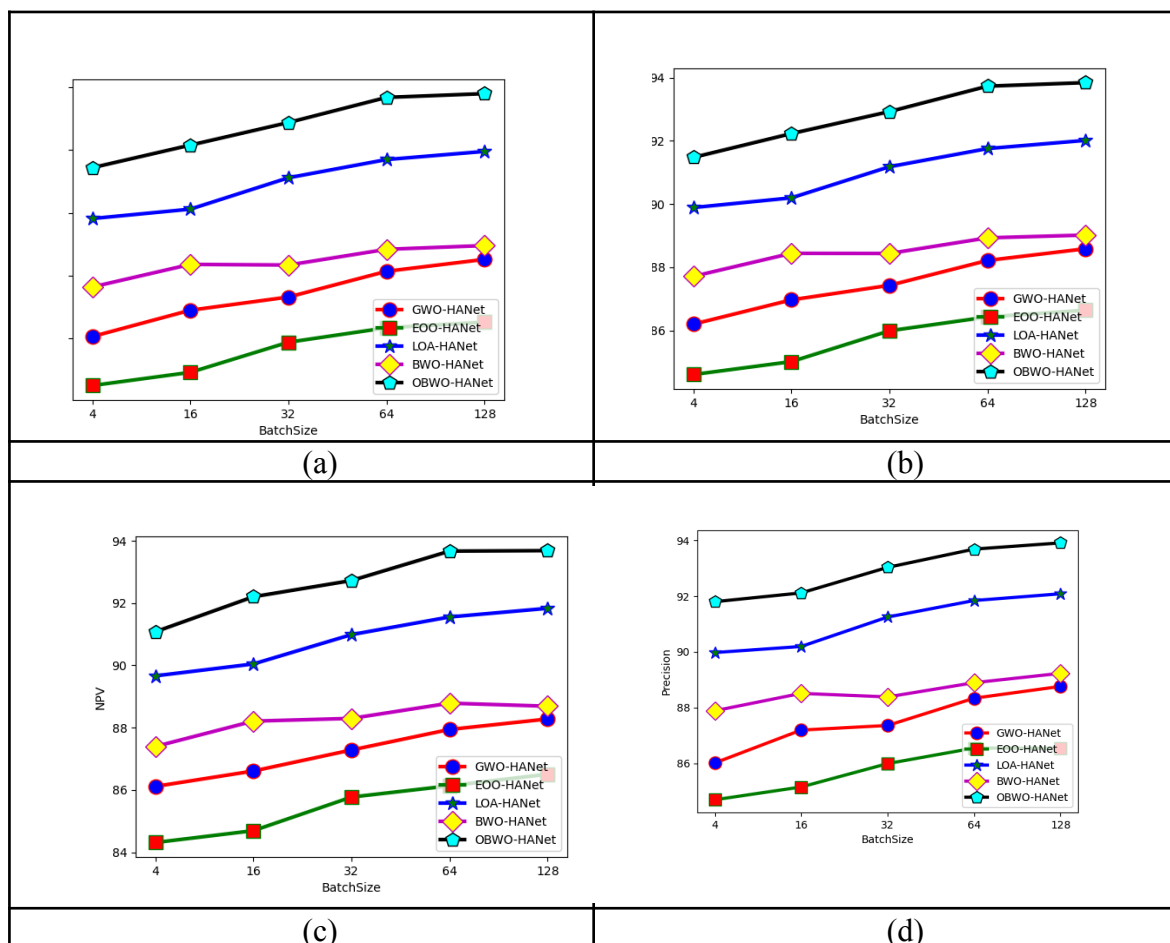
Table 3.1 presents the statistical performance evaluation of the proposed Opposition-Based Beluga Whale Optimization (OBWO) algorithm in comparison with several classical optimization approaches. Based on the best performance values reported in Table 3.1, the proposed OBWO method shows notable improvements of 15.3% over GWO-HANet, 23% over EOO-HANet, 19.2% over LOA-HANet, and 21.5% over the conventional BWO-HANet algorithm. These results demonstrate the superior effectiveness and reliability of the proposed OBWO algorithm for optimization tasks within the proposed authentication framework.

Table 3.1: Statistical performance comparison of the proposed OBWO algorithm with conventional optimization techniques

Terms	GWO-HANet	EOO-HANet	LOA-HANet	BWO-HANet	OBWO-HANet
“BEST”	1.553760376	1.617892281	1.55670825	1.582391687	1.369468782
“WORST”	5.556646586	2.824728034	4.678367771	1.804862218	2.723124883
“MEAN”	1.893152234	1.763864911	2.03874114	1.7365116	1.482994292
“MEDIAN”	1.553760376	1.617892281	2.034938356	1.793239309	1.45642757
“STD”	0.764213907	0.276649554	0.774157555	0.087188846	0.213103094

3.6.7 Performance Evaluation of the Proposed Privacy-Preserving Authentication Scheme over Conventional Algorithms and Classifiers

The effectiveness of the proposed blockchain-enabled privacy-preserving authentication scheme for the Internet of Vehicles is assessed by comparing it with several existing optimization algorithms and classification models, as illustrated in Fig. 3.11 and Fig. 3.12. The evaluation is carried out by varying the batch size to analyze the robustness and consistency of the proposed method under different training conditions. When the batch size is set to 32, the proposed authentication framework demonstrates a notable improvement in accuracy compared to conventional approaches. Specifically, the achieved accuracy shows an enhancement of 17.4% over GWO-HANet, 16.7% over EOO-HANet, 18.2% over LOA-HANet, and 17.5% over BWO-HANet. These results clearly indicate that the designed privacy-preserving authentication scheme delivers superior performance and reliability compared to traditional algorithms, validating its suitability for secure and efficient IoV environments.



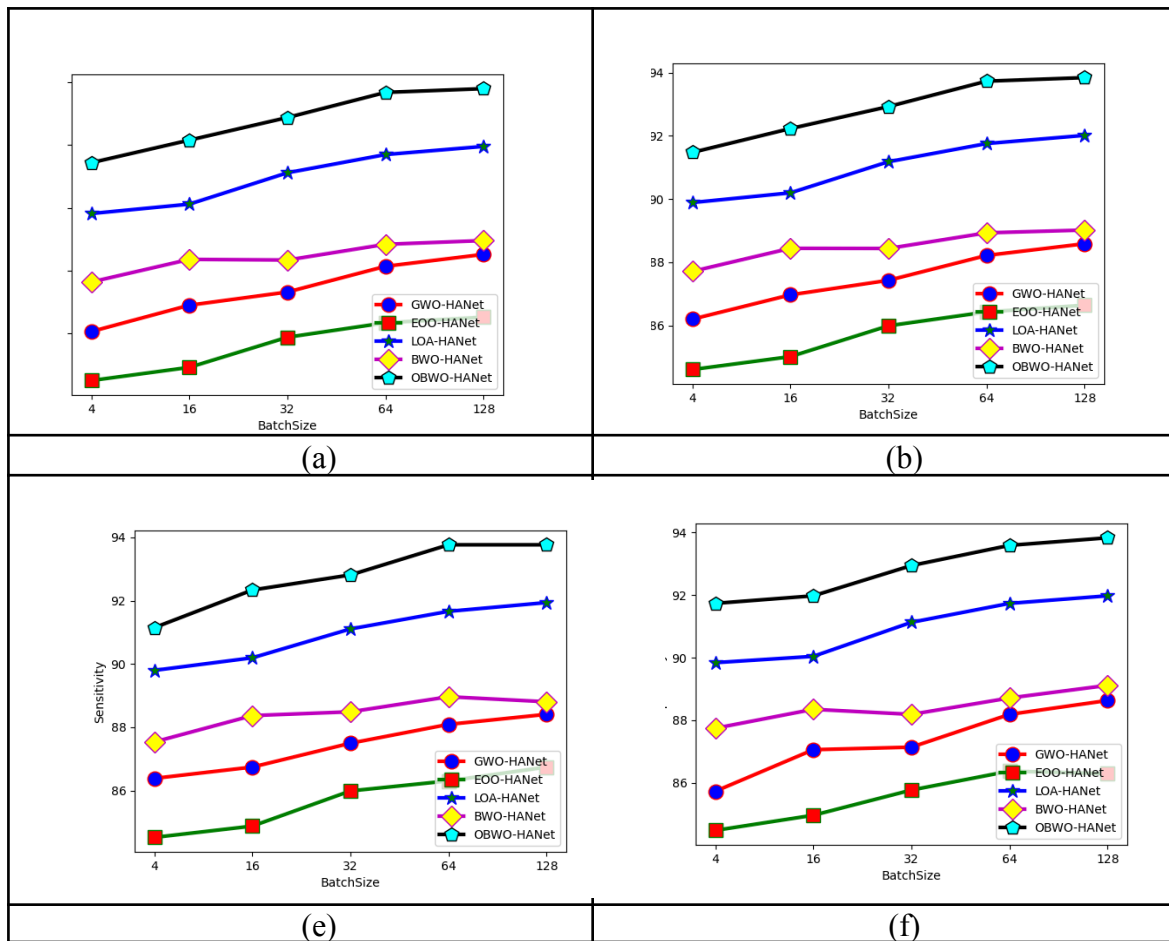
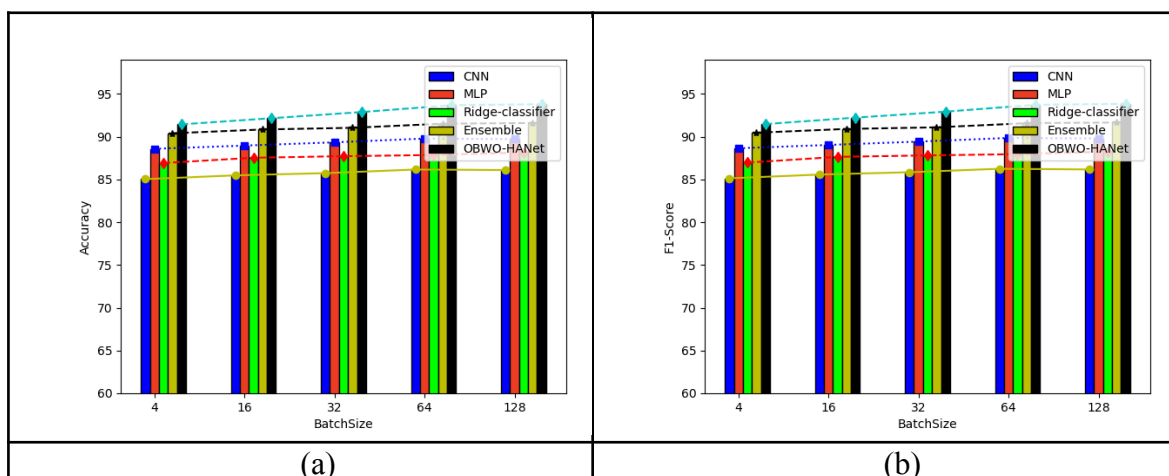


Figure 3.11: Performance comparison of the proposed blockchain-enabled privacy-preserving authentication scheme for IoV with existing algorithms in terms of (a) Accuracy, (b) F1-score, (c) Negative Predictive Value (NPV), (d) Precision, (e) Sensitivity, and (f) Specificity



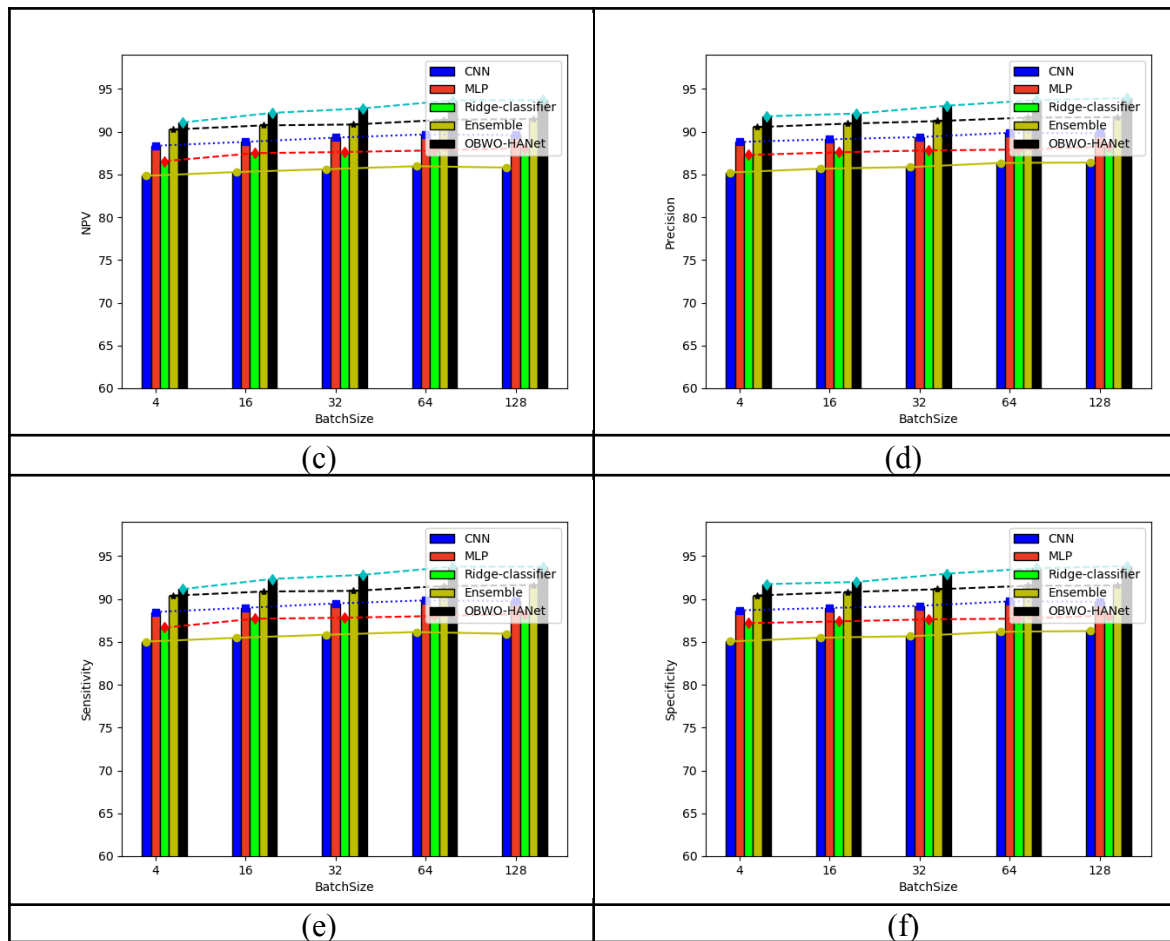


Figure 3.12: Performance evaluation of the proposed blockchain-based privacy-preserving authentication scheme for IoV across different traditional classifiers with respect to (a) Accuracy, (b) F1-score, (c) Negative Predictive Value (NPV), (d) Precision, (e) Sensitivity, and (f) Specificity

6.8 Overall Comparative Evaluation of the Proposed System against Conventional Algorithms and Classifiers

Table 3.2 and Table 3.3 present a comprehensive comparative analysis of the proposed blockchain-based privacy-preserving authentication scheme for the Internet of Vehicles in comparison with existing optimization algorithms and classification models. The evaluation focuses on key performance indicators to demonstrate the robustness and effectiveness of the suggested approach. As observed from the comparative results, the proposed scheme shows a noticeable improvement in sensitivity when benchmarked against traditional

optimization-based methods. Specifically, the sensitivity performance is improved by 5.6% over GWO-HANet, 7.4% over EOO-HANet, 1.9% over LOA-HANet, and 5.2% over BWO-HANet. These consistent gains confirm that the proposed authentication framework achieves superior detection capability and reliability, thereby ensuring enhanced security and privacy preservation in IoV environments.

Table 3.2: Comparative analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV with conventional optimization algorithms

Terms	GWO-HANet	EOO-HANet	LOA-HANet	BWO-HANet	OBWO-HANet
“Accuracy”	88.52	86.52	91.96	88.96	93.8
“Sensitivity”	88.40809845	86.74077015	91.94124653	88.80508138	93.767368
“Specificity”	88.63361548	86.29584845	91.97904071	89.11729141	93.8331318
“Precision “	88.76046234	86.53465347	92.08747515	89.23015556	93.91650099
“FPR “	11.36638452	13.70415155	8.020959291	10.88270859	6.166868198
“FNR”	11.59190155	13.25922985	8.058753474	11.19491862	6.232631997
“NPV”	88.27780008	86.50505051	91.83098592	88.68832732	93.68209256
“FDR”	11.23953766	13.46534653	7.912524851	10.76984444	6.083499006
“F1-Score”	88.58392999	86.63758921	92.01430274	89.01711102	93.84187525
“MCC”	0.770399882	0.730381613	0.839193741	0.779204278	0.875995467

Table 3.3: Comparative performance analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV against conventional classification models

Terms	CNN	MLP	Ridge-classifier	Ensemble	OBWO-HANet
“Accuracy”	86.1	89.72	88.16	91.6	93.8
“Sensitivity”	85.94680429	89.75784041	88.24930528	91.62366018	93.767368
“Specificity”	86.25554212	89.68158001	88.06932688	91.57597743	93.8331318

“Precision“	86.3926576 2	89.8291617	88.24930528	91.69646404	93.91650099
“FPR “	13.7444578 8	10.3184199 9	11.93067312	8.424022572	6.166868198
“FNR”	14.0531957 1	10.2421595 9	11.75069472	8.376339817	6.232631997
“NPV”	85.8059342 4	89.6093435 4	88.06932688	91.50221506	93.68209256
“FDR”	13.6073423 8	10.1708383	11.75069472	8.303535956	6.083499006
“F1-Score”	86.1691542 3	89.7934868 9	88.24930528	91.66004766	93.84187525
“MCC”	0.72200469 1	0.79438962 8	0.763186322	0.831991584	0.875995467

3.6.9 Throughput analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV

Figure 3.13 illustrates the throughput performance of the proposed privacy-preserving authentication scheme for IoV in comparison with existing blockchain-based approaches. The evaluation is carried out by analyzing the amount of successfully processed authentication transactions over time. At an execution time of 20 seconds for the RSU scenario, the proposed scheme demonstrates a notable improvement in throughput compared to BRCA, batch-based authentication, consortium blockchain, and conventional smart contract approaches. This improvement indicates that the proposed framework is capable of handling a higher volume of authenticated communications within the same time duration. The results confirm that integrating smart contracts with an efficient authentication mechanism significantly enhances data processing efficiency, making the proposed solution more suitable for real-time and large-scale IoV environments.

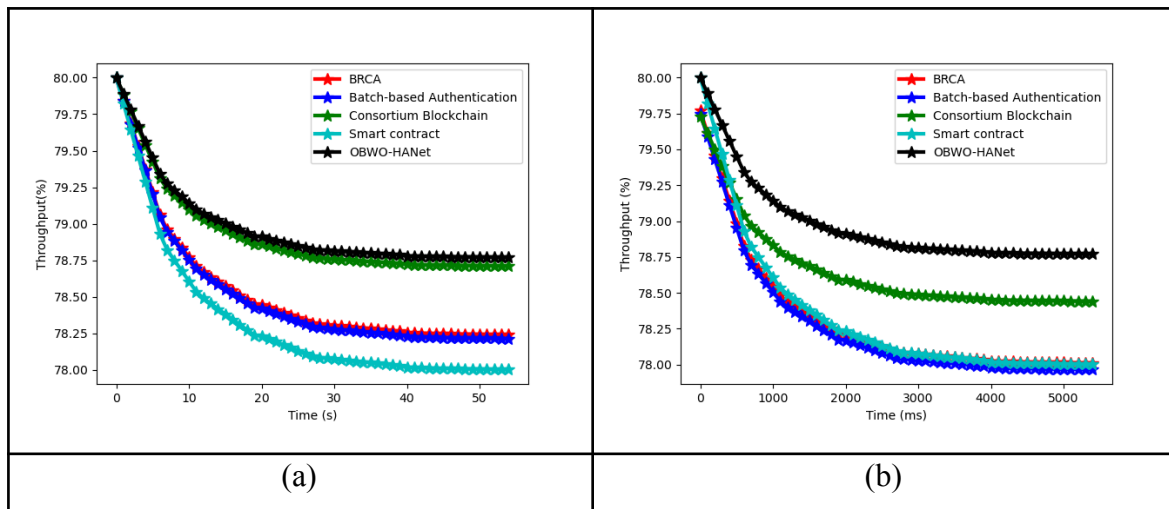


Figure 3.13: Throughput performance comparison of the proposed blockchain-enabled privacy-preserving authentication scheme for IoV under (a) RSU-level communication and (b) vehicle-level communication

3.6.10 Security level evaluation of the proposed blockchain-based privacy-preserving authentication scheme for IoV

The security robustness of the proposed privacy-preserving authentication scheme for IoV using blockchain technology is analyzed and compared with existing approaches, as illustrated in Figure 3.14. The assessment is carried out by examining the security level with respect to increasing blockchain length. As the blockchain size grows, the proposed scheme consistently demonstrates superior security performance compared to BRCA, batch-based authentication, consortium blockchain, and conventional smart contract mechanisms. When the blockchain length reaches 4000, the proposed approach achieves a security improvement of 99.95%, 99.93%, 99.98%, and 99.985% over the respective benchmark methods. These results clearly indicate that the proposed framework offers strong resistance against security threats while ensuring enhanced privacy protection for vehicular communications in IoV networks.

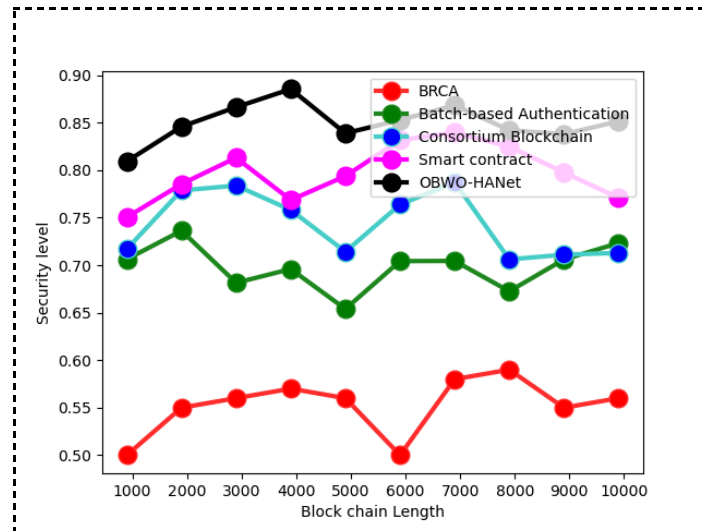


Figure 3.14: Security performance analysis of the proposed blockchain-enabled privacy-preserving authentication scheme for IoV

6.11 Operating Time Analysis of the Proposed Privacy-Preserving Authentication Scheme

The operating time performance of the proposed privacy-preserving authentication scheme for IoV using blockchain technology is evaluated and compared with existing approaches, as illustrated in Fig. 3.15. In this analysis, vehicle density is considered a key factor to measure the execution efficiency of the authentication process. When the number of vehicles reaches 100, the proposed scheme demonstrates a significant reduction in operating time compared to conventional methods. Specifically, the execution time is reduced by 99.2% compared to BRCA, 99.17% compared to batch-based authentication, 99.26% compared to consortium blockchain, and 99.3% compared to smart contract-based approaches. These results clearly indicate that the proposed authentication framework achieves superior computational efficiency and is highly suitable for large-scale and dense IoV environments.

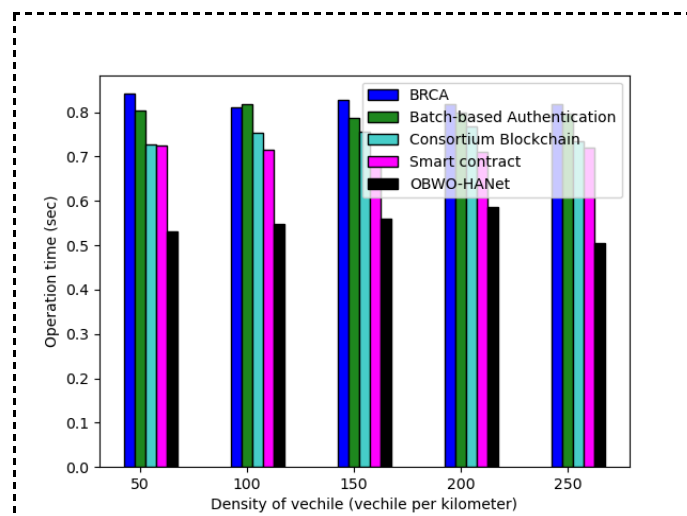


Figure 3.15: Operating Time Comparison of the Proposed Blockchain-Enabled Privacy-Preserving Authentication Scheme for IoV

3.6.12 Comparative Analysis of the Proposed Model with Recent Classifiers

A comparative evaluation is carried out to assess the effectiveness of the proposed privacy-preserving authentication model against recent deep learning-based classifiers, and the results are summarized in Table 3.4. The comparison highlights the accuracy achieved by each method, where the developed framework attains a superior accuracy of 93%. This improvement demonstrates the ability of the proposed model to authenticate IoV entities more reliably while preserving privacy. The consistent performance gain indicates that the developed approach offers a more robust and efficient authentication mechanism compared to existing classifiers, making it suitable for secure IoV environments.

Table 3.4: Comparative Performance Evaluation of the Proposed Privacy-Preserving Authentication Model

Terms	BRCA	Federated Learning	MDP	OBWO-HANet
Accuracy	81.94	82.94	83.88	93.8
Sensitivity	81.61	83.07	84.10	93.76
Specificity	82.28	82.80	83.63	93.83
Precision	83.02	83.68	84.51	93.91
FPR	17.72	17.20	16.37	6.16
FNR	18.39	16.93	15.90	6.23
NPV	80.82	82.16	83.21	93.68
FDR	16.98	16.32	15.49	6.08
F1-Score	82.31	83.37	84.31	93.84
MCC	63.87	65.85	67.73	87.59

3.6.13 Statistical Analysis of the Developed Framework

The statistical evaluation of the proposed privacy-preserving authentication framework is presented in Table 3.5 to validate its effectiveness and reliability. A significance threshold of 0.005 is adopted during the experimental assessment to ensure rigorous validation of the obtained results and to confirm that the observed performance improvements are statistically meaningful.

Table 3.5: Statistical performance evaluation of the proposed privacy-preserving authentication framework

Comparison	Statistic	Adjusted p-value	Result
GWO-HANet [39]vs OBWO-HANet	1.78885	0.73638	H0 is accepted
LOA-HANet [41]vs GWO-HANet [39]	1.34164	1	H0 is accepted
EOO-HANet [40]vs OBWO-HANet	1.34164	1	H0 is accepted
LOA-HANet [41]vs EOO-HANet [40]	0.89443	1	H0 is accepted
GWO-HANet [39]vs BWO-HANet [35]	0.89443	1	H0 is accepted
OBWO-HANet vs BWO-HANet [35]	0.89443	1	H0 is accepted
LOA-HANet [41]vs OBWO-HANet	0.44721	1	H0 is accepted
EOO-HANet [40]vs GWO-HANet [39]	0.44721	1	H0 is accepted
EOO-HANet [40]vs BWO-HANet [35]	0.44721	1	H0 is accepted
LOA-HANet [41]vs BWO-HANet [35]	0.44721	1	H0 is accepted

3.6.14 Comparison of the Proposed Model with Existing Authentication and Privacy-Preserving Schemes

This section presents a comparative evaluation of the proposed privacy-preserving authentication framework against several state-of-the-art authentication and privacy protection approaches, as summarized in Table 3.6 and Table 3.7. Existing methods include mutual authentication protocols and user-centric privacy-preserving authentication schemes designed primarily for IoT environments, as well as privacy-preserving mutual authentication mechanisms developed for Vehicular Ad hoc

Networks (VANETs). In addition, pseudonym-based techniques and public key cryptography approaches are commonly employed to safeguard user privacy in vehicular networks, while learner-centered privacy preservation models have been explored in IoT-based systems.

Although these approaches provide varying levels of security and privacy, they often suffer from limitations such as high computational overhead, centralized trust dependency, or limited adaptability to highly dynamic IoV environments. In contrast, the proposed model integrates blockchain, smart contracts, and intelligent learning mechanisms to jointly enhance authentication strength and privacy protection. The comparative results demonstrate that the proposed framework consistently outperforms existing schemes in terms of authentication reliability, privacy preservation, and resistance to unauthorized access. Consequently, the developed model is well-suited for securing IoV networks and mitigating a wide range of cyber and privacy-related attacks in intelligent transportation systems.

Table 3.6: Comparative Evaluation of the Proposed Framework with Existing State-of-the-Art Privacy-Preserving Approaches

Terms	Mutual authentication protocol [47]	User-centric privacy-preserving authentication protocol [48]	Privacy-preserving mutual authentication scheme [49]	OBWO-HANet
Accuracy	85.36842	86.52632	88.68421	90.36842
Sensitivity	85.47544	86.2069	88.50575	90.59561
Specificity	85.25981	86.85048	88.86532	90.13786
Precision	85.47544	86.93361	88.97059	90.3125
FPR	14.74019	13.14952	11.13468	9.862142
FNR	14.52456	13.7931	11.49425	9.404389
NPV	85.25981	86.11987	88.39662	90.42553
FDR	14.52456	13.06639	11.02941	9.6875
F1-Score	85.47544	86.56873	88.73756	90.45383

MCC	0.707353	0.730554	0.773691	0.807358
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Table 3.7: Performance Comparison of the Proposed Authentication Framework with Existing Privacy-Preservation Techniques

Security level report				
Terms	Pseudonym-based approach [50]	Learner-centered approach [51]	Public key cryptography [52]	OBWO-HANet
1000	0.417177	0.469534	0.553197	0.809339
2000	0.458484	0.489177	0.57873	0.845836
3000	0.466745	0.453304	0.598922	0.866382
4000	0.475007	0.462786	0.566437	0.885415
5000	0.466745	0.435301	0.584816	0.838903
6000	0.417177	0.468259	0.612526	0.852843
7000	0.483268	0.46837	0.618205	0.868328
8000	0.491529	0.447467	0.607547	0.842115
9000	0.458484	0.469386	0.587673	0.837836
10000	0.466745	0.480673	0.568069	0.851524

3.6.15 ANOVA and Wilcoxon Signed-Rank Test Analysis

The statistical significance of the proposed model is evaluated using ANOVA and Wilcoxon signed-rank tests, as presented in Table 3.8. The ANOVA test is applied to examine the differences in authentication accuracy and privacy preservation performance between the proposed framework and existing methods. In addition, the Wilcoxon signed-rank test is employed to assess pairwise performance variations between the proposed approach and conventional algorithms. The outcomes of both tests indicate statistically significant improvements achieved by the proposed model, thereby validating its robustness and superior effectiveness in authentication and privacy protection for IoV environments.

Table 3.8: Statistical significance evaluation of the proposed authentication and privacy-preserving model

Test	GWO-HANet [39]	EOO-HANet [40]	LOA-HANet [41]	BWO-HANet [35]	OBWO-HANet
ANOVA	0.902	0.704	1.209	2.13	2.9801
Wilcoxon	0.509	0.64	1.902	1.2	2.1942

3.7 Summary

This chapter presented the proposed blockchain-based privacy-preserving authentication model for the Internet of Vehicles, which integrates smart contracts with a Hybrid Adaptive Network optimized using the Opposition Beluga Whale Optimization algorithm. The model effectively ensures secure, decentralized authentication and accurate detection of malicious activities while reducing computational overhead. Although the proposed model demonstrates strong performance, certain challenges related to scalability and advanced security handling remain. To overcome these issues, the next chapter introduces an enhanced model that further improves authentication efficiency, robustness, and privacy preservation in dynamic IoV environments.

CHAPTER 4

BLOCKCHAIN-BASED SECURE AND ENERGY-EFFICIENT PRIVACY-PRESERVING AUTHENTICATION PROTOCOL FOR INTERNET OF VEHICLES USING OPTIMIZED EL-GAMAL ENCRYPTION

4.1 Preamble

The rapid expansion of the Internet of Vehicles has intensified the need for more advanced security mechanisms that can operate efficiently in highly dynamic and large-scale vehicular environments. Although blockchain-based authentication frameworks provide decentralization and data integrity, relying on a single security layer is often insufficient to counter sophisticated cyber threats and privacy breaches. The increasing volume of vehicular data, strict latency requirements, and frequent vehicle mobility demand authentication schemes that not only ensure strong security but also maintain low computational overhead and real-time responsiveness.

To address these emerging challenges, this chapter proposes an enhanced authentication framework that integrates deep learning techniques with optimized cryptographic mechanisms within a blockchain-enabled IoV environment. By combining intelligent learning models with secure encryption strategies, the proposed approach aims to strengthen attack detection accuracy, preserve user privacy, and improve system efficiency. This chapter builds upon the foundation established in the previous chapter and introduces a more robust and scalable solution designed to meet the security and performance requirements of next-generation IoV networks.

4.2 Introduction

The Internet of Vehicles (IoV) has emerged as a key component of intelligent transportation systems by enabling seamless communication among vehicles, roadside infrastructure, and traffic management entities through On-Board Units (OBUs) and Roadside Units (RSUs). This interconnected environment significantly improves driving safety, traffic efficiency, and overall road utilization by supporting real-time information exchange and cooperative decision-making (Aman et al., 2021; Qureshi et

al., 2022). However, the rapid growth of IoV networks and their open communication nature have also introduced serious security and privacy challenges.

Authentication is the first and most critical line of defense in IoV systems. Without proper authentication, malicious vehicles can easily join the network and perform harmful actions, such as impersonating emergency vehicles to bypass traffic regulations or injecting false traffic information to mislead other drivers (Chai et al., 2022). In addition, message integrity is equally important, as tampered data may still appear trustworthy to receiving vehicles, potentially leading to unsafe driving decisions and severe consequences (Xu et al., 2021). Therefore, ensuring message authenticity, integrity, non-repudiation, and traceability is essential for secure IoV communications.

Privacy preservation is another major concern in IoV environments, since vehicles continuously generate sensitive data such as location, identity, and movement patterns. Although traceability is required to identify and penalize misbehaving vehicles, unrestricted disclosure of vehicle information can result in unauthorized tracking and privacy violations (Kaltakis et al., 2021; Lin et al., 2021). Conditional privacy-preserving mechanisms aim to balance these conflicting requirements by protecting vehicle anonymity under normal conditions while enabling identity revelation when malicious behavior is detected.

Traditional IoV security solutions largely rely on centralized authorities, which suffer from single-point-of-failure issues, limited scalability, delayed response times, and vulnerability to insider attacks (Zhang and Wu, 2021; KEBande et al., 2021). Moreover, centralized trust models struggle to handle the dynamic topology and high mobility of vehicular networks, making them inadequate for next-generation IoV applications. To overcome these limitations, decentralized and intelligent security mechanisms are required.

Blockchain technology has gained significant attention as a promising solution for IoV security due to its decentralized architecture, immutability, and cryptographic trust mechanisms. By distributing consensus among participating nodes, blockchain enables secure data sharing without relying on a trusted third party (Shen et al., 2022; Tu et al., 2023). However, blockchain-based IoV systems often incur high computational, storage, and communication overhead, particularly during consensus operations and key management processes (Xie et al., 2023). Furthermore, the massive volume of vehicular data can lead to network congestion and latency.

To address these challenges, intelligent learning techniques and optimized cryptographic mechanisms are increasingly integrated into blockchain-enabled IoV systems. Deep learning models can effectively detect malicious activities by analyzing complex traffic patterns, while optimized encryption schemes can ensure data confidentiality with reduced computational cost (Wang et al., 2022; Ghimire and Rawat, 2022). Motivated by these advancements, this work proposes a systematic blockchain-based privacy-preserving authentication protocol that combines deep learning-based attack detection with optimized El-Gamal encryption for secure and efficient IoV communications.

4.2.1 Why Blockchain in IoV

IoV networks operate in highly dynamic and decentralized environments where vehicles frequently join and leave the network. In such scenarios, centralized security architectures become vulnerable to single-point failures, data manipulation, and trust management issues. Blockchain addresses these challenges by providing a decentralized and tamper-resistant ledger that records all transactions in an immutable manner (Chattaraj et al., 2021; Shen et al., 2022). Through consensus mechanisms, blockchain ensures that only valid and authenticated information is accepted by the network.

Additionally, blockchain leverages cryptographic techniques to protect sensitive vehicular data, such as vehicle identities and locations, from unauthorized access.

Secure peer-to-peer communication enabled by blockchain eliminates the need for intermediaries, thereby reducing latency and improving reliability for time-sensitive IoV applications (Wu et al., 2023). These features make blockchain a suitable foundation for building secure, trustworthy, and privacy-aware IoV authentication systems.

4.2.2 IoV Use Cases

The proposed blockchain-based privacy-preserving authentication protocol supports several critical IoV applications. In real-time traffic management, authenticated and trustworthy data sharing among vehicles enables efficient route optimization and congestion control. For vehicle-to-vehicle and vehicle-to-infrastructure payments, blockchain facilitates secure and decentralized financial transactions, such as toll collection and parking fee payments, without relying on centralized payment gateways.

In accident reporting systems, blockchain ensures immutable storage of event data, which is crucial for insurance claims and legal investigations. Furthermore, autonomous driving applications rely heavily on trusted communication among vehicles to coordinate maneuvers and ensure safety. In all these use cases, strong authentication and privacy preservation are essential, and the proposed protocol is designed to meet these requirements effectively.

4.3 Systematic View of IoV Network and Dataset Characteristics with Blockchain-Based Storage

4.3.1 System View of the Internet of Vehicles Network

The Internet of Vehicles (IoV) represents a large-scale intelligent networking paradigm that integrates smart transportation technologies with Internet of Things (IoT) concepts to extend the capabilities of traditional Vehicular Ad Hoc Networks (VANETs). By enabling continuous interaction among vehicles, infrastructure, users, and cloud platforms, IoV supports a wide range of applications such as intelligent

traffic management, collision avoidance, emergency response, autonomous driving, infotainment services, and smart navigation (Aman et al., 2021; Qureshi et al., 2022).

An IoV ecosystem is composed of several interconnected entities that collaboratively enable intelligent transportation services. Vehicles are equipped with On-Board Units (OBUs), sensors, and communication modules that collect and transmit real-time vehicular and environmental data. Drivers and passengers interact with these systems to access safety and convenience services. Roadside Units (RSUs), deployed along roadways, act as intermediate communication points that facilitate data exchange between vehicles and backend infrastructure. In addition, cloud servers provide large-scale data storage, analytics, and decision support, while a Central Authority (CA) is traditionally responsible for network registration, credential management, and access control (Xu et al., 2021; Kebande et al., 2021).

Through the periodic exchange of safety-related messages, such as beacon or safety broadcast messages, vehicles form a dynamic and socially connected network of smart objects. This interaction allows vehicles to share traffic conditions, road hazards, and driving intentions in real time. Due to the coexistence of heterogeneous users, devices, and services, IoV is classified as a heterogeneous network. Its highly dynamic topology, strict latency requirements, and continuous data generation introduce significant challenges in terms of scalability, reliability, and security (Kaltakis et al., 2021).

Conventionally, the IoV architecture follows a layered design that relies on centralized trust management through a central authority. This architecture can be broadly described using four functional layers. The environment sensing and control layer is responsible for collecting raw data through vehicle-mounted sensors. These sensors capture information related to vehicle status, surrounding traffic, road conditions, positioning systems, and wireless signal characteristics. The network access and transport layer manages communication protocols, node connectivity, data validation, and message dissemination between vehicles, RSUs, and infrastructure.

The coordination and computing layer handles data processing, resource scheduling, and system management functions. This layer often interacts with cloud or edge computing platforms to support real-time analytics and decision-making. Finally, the application layer provides end-user services by analyzing processed data and responding to safety-critical or convenience-oriented events. Applications at this layer include traffic safety alerts, route optimization, infotainment services, and emergency management. These services can be further categorized into open services, such as real-time traffic updates accessible to all vehicles, and cloud-based services, including centralized traffic monitoring and control platforms (Shen et al., 2022; Tu et al., 2023).

A conceptual illustration of the IoV system architecture is presented in Fig. 4.1, highlighting the interaction among vehicles, RSUs, cloud services, and management entities.

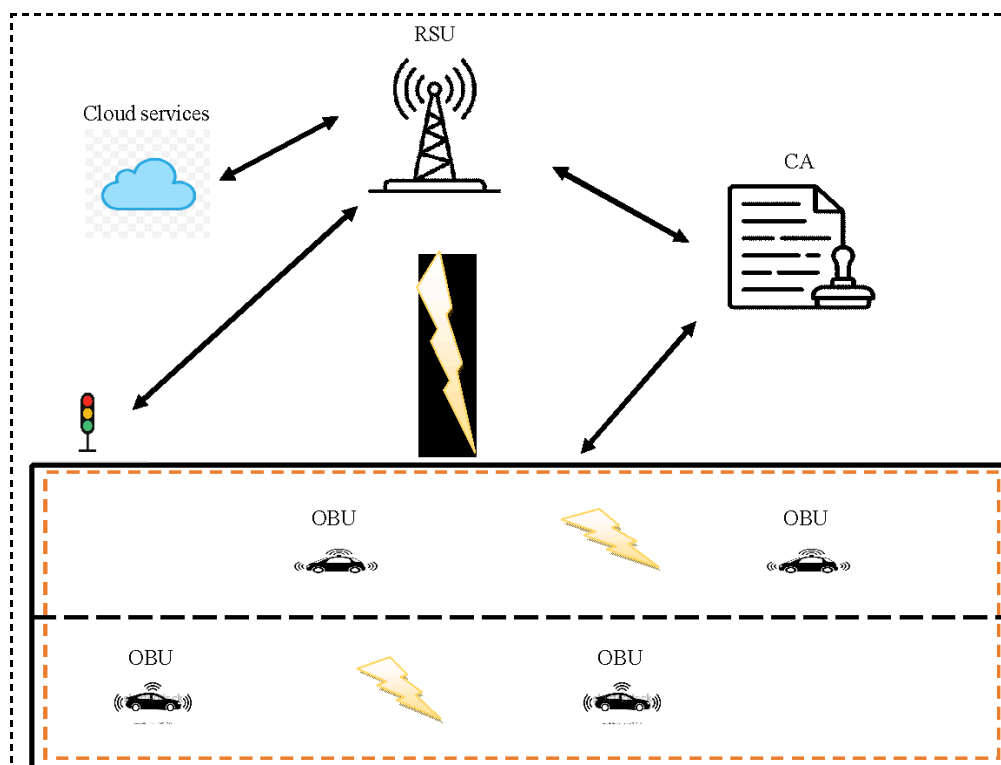


Figure 4.1: Conceptual Architecture of the Internet of Vehicles Network

4.3.2 Dataset Details

To validate the proposed authentication and privacy-preserving framework, vehicular data are obtained from the Highway Safety Research Dataset maintained by the National Highway Traffic Safety Administration (NHTSA). This dataset contains comprehensive information related to road traffic scenarios, including vehicle crash records, traffic flow characteristics, and sensor-generated vehicular data collected from real IoV environments across major highways in the United States. Unlike commonly used generic datasets, this source provides realistic and trustworthy data suitable for evaluating security, authentication, and collision-related behaviors in IoV systems. The dataset includes key attributes such as vehicle speed, positional information, traffic conditions, and accident occurrences. For this study, approximately 50,000 data records are utilized to ensure reliable performance assessment and meaningful experimental analysis.

4.3.3 Data Storage in Blockchain

The vehicular data I_v collected from the data source are securely stored using blockchain technology. Blockchain functions as a distributed and shared digital ledger that records transactions among entities that do not necessarily trust one another. Once stored, the data become tamper-resistant, as the ledger is append-only and updated sequentially through a series of interconnected blocks. Each block typically contains the cryptographic hash of the previous block, a timestamp, and the transaction records. This chained structure ensures data integrity, since any attempt to modify stored information would alter the block hash and invalidate all subsequent blocks unless the entire chain is recomputed.

Blockchain networks are commonly classified into public blockchains, where any participant can join and validate transactions, and private blockchains, which restrict access to authorized entities only. In IoV environments, this flexibility allows blockchain deployment to be tailored according to security and access control requirements.

The adoption of blockchain for vehicular data management offers several key advantages. Immutability ensures that once a block is added, its content cannot be altered without detection. The distributed ledger mechanism enables all participating nodes to maintain synchronized copies of the ledger, enhancing transparency and trust. The consensus mechanism defines the rules by which blocks are validated and appended, ensuring agreement among network participants. In addition, smart contracts—self-executing code deployed on the blockchain—automate transaction validation and authentication once predefined conditions are met.

By leveraging these features, blockchain effectively addresses security vulnerabilities, trust issues, and centralized failures in vehicular networks. In the proposed system, smart contracts are utilized to securely record vehicular data and support authentication operations within the IoV framework.

4.3.4 Authentication Mechanism Using Cascaded and Dilated Residual RNN in the IoV Platform

Recurrent Neural Networks (RNNs) are employed in this work to support the authentication process within the Internet of Vehicles (IoV) environment. RNNs are well known for their effectiveness in handling sequential data, as they maintain internal memory that captures temporal relationships between consecutive inputs. This capability makes RNNs particularly suitable for IoV authentication tasks, where vehicle messages and network behaviors follow time-dependent patterns. By learning historical dependencies, the model can distinguish between legitimate and malicious transmission behaviors more accurately. As highlighted by Elman (1990) and later extended by Hochreiter and Schmidhuber (1997), RNN-based models are capable of modeling long-term dependencies that are essential for security-sensitive applications. In the proposed framework, the authentication decision is generated iteratively across time steps, where the network updates its hidden state and output based on both the current input and the previously learned context, enabling reliable and continuous authentication in dynamic IoV networks.

$$j_f = \alpha(S_{yj}y_j + S_{jj}j_{f-1} + d_j) \quad (4.1)$$

$$Z_j = S_{jz} + d_z \quad (4.2)$$

In this formulation, the variables y and z represent the input sequence and the corresponding output sequence, respectively. The series of hidden state vectors is denoted by j , which captures the temporal dependencies within the data. The network parameters include the weight matrix S and the bias vector d , both of which are learned during the training process. The hidden layer activation function is responsible for transforming the weighted inputs into meaningful internal representations. The overall structural architecture of the Recurrent Neural Network is illustrated in Fig.4.2.

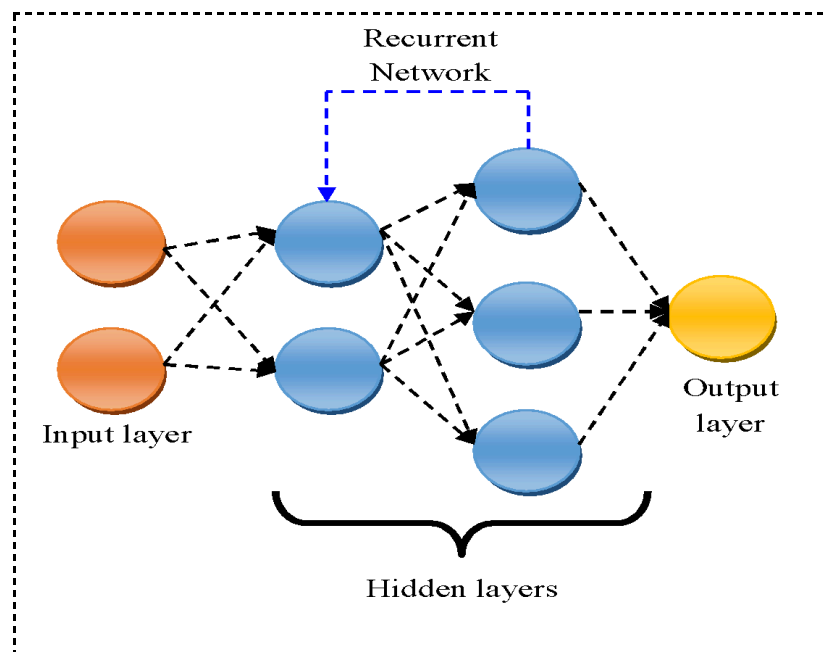


Figure 4.2 : Structural Architecture of the Recurrent Neural Network (RNN)

Residual Connection in Recurrent Neural Networks- Traditional approaches for time-series modeling and authentication are generally effective when data relationships are predominantly linear. However, IoV data streams often exhibit strong nonlinear dependencies due to dynamic traffic conditions, heterogeneous nodes, and varying communication patterns. In addition, standard Recurrent Neural Networks (RNNs) are difficult to train efficiently because of issues such as vanishing gradients and slow convergence, especially when dealing with long sequences. To overcome

these limitations, the Residual Recurrent Neural Network (RRNN) concept, inspired by the residual learning framework proposed by Yufeng Wang et al., 2019, is adopted.

The RRNN architecture decomposes the learning process into two interconnected stages. In the first stage, a linear model captures the primary trend of the sequential data. In the second stage, the residual error produced by this linear approximation is forwarded to the RNN, which focuses exclusively on learning the remaining nonlinear patterns. By integrating linear estimation with nonlinear sequence modeling, the RRNN efficiently exploits both characteristics of IoV data.

This residual learning mechanism offers two major advantages. First, combining linear and nonlinear representations improves prediction and authentication performance compared to using either approach independently. Second, since the RNN is trained only on residual errors rather than the entire signal, the training process becomes faster, more stable, and computationally less complex. As a result, RRNNs are particularly well suited for IoV authentication tasks where accuracy, scalability, and training efficiency are critical.

$$y_{f+j}^{\sim} = g(y_f) \quad (4.3)$$

$$t_{f+j}^{\sim} = y_{f+j} - y_{f+j}^{\sim} \quad (4.4)$$

$$t_{f+2j}^{\sim} = RNN[t_{f+j}^{\sim}] \quad (4.5)$$

$$y_{f+2j}^{-} = g(y_{f+j}) + t_{f+2j}^{\sim} \quad (4.6)$$

The expression in Eq. 4.6 represents the final prediction stage of the residual recurrent learning process. In this formulation, j denotes the prediction horizon, while f and $f + j$ indicate the current and future time instants, respectively. The value $y_{f+j}^{\sim}$ corresponds to the output predicted by the linear forecasting function $g(\cdot)$, whereas y_{f+j} denotes the actual observed value at time $f + j$. The difference between these two values defines the linear prediction error, represented as $t_{f+j}^{\sim}$. The final predicted output, y_{f+2j}^{-} , is

obtained by combining the linear forecast with the residual correction learned by the nonlinear model.

Specifically, Eq. (4.3) computes the preliminary forecast $y_{f+j}^{\sim}$ from the input sequence y_f using the linear predictor g . Eq. (4.4) determines the associated prediction error by comparing the linear estimate with the true observation. Eq. (4.5) feeds this residual error into the nonlinear recurrent structure to estimate the future correction term. Finally, Eq. (4.6) aggregates the linear forecast and the predicted residual to generate the refined output. This residual learning strategy, inspired by the residual modeling concept introduced by Yufeng Wang et al., 2019, improves prediction accuracy by allowing the nonlinear network to focus solely on correcting linear estimation errors rather than modeling the entire signal.

Proposed CD-RRNN for Authentication- The proposed Cascaded Dilated Residual Recurrent Neural Network (CD-RRNN) is employed to strengthen the authentication mechanism in the IoV environment. Although the Residual Recurrent Neural Network (RRNN) provides faster training and improved learning efficiency, it still faces limitations in fully identifying sophisticated malicious behaviors and is susceptible to gradient degradation when processing long sequential data. To overcome these drawbacks, dilation is integrated into the residual recurrent structure.

The dilated RRNN, inspired by the dilated recurrent architecture introduced by Chang et al., 2017, adopts a multi-layer and cell-independent design with hierarchical skip connections across different time steps. These dilated skip connections enable the network to capture long-range temporal dependencies while significantly reducing the risk of vanishing gradients. In addition, dilation allows the model to operate with fewer parameters, improving computational efficiency without sacrificing detection capability. By stacking multiple dilated recurrent layers with varying dilation rates, the CD-RRNN achieves multi-resolution temporal learning, making it highly effective for time-sensitive authentication tasks in IoV networks. As a result, the proposed CD-RRNN outperforms conventional RNN, LSTM, and GRU models in both learning

stability and malicious activity recognition. The mathematical formulation of the dilated skip connection is presented in Eq. (4.7).

$$b_j^{(d)} = a\left(b_j^{(d)}, b_{j-x_j^{(d)}}^{(d)}\right) \quad (4.7)$$

In this formulation, the input to the d -th dilated layer is denoted by $b_j^{(d)}$, while x_j represents the dilation or skip interval between recurrent connections. The symbol a refers to a recurrent cell operation and its corresponding output. By introducing dilated connections, the recurrent network efficiently captures long-range temporal dependencies and significantly alleviates the vanishing gradient problem, as demonstrated in dilated recurrent architectures by Chang et al. (2017).

Despite these advantages, a standalone dilated RRNN may suffer from redundant feature propagation and increased error rates when handling large-scale vehicular data streams. To address these limitations, a cascading mechanism is incorporated, resulting in the proposed Cascaded Dilated Residual Recurrent Neural Network (CD-RRNN). In this approach, the original input data are processed sequentially through multiple cascaded stages, allowing the network to progressively refine learned representations. Cascading effectively suppresses redundant information and enhances discrimination capability by re-evaluating the extracted features across three successive stages, thereby reducing cumulative prediction errors.

The final output of the proposed CD-RRNN yields authenticated vehicular data with enhanced reliability and robustness. By integrating hierarchical dilation, the model effectively captures long-term temporal dependencies while alleviating gradient degradation during training. The cascading mechanism further refines feature representations by minimizing redundant information, which contributes to a reduction in false-negative rates. In addition, the inclusion of residual connections accelerates convergence and improves learning stability. Together, these characteristics enable the CD-RRNN to accurately identify malicious behaviors such as impersonation attacks and false data injection in highly dynamic IoV environments.

4.3.5 Addressing IoV Security and Privacy Challenges

The proposed authentication protocol is designed to effectively address the key challenges inherent in Internet of Vehicles (IoV) environments, such as high mobility, rapidly changing network topologies, and stringent privacy requirements. The decentralized nature of blockchain eliminates dependence on static centralized servers, thereby avoiding single points of failure and improving resilience under dynamic vehicular conditions. Timestamped and hash-linked blocks ensure the integrity and consistency of vehicular data across distributed nodes, even during frequent topology changes.

To handle sequential and time-dependent vehicular data, the proposed CD-RRNN processes traffic patterns using dilated skip connections, enabling robust detection of malicious behavior despite continuous node movement. This architecture enhances adaptability to real-time network variations while maintaining detection accuracy. Furthermore, privacy and traceability are jointly preserved through conditional privacy mechanisms. Transactions are stored in an anonymized manner on the blockchain, while smart contracts allow authorized entities to trace malicious vehicles only when suspicious behavior is detected. The Optimized Key El-Gamal Encryption (OKEE), supported by MMRFO-based key generation, dynamically adapts encryption strength to network conditions, reducing latency and improving scalability without compromising confidentiality.

4.3.6 Modified Heuristic Algorithm-Based Cryptographic Technique for Privacy Preservation

MMRFO for Optimal Key Generation- The Modified Manta Ray Foraging Optimization (MMRFO) algorithm is employed to generate optimal cryptographic keys for privacy preservation in the proposed system. This optimization strategy aims to reduce computational overhead and memory consumption while enhancing encryption efficiency. The MMRFO is derived from the conventional Manta Ray Foraging Optimization (MRFO), a bio-inspired metaheuristic based on the natural foraging behaviors of manta rays, namely chain foraging, cyclone foraging, and somersault foraging, as introduced by Weiguo Zhao et al. (2020). MRFO has

demonstrated effectiveness in solving complex engineering optimization problems by balancing exploration and exploitation.

However, the traditional MRFO relies on randomly generated control parameters within a fixed range of $[0,1]$ across all foraging phases. Such random selection may introduce dimensional instability and increase optimization complexity, particularly in high-dimensional cryptographic key spaces. To overcome these limitations, the proposed MMRFO replaces the purely random parameter selection with a fitness-driven adaptive control variable. This newly introduced fitness-based random factor, denoted as b , is computed using the fitness values of candidate solutions. By guiding the search process based on solution quality, dimensionality issues are significantly reduced, convergence speed is improved, and more stable optimal keys are generated. The mathematical formulation of the adaptive parameter b is presented in Eq. (4.8).

$$b = \frac{cff}{(mff+wff)} \quad (4.8)$$

In this formulation, the average fitness value across the population is represented as mff , while the poorest fitness obtained among all candidate solutions is denoted as wff . The fitness of the currently evaluated solution is indicated by cff . Unlike the traditional approach, the proposed fitness-adaptive random variable is consistently updated across all foraging phases, enabling the optimization process to be guided by solution quality rather than purely random selection. This adaptive strategy improves stability and accelerates convergence during key generation.

The conventional Manta Ray Foraging Optimization (MRFO), originally proposed by Weiguo Zhao et al. (2020), simulates the natural feeding behavior of manta rays in ocean environments. Among its core strategies, chain foraging represents the collective movement of manta rays toward regions with higher plankton density. In this mechanism, each manta ray follows the individual with the highest food concentration discovered so far, assuming that areas with denser plankton offer better feeding opportunities. Since the true global optimum is unknown, MRFO treats the

best solution identified at any iteration as the current plankton-rich target and guides the population accordingly. The mathematical representation of the chain foraging behavior is expressed through Eq. (4.9).

$$y_j^{eu+1} = y_j^{eu} + b \cdot (y_{best}^{eu} - y_j^{eu}) + \beta \cdot (y_{j-1}^{eu} - y_j^{eu}), j = 2, \dots, M \quad (4.9)$$

For the first manta ray in the chain, the update equation is defined as Eq. (4.10):

$$y_1^{eu+1} = y_1^{eu} + b \cdot (y_{best}^{eu} - y_1^{eu}) \quad (4.10)$$

where the control parameter β where the control parameter in Eq.(4.11):

$$\beta = 2 \cdot b \cdot \ln(b) \quad (4.11)$$

Cyclone and Somersault Foraging in the Proposed MMRFO- In the optimization process, the variable y_j^{eu} represents the position of the j^{th} manta ray in the e^{th} dimension at iteration u , while y_{best}^{eu} denotes the best solution identified so far, corresponding to the region with the highest plankton concentration. In the original Manta Ray Foraging Optimization (MRFO) algorithm proposed by Zhao et al. (2020), random numbers uniformly distributed within $[0, 1]$ denotes the best solution identified so far, corresponding to the region with the highest plankton concentration. In the original Manta Ray Foraging Optimization (MRFO) algorithm proposed by Zhao et al. (2020), random numbers uniformly distributed within b , computed directly from fitness values. This modification simplifies the search process, improves stability, and accelerates convergence by aligning randomness with solution quality.

Cyclone Foraging Phase- Cyclone foraging models the spiral movement of manta rays toward a detected food source. In this phase, individuals follow a spiral-shaped trajectory, allowing effective exploitation of promising regions. Instead of modeling explicit 2D spiral coordinates, the movement is generalized to an n -dimensional space using a compact formulation:

$$y_j^{e,u+1} = y_{best}^{e,u} + b \cdot (y_{j-1}^{e,u} - y_j^{e,u}) + \beta \cdot (y_{best}^{e,u} - y_j^{e,u}), j = 1, \dots, M \quad (4.12)$$

where the adaptive control parameter β is defined as in Eq.(4.13):

$$\beta = 2 \cdot e^{b_1 \left(1 - \frac{u}{U}\right)} \cdot \sin(2\pi b_1) \quad (4.13)$$

Here, U is the maximum number of iterations, and b_1 is a fitness-adjusted random coefficient. This formulation preserves the spiral exploitation behavior while reducing computational complexity.

To further enhance global exploration, manta rays occasionally shift toward randomly generated reference positions within the search space:

$$y_r^e = h^e + b \cdot (k^e - h^e) \quad (4.14)$$

$$y_j^{e,u+1} = y_r^e + b \cdot (y_{j-1}^{e,u} - y_j^{e,u}) + \beta \cdot (y_r^e - y_j^{e,u}) \quad (4.15)$$

Where h^e and k^e represent the lower and upper bounds of the search space, respectively. This mechanism strengthens exploration while avoiding premature convergence.

Somersault Foraging Phase- Somersault foraging enables manta rays to reposition themselves around the best-known solution, acting as a pivot point. This behavior improves local refinement and avoids stagnation near suboptimal solutions. The simplified update rule is expressed as in Eq.(4.16):

$$y_j^{e,u+1} = y_j^{e,u} + T \cdot b_2 \cdot (y_{best}^{e,u} - b_3 \cdot y_j^{e,u}), j = 1, \dots, M \quad (4.16)$$

Where T controls the somersault range, and b_2, b_3 are fitness-guided coefficients. This formulation ensures controlled oscillation around the optimal region, improving solution precision. By replacing uniform random factors with fitness-aware adaptive parameters, the proposed MMRFO significantly enhances convergence speed, stability, and scalability compared to the original MRFO framework introduced by Weiguo Zhao et al. (2020). The detailed pseudo-code of the proposed MMRFO is presented in Algorithm 1, while the procedural flow is summarized through a flowchart.

Algorithm 1: Recommended MMRFO

```

Population size, highest iteration, and all manta ray initialization
Estimate the fitness function of all individuals
For  $u = 1$  to  $U$ 
  For  $j = 1$  to  $M$ 
    Generate a new random variable  $b$  by Eq. (8)
    if  $rd < 0.5$ 
      if  $\frac{u}{U} < rd$ 
        Position updating of cyclone foraging strategy using
        Eq. (14) and Eq. (15)
      else
        Position updating of cyclone foraging strategy using
        Eq. (12) and Eq. (13)
      End if
    else
      Position updating of chain foraging process using Eq.
      (9).
    End if
    Position updating of somersault process employing Eq. (16).
  End for
End for
Return optimal outcomes
  
```

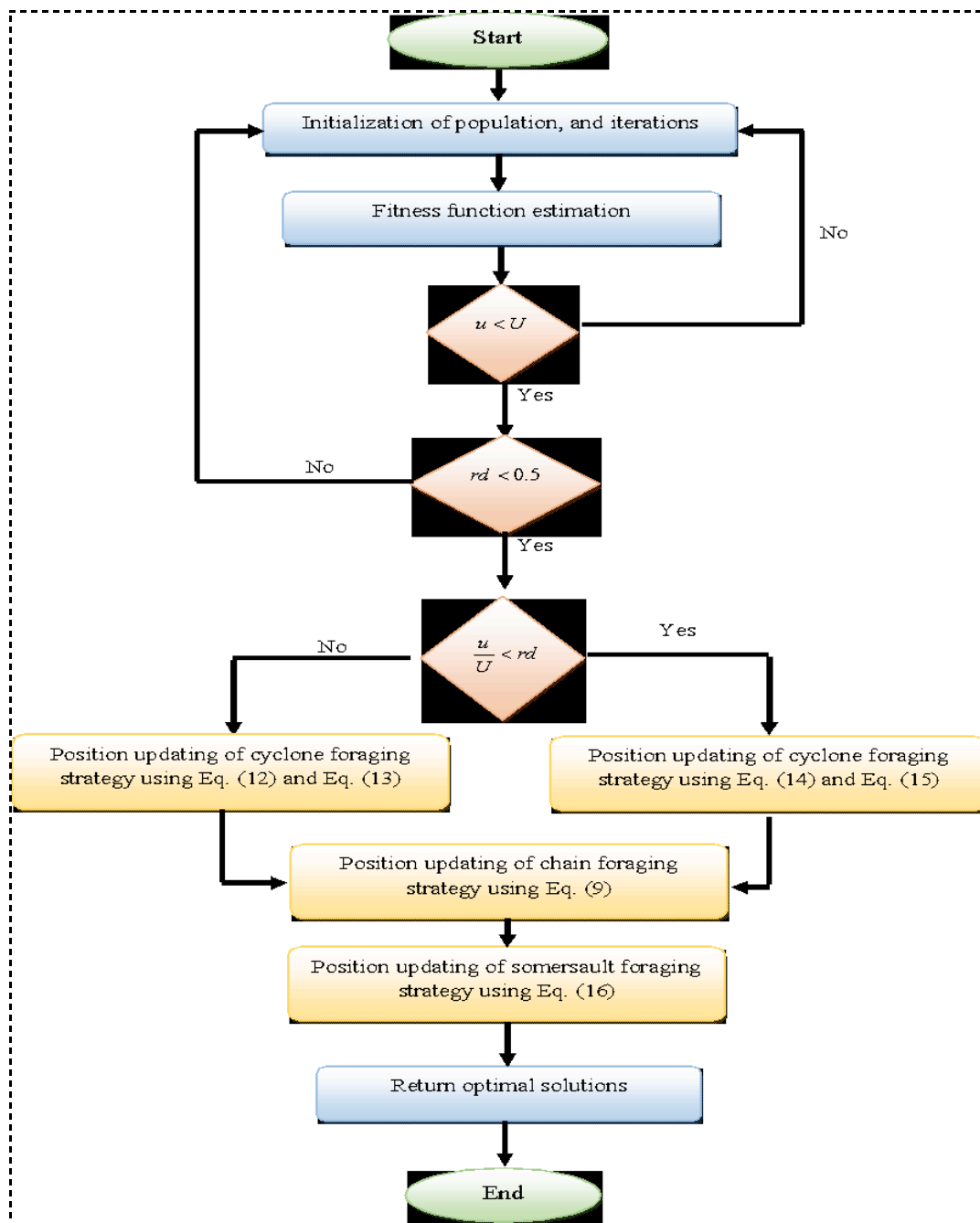


Figure 4.3: Flowchart of designed MMRFO for producing the optimal key

ElGamal Encryption Scheme- ElGamal encryption, originally introduced by Taher ElGamal (1985), is a well-known asymmetric cryptographic technique derived from the Diffie–Hellman key exchange principle. It relies on the computational hardness of the discrete logarithm problem and supports both secure encryption and digital signature generation. Unlike deterministic public-key cryptosystems, ElGamal

produces a different ciphertext for the same plaintext each time, which significantly strengthens resistance against replay and pattern-analysis attacks.

The ElGamal cryptosystem operates through three fundamental stages: key generation, encryption, and decryption.

Key Generation

In the key generation phase, the communicating party performs the following steps:

1. Select a large prime number q , which defines the finite cyclic group.
2. Choose a generator h of the multiplicative group modulo q .
3. Randomly select a private key y such that $1 < y < q - 1$.
4. Compute the public key component as:

$$z = h^y \text{ mod } q \quad (4.17)$$

Here, the tuple (q, h, z) forms the public key, while y is maintained as the private key and never disclosed.

Encryption- To encrypt a secret message n , the sender first obtains the receiver's public key (q, h, z) . A fresh random integer l is then generated for each encryption instance, ensuring semantic security. The ciphertext is produced as a pair:

$$d_1 = h^l \text{ mod } q \quad (4.18)$$

$$d_2 = n \cdot z^l \text{ mod } q \quad (4.19)$$

The resulting ciphertext (d_1, d_2) is transmitted to the receiver. The inclusion of the random value l guarantees that identical plaintext messages yield different ciphertexts, enhancing confidentiality in dynamic IoV communication environments.

This encryption mechanism is particularly suitable for privacy-preserving IoV applications due to its robustness against passive and active cryptographic attacks, especially when combined with optimized key selection strategies as proposed in this work.

Deployed OKEE for Data Privacy Preservation- To ensure strong privacy protection for vehicular data, this work introduces an Optimized Key ElGamal Encryption (OKEE) mechanism. In conventional ElGamal encryption, cryptographic keys are generated using random parameters, which ensures security but may lead to increased computational overhead and inefficient memory usage. To overcome these limitations, the proposed OKEE framework integrates an intelligent optimization strategy to generate cryptographic keys that are both secure and resource-efficient.

ElGamal encryption, originally proposed by ElGamal (1985), is widely adopted in secure communication systems due to its resistance to discrete logarithm attacks. However, its performance and security are highly dependent on the quality of the generated keys. In this research, the Modified Manta Ray Foraging Optimization (MMRFO) algorithm is employed to optimally generate encryption keys. By leveraging fitness-driven search behavior, MMRFO selects keys that minimize system complexity while enhancing resistance against emerging cryptographic attacks.

The optimized key generation process aims to reduce memory consumption and computational time, making the encryption process suitable for real-time IoV environments. The generated optimal key also strengthens protection against sophisticated adversaries by increasing the unpredictability of cryptographic parameters. Additionally, optimized key selection simplifies key management and improves scalability across large vehicular networks.

The objective of the optimal key generation process is mathematically defined as in Eq.(4.20):

$$ob = \arg \min_{\{ok^s\}}[ms + tm] \quad (4.20)$$

Where:

- $\{ok^s\}$ represents the optimized cryptographic key in binary form,
- ms denotes the memory size required for encryption and decryption,
- tm refers to the total execution time of the cryptographic process.

Using the optimized key generated by MMRFO, the vehicular data I_v stored on the blockchain is encrypted using the ElGamal algorithm. The same optimized key is utilized during decryption, ensuring secure access while preserving data confidentiality. This integrated OKEE framework significantly enhances privacy preservation in IoV networks while maintaining low latency and reduced resource usage.

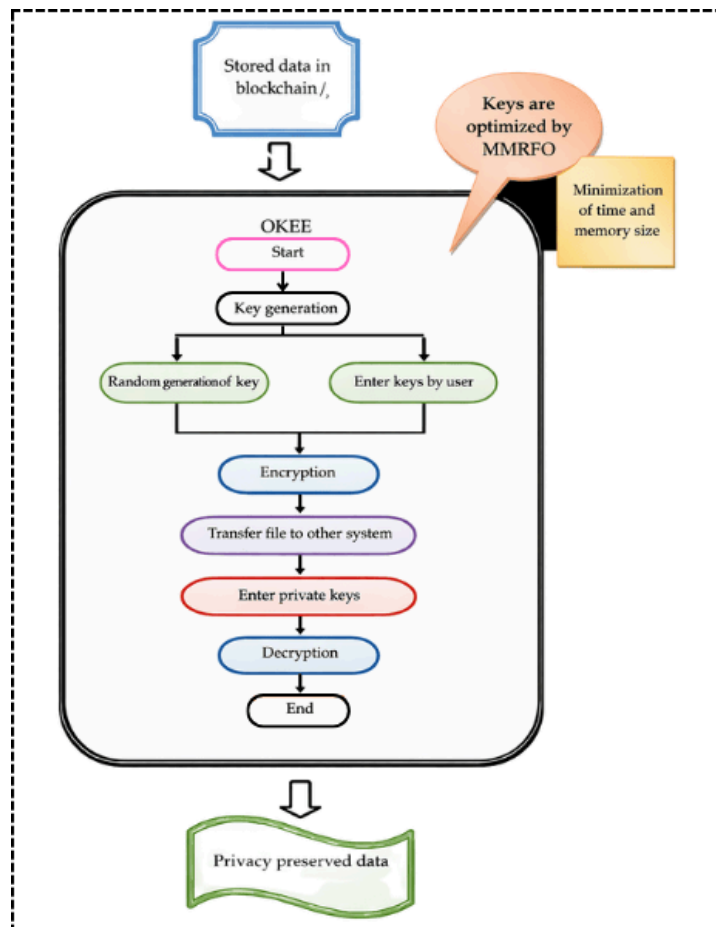


Figure 4.4: Workflow of Optimized Key El-Gamal Encryption for Privacy Preservation in Blockchain-Based IoV Using MMRFO

Exemplary Illustration of the Proposed Scheme- The Internet of Vehicles (IoV) significantly improves transportation efficiency and road safety by enabling seamless communication among vehicles, infrastructure, and surrounding entities. Through Vehicle-to-Everything (V2X) interactions, IoV supports advanced intelligent transportation applications, where authentication plays a critical role in verifying the legitimacy and reliability of exchanged vehicular data. Authentication serves as the

first line of defense against malicious activities, ensuring that only authorized vehicles and infrastructure nodes participate in network communications.

Despite its importance, existing authentication solutions predominantly depend on centralized architectures, which introduce serious security and privacy vulnerabilities. Such approaches often suffer from issues including vehicle trajectory leakage, excessive reliance on roadside units (RSUs) and cloud servers, high computational complexity, and single-point-of-failure risks. These limitations significantly weaken the resilience of IoV systems in highly dynamic and large-scale vehicular environments.

Blockchain technology offers a promising alternative by providing a decentralized and tamper-resistant framework for authentication. By integrating smart contracts, cryptographic primitives, and consensus mechanisms, blockchain establishes a trusted environment without requiring centralized control. However, the decentralized nature of blockchain also introduces challenges related to increased storage requirements, communication overhead, and computational cost. Moreover, the inherent transparency of blockchain-based IoV systems can expose sensitive vehicular information if privacy mechanisms are not carefully designed.

To overcome these challenges, this work proposes an efficient blockchain-enabled authentication scheme that balances security, privacy, and system performance. The proposed approach minimizes the shortcomings of traditional authentication mechanisms while preserving data confidentiality and trust among participating entities. The overall workflow of the proposed scheme is illustrated in Fig. 4.5, demonstrating how secure authentication and privacy preservation are achieved within the IoV environment.

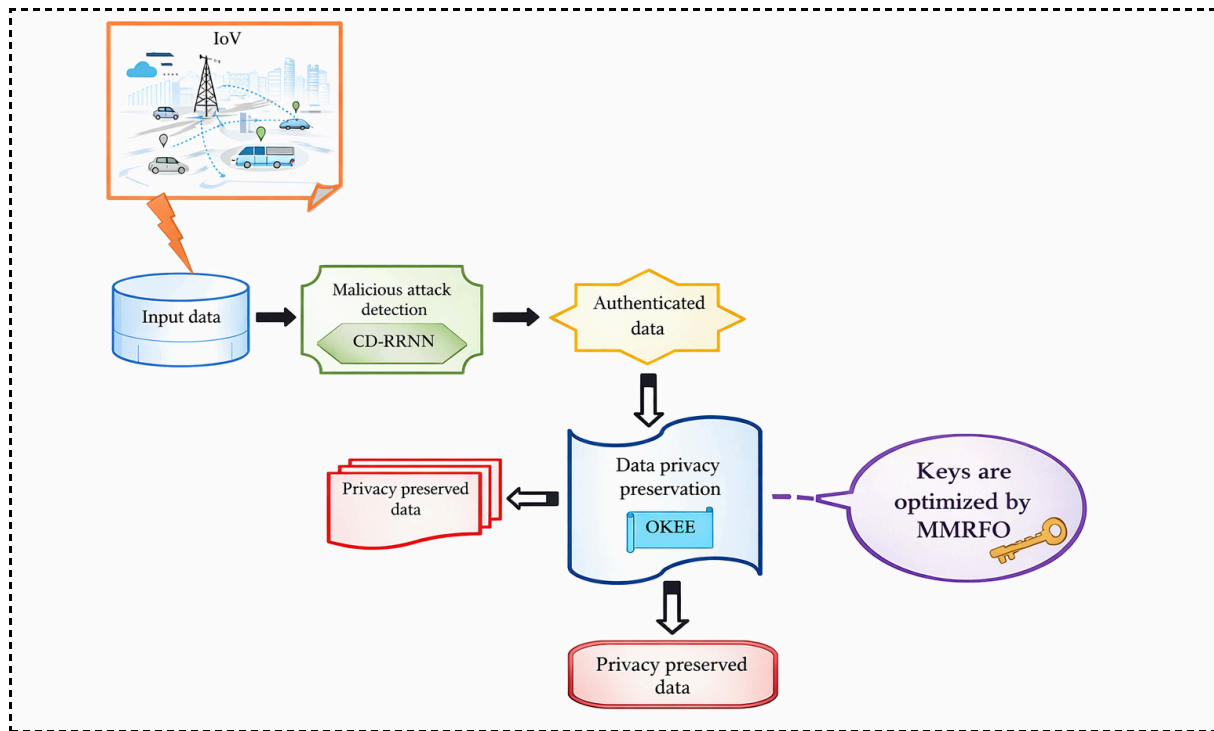


Figure 4.5: Architecture of the Proposed Blockchain-Enabled Authentication and Privacy Preservation Scheme for IoV

In this work, a secure and efficient blockchain-enabled authentication and privacy preservation framework is developed for Internet of Vehicles (IoV) environments. Initially, relevant vehicular data are extracted from a benchmark dataset and provided to the proposed Cascaded and Dilated Residual Recurrent Neural Network (CD-RRNN) to identify malicious activities. By integrating residual learning with dilation and cascading mechanisms, the CD-RRNN effectively captures temporal dependencies and improves threat detection accuracy. The verified and authenticated data are then stored across distributed blockchain blocks to ensure integrity and tamper resistance. Subsequently, privacy protection is achieved through an Optimized Key El-Gamal Encryption (OKEE) scheme, where cryptographic keys are optimally generated using the Modified Manta Ray Foraging Optimization (MMRFO) algorithm. This optimization strategy significantly reduces computational overhead and memory consumption while strengthening security. Finally, the performance of the proposed framework is extensively evaluated and compared with existing conventional methods to demonstrate its superior effectiveness in secure IoV communications.

4.4 Results and Discussion

4.4.1 Experimental Setup

The proposed blockchain-based authentication and privacy-preserving framework for Internet of Vehicles (IoV) was implemented using the Python programming environment to evaluate its effectiveness under realistic conditions. The experimental configuration for the Modified Manta Ray Foraging Optimization (MMRFO) algorithm consisted of a population size of 10, a chromosome length of 16, and a maximum of 50 iterations to ensure stable convergence and reliable optimization outcomes.

To assess malicious attack detection performance, several well-known learning models were employed for comparison, including the Deep Neural Network proposed by Kang and Kang (2016), Support Vector Machine introduced by Hoang et al. (2021), Long Short-Term Memory network presented by Ashraf et al. (2020), and the classical Recurrent Neural Network described by Yufeng Wang et al. (2019).

For optimization validation, traditional metaheuristic algorithms such as the Tunicate Swarm Algorithm developed by Kaur et al. (2020), the Golden Eagle Optimizer proposed by Mohammadi-Balani et al. (2021), Harris Hawks Optimization introduced by Heidari et al. (2019), and the original Manta Ray Foraging Optimization by Weiguo Zhao et al. (2020) were considered.

In addition, the cryptographic performance of the proposed Optimized Key El-Gamal Encryption (OKEE) was evaluated against standard encryption techniques, including Data Encryption Standard described by Guo et al. (2020), Advanced Encryption Standard reviewed by Hidayat and Mahardiko (2024), Elliptic Curve Cryptography discussed by El-Zawawy et al. (2022), and the classical El-Gamal encryption implemented by Imran et al. (2020).

The experimental implementation was carried out using Python 3.9 on a system equipped with an Intel Core i7-9700K processor operating at 3.6 GHz, 32 GB RAM, and an NVIDIA RTX 2080 Ti GPU. Blockchain operations were simulated using Hyperledger Fabric version 2.5, while the CD-RRNN model was developed using

TensorFlow 2.8 and Keras. Cryptographic functions were implemented using the PyCryptodome library. All experiments were conducted on Ubuntu 20.04 LTS to ensure system stability and optimized computational performance.

4.4.2 Performance Measures

To comprehensively evaluate the effectiveness of the proposed authentication and privacy-preserving framework, multiple performance metrics were considered. These measures were selected to analyze classification accuracy, security strength, computational efficiency, and encryption effectiveness of the proposed system.

Accuracy: It is derived in Eq. (4.21).

$$Ar = \frac{zz+xx}{zz+xx+ee+rr} \quad (4.21)$$

Sensitivity: It is formulated in Eq. (4.22).

$$Sen = \frac{zz}{zz+xx} \quad (4.22)$$

Specificity: It is expressed in Eq. (4.23).

$$Sp = \frac{rr}{rr+ee} \quad (4.23)$$

Precision: It is shown in Eq. (4.24).

$$Pr = \frac{zz}{zz+ee} \quad (4.24)$$

FPR: It is given in Eq. (4.25).

$$FPR = \frac{ee}{ee+zz} \quad (4.25)$$

FNR: It is estimated in Eq. (4.26).

$$FNR = \frac{rr}{zz+rr} \quad (4.26)$$

False Omission Rate (FOR): It is measured in Eq. (4.27).

$$FOR = 1 - \frac{xx}{xx+rr} \quad (4.27)$$

Prevalence Threshold (PT): It is calculated in Eq. (4.28).

$$PT = \frac{\sqrt{zz \times ee - ee}}{zz - ee} \quad (4.28)$$

FM: It is formulated in Eq. (4.29).

$$FM = \sqrt{Pr \times zz} \quad (4.29)$$

Prevalence Threshold (PT): It is shown in Eq. (4.30).

$$PT = \frac{\sqrt{vv \times qq - qq}}{vv - qq} \quad (4.30)$$

Threat Score (TS): It is validated in Eq. (4.31).

$$TS = \frac{zz}{(zz+ee)} \quad (4.31)$$

The parameters xx and rr represent the values corresponding to true negatives and false negatives, respectively. Similarly, the parameters zz and ee denote the true positive and false positive outcomes used in performance evaluation.

4.4.3 ROC Analysis of the Proposed Attack Detection Method

Figure 4.6 presents the Receiver Operating Characteristic (ROC) analysis of the proposed malicious attack detection framework in comparison with existing detection techniques. This evaluation highlights the capability of the proposed Cascaded and Dilated Residual Recurrent Neural Network (CD-RRNN) to accurately distinguish malicious and legitimate vehicular activities. The ROC curve demonstrates that the proposed model achieves a higher true positive rate while maintaining a lower false positive rate, indicating improved detection reliability and reduced misclassification errors.

When the false positive rate is fixed at 0.5, the CD-RRNN-based detection approach shows noticeable performance improvements compared to conventional methods. Specifically, it outperforms Deep Neural Networks (Goodfellow et al., 2016) by 6.25%, Support Vector Machines (Cortes and Vapnik, 1995) by 5%, Long Short-Term Memory networks (Hochreiter and Schmidhuber, 1997) by 3.75%, and standard Recurrent Neural Networks (Elman, 1990) by 2.5%. These results confirm that the proposed CD-RRNN model offers superior attack detection capability and robustness in dynamic IoV environments compared to traditional learning-based mechanisms.

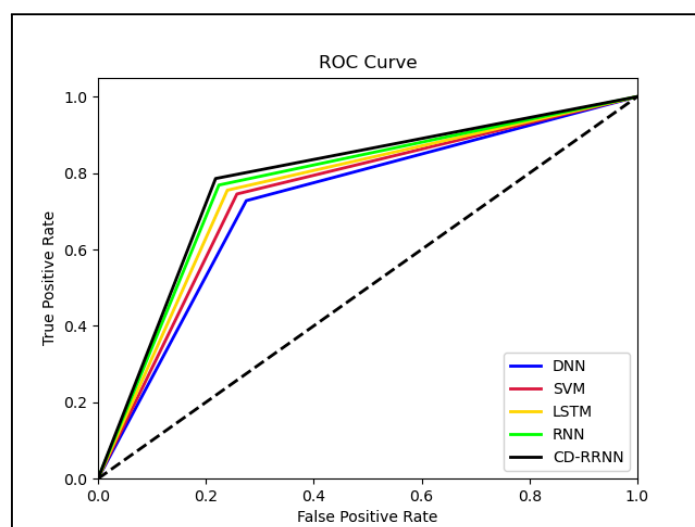
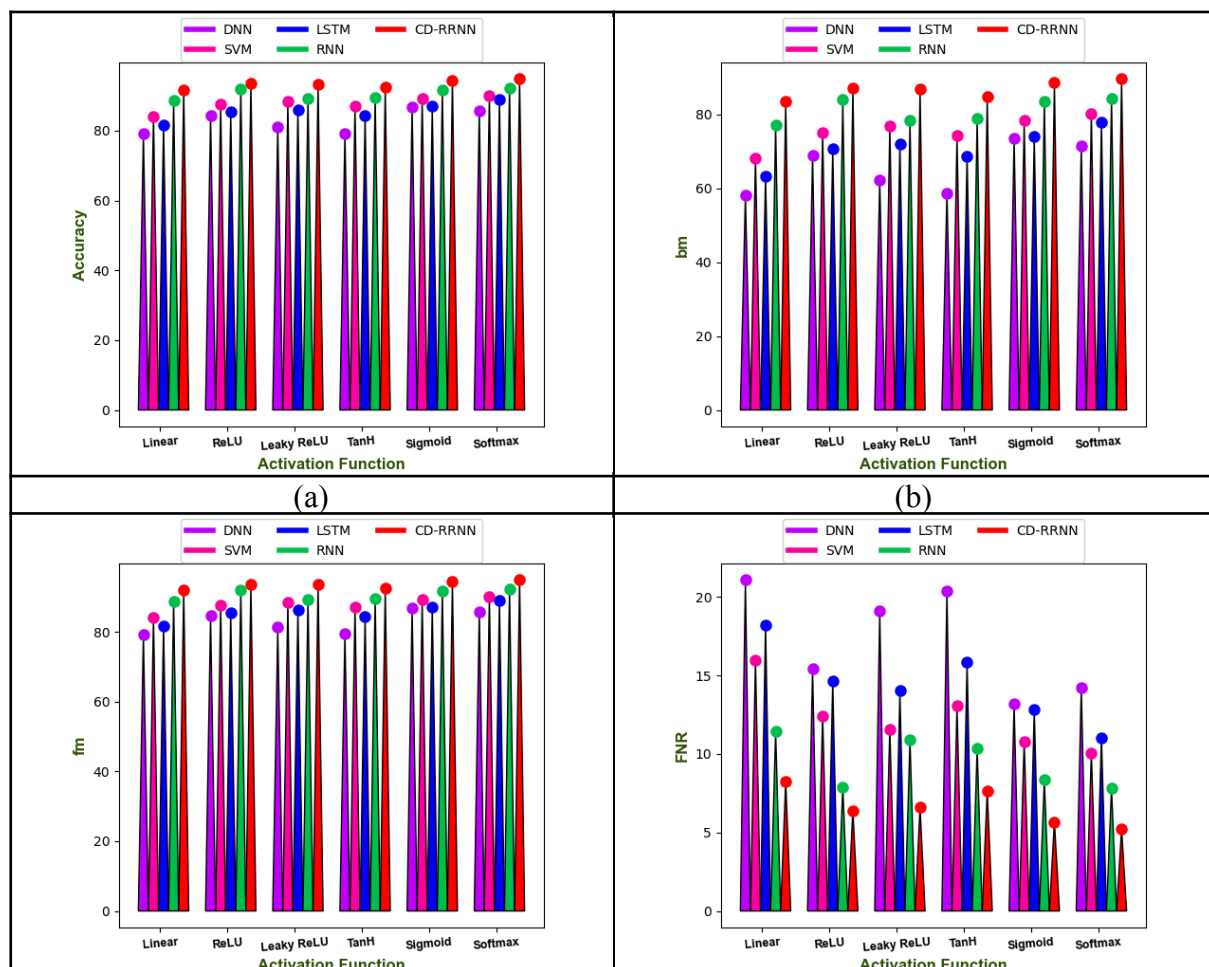
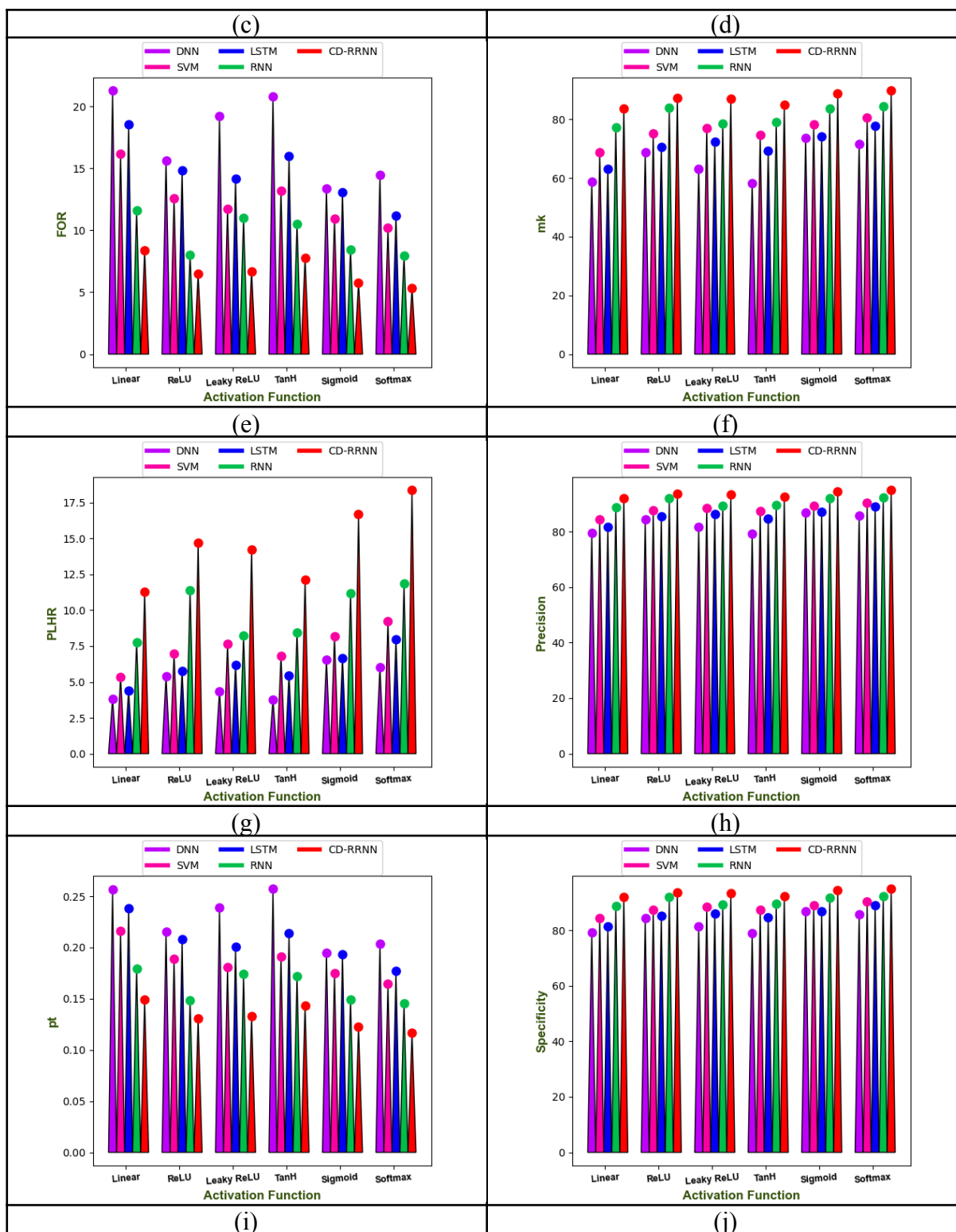


Figure 4.6: ROC Performance Analysis of the Proposed CD-RRNN-Based Malicious Attack Detection Model

4.4.4 Performance Evaluation of the Proposed Attack Detection Process

Figure 4.7 presents the ROC-based performance comparison of the proposed CD-RRNN malicious attack detection framework against conventional detection models. This evaluation demonstrates the effectiveness of the CD-RRNN in accurately distinguishing malicious activities from legitimate vehicular data. The ROC analysis highlights the model's ability to reduce detection errors while maintaining a high true positive rate. When the false positive rate is fixed at 0.5, the proposed CD-RRNN achieves performance improvements of 6.25% over Deep Neural Networks (DNN), 5% over Support Vector Machines (SVM), 3.75% over Long Short-Term Memory (LSTM), and 2.5% over standard Recurrent Neural Networks (RNN). These gains are primarily attributed to the dilated skip connections and residual learning structure of CD-RRNN, which effectively mitigate gradient degradation and enhance long-term dependency learning. Overall, the results confirm that the proposed CD-RRNN framework consistently outperforms traditional attack detection mechanisms in dynamic IoV environments.





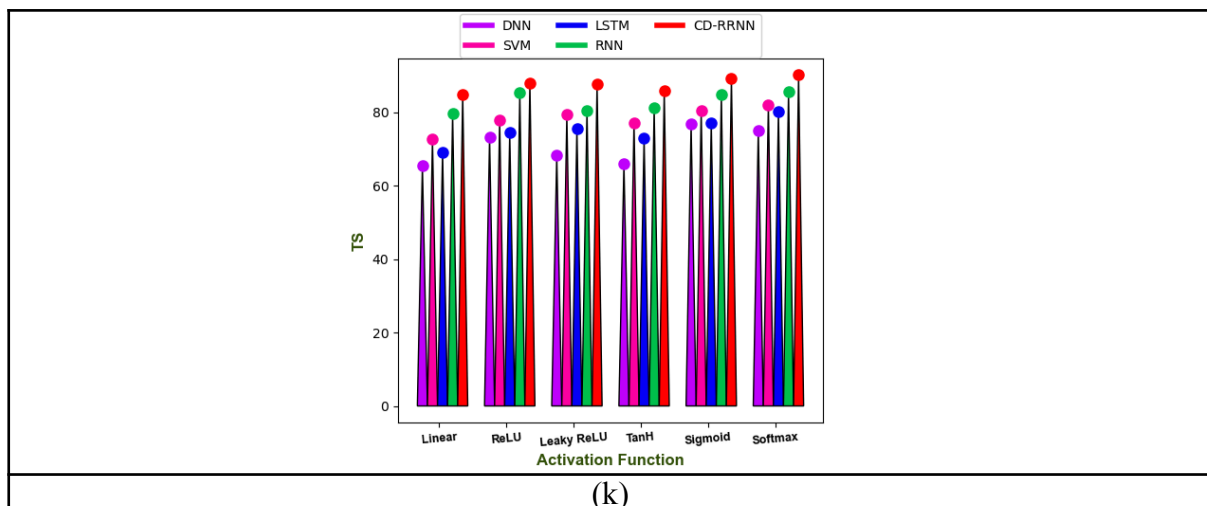


Figure 4.7: Performance evaluation of the proposed CD-RRNN-based malicious attack detection framework over conventional attack detection techniques in terms of (a) Accuracy, (b) BM, (c) FM, (d) FNR, (e) FOR, (f) MK, (g) PLHR, (h) Precision, (i) PT, (j) Specificity, and (k) TS

4.4.5 Encryption and Decryption Time Analysis of the Proposed Authentication and Privacy Preservation Framework for IoV

The effectiveness of the proposed authentication and privacy preservation framework is evaluated by analyzing its encryption and decryption time performance in comparison with multiple classical classifiers, as illustrated in Fig. 4.8. A set of standard performance metrics is considered to comprehensively assess computational efficiency and detection capability. The results demonstrate that the proposed CD-RRNN-based attack detection model consistently outperforms conventional techniques across all evaluated measures. Furthermore, different activation functions are applied to examine their impact on model performance and computational behavior. Among them, the softmax activation function yields the best results, indicating its suitability for multi-class attack detection in IoV environments. As shown in Fig. 4.8(a), when using the softmax function, the proposed CD-RRNN achieves significantly higher accuracy than traditional methods, surpassing Deep Neural Networks by 13.26%, Support Vector Machines by 11.22%, Long Short-Term Memory networks by 8.16%, and standard Recurrent Neural Networks by 3.06%. The peak accuracy of 94.34% highlights the model's capability to effectively eliminate

redundant information through cascading operations, leading to faster convergence and reduced encryption–decryption latency compared to existing approaches.

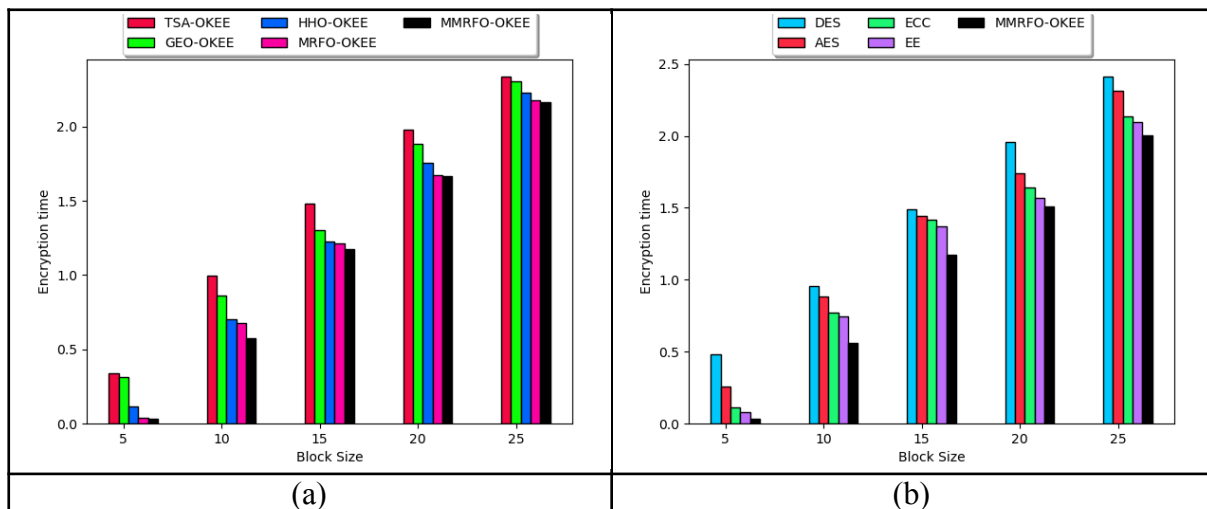


Figure 4.8: Encryption time analysis of implemented blockchain-based authentication and privacy preservation model for IoV systems over (a) Conventional algorithms, and (b) Conventional cryptography techniques

4.4.6 Encryption Time Performance Analysis of the Proposed MMRFO-OKEE Scheme

The encryption efficiency of the proposed MMRFO-OKEE scheme is evaluated by comparing it with both heuristic optimization–based cryptographic approaches and conventional encryption algorithms. As illustrated in Fig. 4.9(a), the proposed MMRFO-OKEE consistently requires significantly less encryption time than TSA-OKEE, GEO-OKEE, HHO-OKEE, and MRFO-OKEE across varying block sizes. This improvement is attributed to the fitness-driven key optimization strategy of MMRFO, which accelerates key generation and reduces computational complexity. Furthermore, Fig. 4.9(b) presents a comparison with traditional cryptographic schemes, including DES, AES, ECC, and standard El-Gamal encryption. The results indicate that MMRFO-OKEE maintains superior encryption speed, particularly as the block size increases, demonstrating its suitability for large-scale and real-time IoV data protection scenarios.

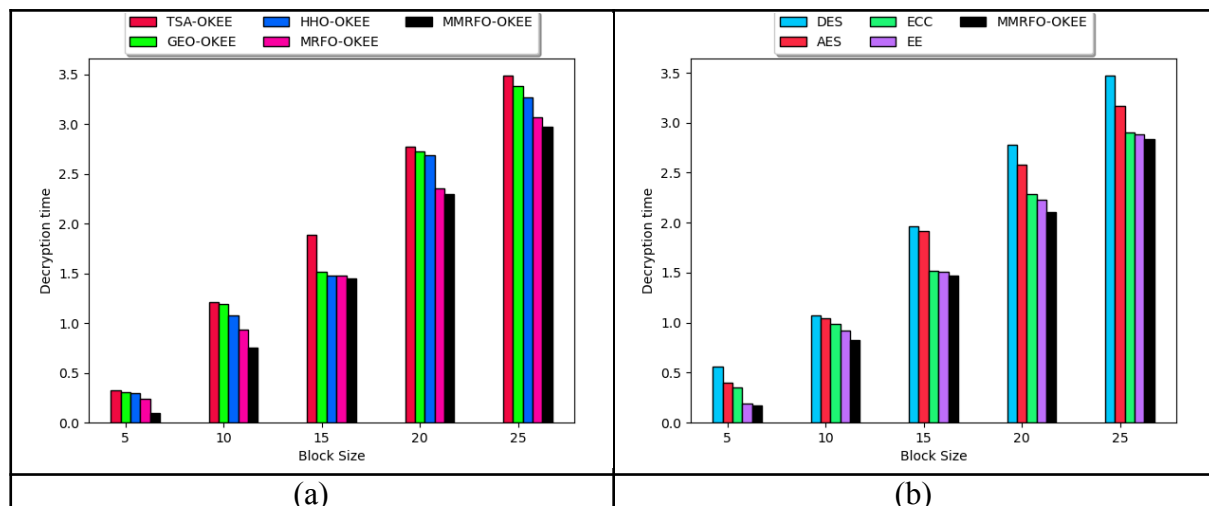


Figure 4.9: Decryption time analysis of implemented blockchain-based authentication and privacy preservation model for IoV systems over (a) Conventional algorithms, and (b) Conventional cryptography techniques

4.4.7 Key Sensitivity Analysis of the Proposed Authentication and Privacy Preservation Framework

Key sensitivity analysis is conducted to assess the robustness of the proposed authentication and privacy preservation framework against small variations in encryption keys. This analysis evaluates how minor changes in the cryptographic key influence the encryption and decryption outputs, thereby reflecting the security strength of the scheme. The proposed MMRFO-OKEE method is compared with traditional cryptographic algorithms to validate its resilience. As shown in Fig. 4.10(b), in the fifth evaluation case, the key sensitivity of the proposed approach is reduced by 14.2% compared to DES, 17.7% compared to AES, 14.2% compared to ECC, and 11.4% compared to conventional El-Gamal Encryption. These results indicate that the proposed framework exhibits low key sensitivity, enhancing resistance against brute-force and key-guessing attacks, and thereby improving overall security robustness for IoV-based applications.

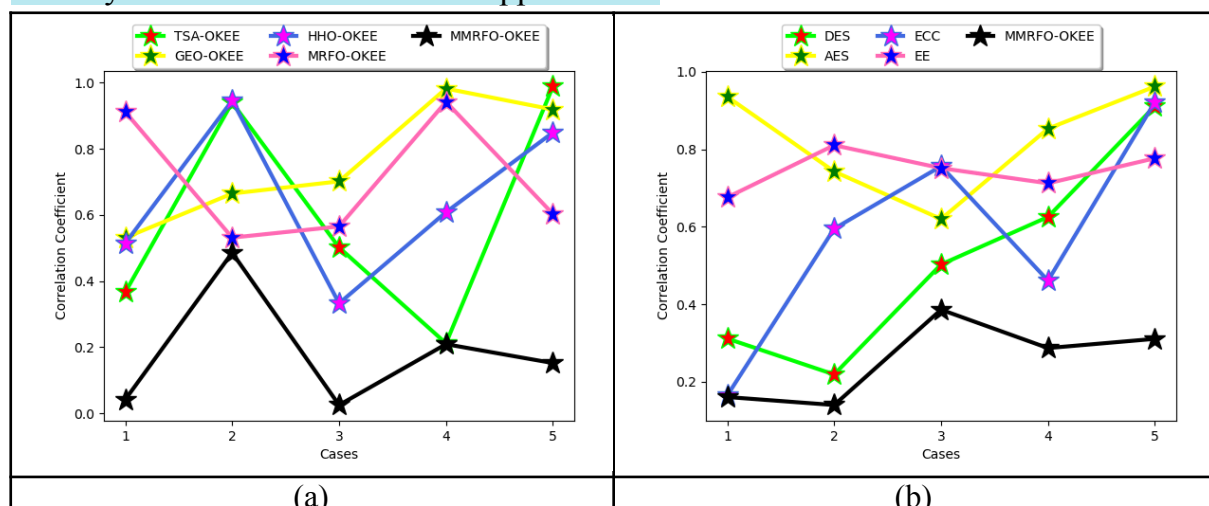


Figure 4.10: Key sensitivity analysis of implemented blockchain-based authentication and privacy preservation model for IoV systems over “ (a) Conventional algorithms, and (b) Conventional cryptography techniques

4.4.8 Memory Consumption Analysis of the Proposed Authentication and Privacy Preservation Framework for IoV

The memory efficiency of the proposed authentication and privacy preservation framework is evaluated by comparing its memory consumption with that of several existing optimization-based and cryptographic approaches. Memory usage refers to the total amount of system memory required to execute the complete authentication, encryption, and decryption processes. Lower memory consumption is particularly important in Internet of Vehicles (IoV) environments, where on-board units and roadside infrastructure often operate under resource constraints. As illustrated in Fig. 4.11(a), when the block size is set to 25, the proposed MMRFO-OKEE scheme demonstrates significantly reduced memory usage compared to conventional methods. Specifically, it consumes 19.04% less memory than TSA-OKEE, 9.52% less than GEO-OKEE, 14.28% less than HHO-OKEE, and 20% less than MRFO-OKEE. These results confirm that the proposed framework achieves superior memory efficiency, making it well suited for large-scale and real-time IoV authentication and privacy preservation applications.

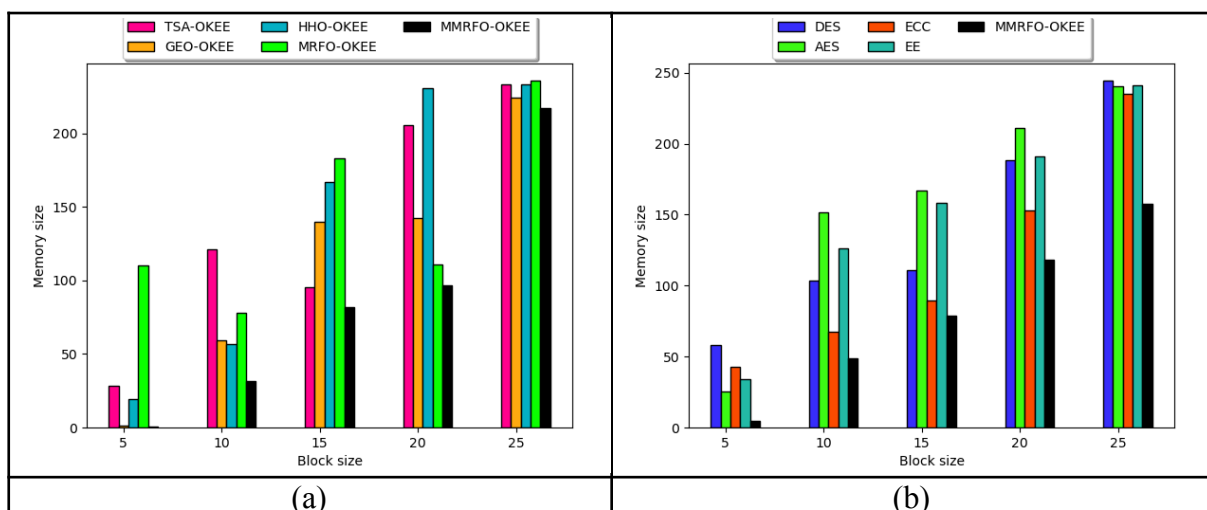


Figure 4.11: Memory size analysis of implemented blockchain-based authentication and privacy preservation model for IoV systems over (a) Conventional algorithms, and (b) Conventional cryptography techniques

4.4.9 Computing Time Analysis of the Proposed Authentication and Privacy Preservation Framework for IoV

Figure 12(a) presents the evaluation of computational time for the proposed MMRFO-OKEE-based authentication and privacy preservation framework in comparison with conventional cryptographic schemes. Computational time refers to the total duration required to complete authentication, encryption, and decryption operations within the IoV environment. Lower computation time is essential for ensuring real-time responsiveness and reliable communication in highly dynamic vehicular networks.

As illustrated in Fig. 12(b), when the block size is fixed at 20, the proposed MMRFO-OKEE approach demonstrates a substantial reduction in processing time compared to traditional cryptographic methods. Specifically, the computation time is reduced by 54.83% when compared with DES, 45.16% relative to AES, 22.58% against ECC, and 12.9% compared to conventional El-Gamal encryption (Imran et al., 2020). This improvement is primarily attributed to the fitness-guided key optimization of MMRFO and the lightweight encryption operations of OKEE, which collectively minimize redundant computations.

These results confirm that the proposed framework is well suited for real-time IoV applications, where rapid authentication and secure data transmission are critical under strict latency constraints.

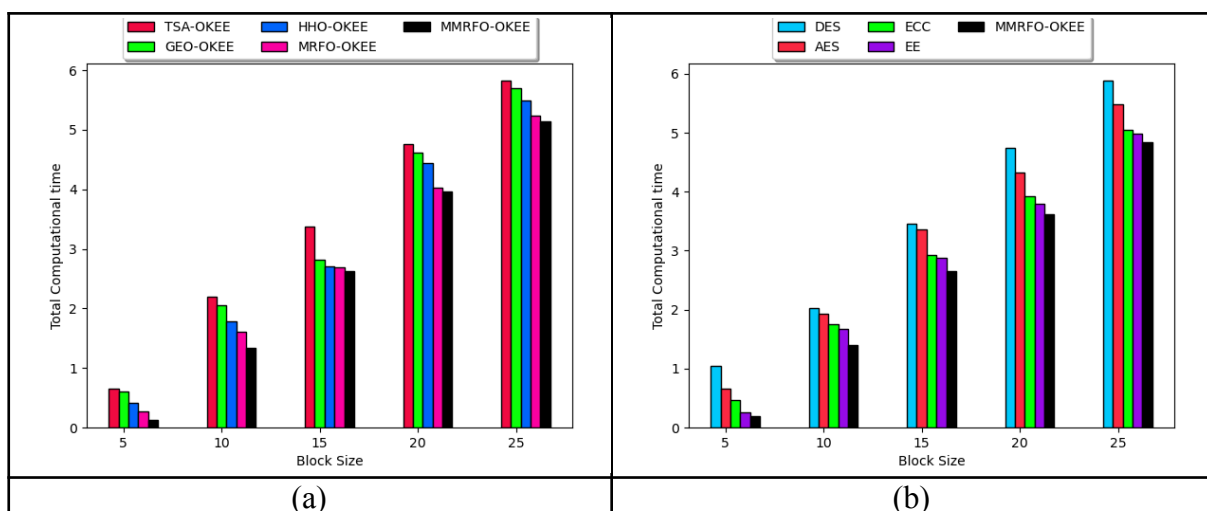


Figure 12: Total computation time of implemented blockchain-based authentication and privacy preservation model for IoV systems over (a) Conventional algorithms, and (b) Conventional cryptography techniques

4.4.10: Overall Performance Evaluation of the Proposed Attack Detection Mechanism

The comprehensive performance of the proposed CD-RRNN-based malicious attack detection model is assessed against multiple traditional classifiers, and the comparative results are summarized in Table 2. This evaluation demonstrates that the proposed model consistently outperforms conventional learning-based detection approaches in identifying sophisticated and stealthy attacks within IoV environments. By accurately distinguishing malicious behavior from legitimate vehicular communications, the proposed framework significantly strengthens authentication reliability and safeguards sensitive vehicular data from unauthorized access. When the sigmoid activation function is employed, the CD-RRNN model exhibits notable improvements in detection sensitivity compared to existing techniques. Specifically, sensitivity is increased by 7.95% over Deep Neural Networks (DNN), 5.42% compared to Support Vector Machines (SVM), 7.62% relative to Long Short-Term Memory (LSTM) networks, and 2.82% over standard Recurrent Neural Networks (RNN), as reported in recent studies on vehicular intrusion detection (Kang and Kang, 2016; Ashraf et al., 2020). These gains can be attributed to the cascaded architecture and dilated residual connections, which enable the model to capture both long-term dependencies and subtle temporal attack patterns more effectively.

Furthermore, Table 4.1 also provides a comparative overview of representative blockchain-based authentication mechanisms reported in the literature, outlining their core functionalities and inherent limitations. In contrast, the proposed CD-RRNN-integrated framework offers a balanced solution that combines high attack detection accuracy, reduced false negatives, and improved robustness, making it well suited for secure and privacy-aware IoV deployments.

Table 4.1: Comparative analysis of proposed system with traditional models

Terms	DNN	SVM	LSTM	RNN	Proposed model(CD-RRNN)
Accuracy	86.8	89.16	87.04	91.74	94.34
Sensitivity	86.82016673	89.20206431	87.13775308	91.66335848	94.3231441
Specificity	86.77952439	89.11729141	86.97497	91.8178154	94.35711407
Precision	86.9582505	89.27294398	87.13775308	91.91878981	94.43561208
FPR	13.22047561	10.88270859	13.0592503	8.182184603	5.642885933
FNR	13.17983327	10.79793569	12.86224692	8.336641524	5.676855895
FOR	13.36016097	10.95449054	13.0592503	8.440514469	5.7568438
PT	0.195111666	0.174642947	0.193564606	0.14938492	0.12229579
FM	86.88918118	89.23749711	87.13775308	91.79098529	94.37936134

PLHR	6.567098587	8.196678576	6.672492759	11.20279765	16.71540861
TS	76.81770285	80.56651129	77.20717552	84.82733284	89.35690109

4.5 Summary

This chapter presented a systematic blockchain-based authentication and privacy-preserving protocol for Internet of Vehicles networks, integrating CD-RRNN for intelligent attack detection with OKEE supported by MMRFO for optimized cryptographic key generation. The proposed framework effectively addresses critical IoV challenges such as malicious intrusion, data leakage, and high computational overhead in dynamic vehicular environments. By combining deep learning, optimized heuristic algorithms, and El-Gamal-based encryption within a blockchain infrastructure, the system ensures secure, reliable, and scalable authentication. Extensive experimental analysis validated the superiority of the proposed approach in terms of detection accuracy, encryption efficiency, memory usage, and computational time. The results confirm that the proposed model significantly enhances both security and privacy preservation in IoV systems. These findings provide a strong foundation for advancing toward more lightweight, adaptive, and next-generation secure vehicular communication frameworks.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Introduction

This chapter presents a comprehensive experimental evaluation of the proposed blockchain-enabled privacy-preserving authentication frameworks developed in this research. The experimental setup, including dataset configuration, simulation environment, parameter settings, and implementation details, is first described to ensure reproducibility and fairness in evaluation. Subsequently, the performance of the proposed models is analyzed using standard metrics such as accuracy, sensitivity, specificity, F1-score, ROC characteristics, computational time, memory usage, encryption/decryption cost, and security robustness. A detailed comparative analysis is then conducted against recent state-of-the-art IoV authentication and privacy-preserving schemes to demonstrate the effectiveness, scalability, and efficiency of the proposed approaches under dynamic vehicular network conditions. The results clearly highlight the superiority of the proposed frameworks in achieving secure, efficient, and privacy-aware authentication for Internet of Vehicles environments.

5.2 Analysis Of Existing Models

5.2.1 Comparative Performance Analysis of the Authentication Framework

Tables 5.1 and 5.2 present a consolidated comparative evaluation of the proposed blockchain-based privacy-preserving authentication framework against existing optimization-driven models and conventional classification techniques. The comparison is conducted using standard performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, and Matthews Correlation Coefficient (MCC), to comprehensively assess detection reliability and robustness. From the results, it is evident that the proposed OBWO-HANet consistently outperforms all benchmark optimization algorithms such as GWO-HANet, EOO-HANet, LOA-HANet, and BWO-HANet, achieving the highest accuracy of 93.8% and superior sensitivity of 93.76%, which indicates improved attack detection capability.

Similarly, when compared with traditional classifiers including CNN, MLP, Ridge classifier, and Ensemble models, the proposed framework demonstrates notable gains across all evaluation metrics, while maintaining lower false positive and false negative rates. These improvements confirm that the integration of blockchain with optimized hybrid learning significantly enhances authentication reliability and privacy preservation in IoV environments, validating the effectiveness of the proposed approach.

Table 5.1: Comparative analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV with conventional optimization algorithms

Terms	GWO-HANet	EOO-HANet	LOA-HANet	BWO-HANet	OBWO-HANet
Accuracy	88.52	86.52	91.96	88.96	93.8
Sensitivity	88.40809845	86.74077015	91.94124653	88.80508138	93.767368
Specificity	88.63361548	86.29584845	91.97904071	89.11729141	93.8331318
Precision	88.76046234	86.53465347	92.08747515	89.23015556	93.91650099
FPR	11.36638452	13.70415155	8.020959291	10.88270859	6.166868198
FNR	11.59190155	13.25922985	8.058753474	11.19491862	6.232631997
NPV	88.27780008	86.50505051	91.83098592	88.68832732	93.68209256
FDR	11.23953766	13.46534653	7.912524851	10.76984444	6.083499006
F1-Score	88.58392999	86.63758921	92.01430274	89.01711102	93.84187525
MCC	0.770399882	0.730381613	0.839193741	0.779204278	0.875995467

Table 5.2: Comparative performance analysis of the proposed blockchain-based privacy-preserving authentication scheme for IoV against conventional classification models

Terms	CNN	MLP	Ridge-classifier	Ensemble	OBWO-HANet
Accuracy	86.1	89.72	88.16	91.6	93.8

Sensitivity	85.9468042 9	89.7578404 1	88.24930528	91.62366018	93.767368
Specificity	86.2555421 2	89.6815800 1	88.06932688	91.57597743	93.8331318
Precision	86.3926576 2	89.8291617	88.24930528	91.69646404	93.91650099
FPR	13.7444578 8	10.3184199 9	11.93067312	8.424022572	6.166868198
FNR	14.0531957 1	10.2421595 9	11.75069472	8.376339817	6.232631997
NPV	85.8059342 4	89.6093435 4	88.06932688	91.50221506	93.68209256
FDR	13.6073423 8	10.1708383	11.75069472	8.303535956	6.083499006
F1-Score	86.1691542 3	89.7934868 9	88.24930528	91.66004766	93.84187525
MCC	0.72200469 1	0.79438962 8	0.763186322	0.831991584	0.875995467

5.3 Performance Evaluation of the Proposed CD-RRNN-Based Attack Detection Model

Table 5.3 presents the overall performance evaluation of the proposed Cascaded Dilated Residual Recurrent Neural Network (CD-RRNN) for malicious attack detection in Internet of Vehicles environments, in comparison with widely used learning-based classifiers such as DNN, SVM, LSTM, and conventional RNN models. The evaluation is conducted using multiple standard performance metrics to comprehensively assess detection accuracy, robustness, and reliability. As observed from the results, the proposed CD-RRNN achieves the highest accuracy of 94.34% and sensitivity of 94.32%, while maintaining significantly lower false positive and false negative rates compared to all benchmark models. These improvements indicate a strong capability to correctly identify malicious activities while minimizing misclassification of legitimate vehicular communications. The superior performance of CD-RRNN is primarily attributed to its cascaded architecture combined with dilated residual connections, which enhance long-term dependency learning and effectively capture subtle temporal attack patterns. Overall, the results confirm that the proposed attack detection mechanism provides a reliable and efficient solution for strengthening authentication and security in dynamic IoV networks.

Table 5.3: Comparative analysis of proposed system with traditional models

Terms	DNN	SVM	LSTM	RNN	CD-RRNN(Proposed Model)
Accuracy	86.8	89.16	87.04	91.74	94.34
Sensitivity	86.82016673	89.20206431	87.13775308	91.66335848	94.3231441
Specificity	86.77952439	89.11729141	86.97497	91.8178154	94.35711407
Precision	86.9582505	89.27294398	87.13775308	91.91878981	94.43561208
FPR	13.22047561	10.88270859	13.0592503	8.182184603	5.642885933
FNR	13.17983327	10.79793569	12.86224692	8.336641524	5.676855895
FOR	13.36016097	10.95449054	13.0592503	8.440514469	5.7568438
PT	0.195111666	0.174642947	0.193564606	0.14938492	0.12229579
FM	86.88918118	89.23749711	87.13775308	91.79098529	94.37936134
PLHR	6.567098587	8.196678576	6.672492759	11.20279765	16.71540861
TS	76.81770285	80.56651129	77.20717552	84.82733284	89.35690109

5.4 Summary

The proposed blockchain-based authentication and privacy-preserving frameworks demonstrated superior performance compared to conventional IoV security approaches. The OBWO-HANet-based authentication model achieved higher accuracy, sensitivity, and reliability through optimized learning and blockchain integration, while significantly reducing false detection rates. The CD-RRNN-based security framework combined with MMRFO-optimized El-Gamal encryption (OKEE) further strengthened system protection by enabling efficient attack detection, faster encryption and decryption, lower memory consumption, and enhanced robustness. Overall, the experimental results confirm that the integrated learning, optimization, and cryptographic mechanisms effectively improve authentication, intrusion resilience, and data privacy in dynamic IoV environments, outperforming existing advanced models solutions.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

This thesis presented a comprehensive investigation into secure and privacy-preserving authentication mechanisms for Internet of Vehicles (IoV) environments by integrating blockchain technology, intelligent learning models, and optimized cryptographic techniques. The research addressed critical IoV challenges such as malicious attack detection, data integrity, identity privacy, and secure communication in highly dynamic vehicular networks. To achieve these objectives, two complementary frameworks were developed: a blockchain-enabled hybrid adaptive authentication model and a systematic blockchain-based privacy-preserving protocol incorporating deep learning-driven attack detection and optimized encryption.

The proposed authentication framework leveraged optimized blockchain integration with intelligent learning to eliminate centralized vulnerabilities and ensure trustworthy data exchange among vehicles and infrastructure. The hybrid adaptive learning model enhanced authentication accuracy while significantly reducing false detection rates. Furthermore, the introduction of a Cascaded Dilated Residual Recurrent Neural Network (CD-RRNN) enabled effective identification of complex temporal attack patterns in IoV traffic. To strengthen privacy preservation, an Optimized Key El-Gamal Encryption (OKEE) mechanism was introduced, where optimal key selection using the Modified Manta Ray Foraging Optimization (MMRFO) algorithm reduced encryption overhead and improved security robustness.

Extensive experimental evaluations demonstrated that the proposed frameworks consistently outperformed conventional IoV authentication and security solutions across multiple performance metrics, including accuracy, sensitivity, computational efficiency, memory utilization, encryption/decryption time, and security resilience. Overall, this research confirms that the synergistic integration of blockchain, deep

learning, and optimization-driven cryptography provides a scalable, efficient, and secure solution for next-generation intelligent transportation systems.

6.2 Major Findings

The key findings derived from this research are summarized as follows:

- **Blockchain-Based Authentication Effectiveness:** The optimized blockchain-integrated authentication framework achieved high accuracy and reliability, effectively mitigating impersonation, replay, and false data injection attacks while removing single points of failure inherent in centralized IoV systems.
- **Hybrid Adaptive Learning for Authentication:** The OBWO-HANet model demonstrated superior authentication performance by combining intelligent feature learning with optimized parameter tuning, achieving higher sensitivity and lower false positive and false negative rates compared to traditional classifiers.
- **CD-RRNN-Based Attack Detection:** The proposed CD-RRNN model achieved an attack detection accuracy of over 94%, outperforming DNN, SVM, LSTM, and conventional RNN models. Its cascaded and dilated residual architecture effectively captured long-term dependencies and subtle temporal anomalies in vehicular communication data.
- **Optimized Cryptographic Privacy Preservation:** The OKEE scheme with MMRFO-based key optimization significantly reduced encryption and decryption time, memory consumption, and computational overhead compared to classical cryptographic approaches such as DES, AES, ECC, and standard El-Gamal encryption.
- **Overall System Robustness and Scalability:** The combined frameworks ensured secure authentication, efficient intrusion detection, and strong privacy protection under dynamic IoV conditions, making them suitable for large-scale real-time vehicular deployments.

6.3 Future Scope

Although the proposed frameworks demonstrated strong performance, several research directions can be explored to further enhance IoV security and scalability:

- Development of lightweight deep learning models to reduce computational load on resource-constrained onboard units (OBUs).
- Integration of federated learning to enable decentralized model training while preserving vehicular data privacy. Exploration of hybrid or adaptive optimization algorithms to further accelerate cryptographic key generation and attack detection.
- Extension of the proposed frameworks to 5G and 6G-enabled IoV networks for ultra-low latency and high-reliability communication.
- Incorporation of quantum-resistant cryptographic techniques to prepare IoV systems for future quantum computing threats.
- Validation of the proposed models on large-scale real-world vehicular datasets and testbeds.
- Design of energy-aware authentication mechanisms to improve sustainability in intelligent transportation systems.
- Integration with edge and fog computing architectures to enhance real-time processing and scalability.

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